

Machine-Readable Economics Papers for Agentic Research

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Abstract

AI research systems read papers to retrieve literature, evaluate manuscripts, write code, and attempt reproduction, but the papers themselves remain mostly PDFs: readable by people, but hard to query as evidence. This paper builds an economics-specific pipeline that converts full economics-paper PDFs, including in-PDF appendices, into structured records linked to their source locations. Machine-readable here means more than LLM-ready text: each paper becomes records that can be stored in databases, queried directly, compared across papers, analysed with standard algorithms, and selectively returned to an LLM with source evidence. The pipeline extracts seven information types: data sources, methods, findings, limitations, open questions, citation evidence, and replication metadata. The same records are exposed as SQL tables, semantic-search indexes, and graph objects. The result is both a queryable database of paper evidence and a dataset about the structure of economics research. Applied to 1,608 selected 2025–2026 NBER working papers, the first build shows that this representation can be populated beyond a small coded sample and used for source-grounded search, cross-paper measurement, and graph-based research workflows.¹

Keywords: machine-readable papers; economics literature; source-linked extraction; knowledge graphs; hybrid search; retrieval-augmented generation; AI research agents; research infrastructure.

JEL codes: C55, C81, C88, D83, O33.

1 Introduction

AI research tools increasingly retrieve literature, evaluate manuscripts, write code, and attempt empirical reproduction from papers (Kohler et al., 2026; Zheng et al., 2025). The scientific object these systems need is the paper, but most papers remain PDFs: readable by people, hard to query as evidence. The bottleneck is not unique to economics, but economics is a useful first setting because working papers are central research objects and their evidence often depends on datasets, methods, estimates, identification designs, and appendix material that are not visible from metadata or abstracts alone. This paper treats the problem as one of representation, not

*Claude Code (Opus 4.6 and 4.7) and Codex (GPT-5.4 and GPT-5.5) were used to assist with drafting, editing, coding, extraction review, table and figure preparation, and cluster-label review. I reviewed the outputs and take responsibility for all statements, analysis, and errors.

¹The release reports coverage descriptives and a preliminary single-reviewer validation review in Appendix G; future versions will improve automatic promotion, validate source spans and typed links, and benchmark retrieval and downstream tasks before larger-corpus release.

only retrieval: economics papers need to be converted into structured research objects that can be queried, compared, and audited. Existing tools often treat papers as retrievable documents; this paper treats structured records over papers as the unit that researchers can search, measure, link, and audit.

The paper builds this layer for the 2025–2026 NBER working-paper corpus. The pipeline reads the full working-paper PDF, including appendices contained inside the PDF, and extracts seven kinds of paper information: data sources, methods, findings, limitations, open questions, citation evidence, and replication metadata. Each item keeps its source location: page, section, table, figure, equation, or passage. The same records are exposed through structured JSON, SQL tables, semantic-search indexes, and graph objects, so a researcher or agent can combine structured filters, keyword search, semantic retrieval, and link-following before returning to the source evidence. When a working paper later has a published version, this build still processes the NBER working-paper PDF; published versions enter only when they are cited by a processed paper or when a cited-work resolver links a reference to a published DOI. Existing systems already cover important parts of this space. Full-text search is an important first step: it lets researchers search across papers and return to the relevant passages, tables, figures, or sections. The layer built here is complementary: it converts economics papers into source-linked records for methods, data sources, findings, limitations, open questions, citations, and replication metadata. Those records can support search, but they can also serve as a data source for measuring patterns across economics papers. The two uses reinforce each other: structured extraction makes search more precise by turning recurring paper concepts into typed, source-linked records, while the database makes those records reusable across many later queries rather than one-off prompt outputs. Retrieval-augmented agents can search and cite full-text passages, and scholarly knowledge-graph projects such as ORKG show how research contributions can be made machine-actionable (Jaradeh et al., 2019; Lála et al., 2023). Section 2 and Appendix A locate this paper relative to those systems.

The economics-specific part is the ontology and provenance layer, not the PDF format itself. In practical terms, the pipeline gives each paper a field-specific coding guide: different ways of describing the same method, dataset, citation role, limitation, or open question are mapped to common labels, while each label remains linked to the source passage and extraction version.² The output should therefore be read both as infrastructure and as data: a database for querying paper evidence, and a structured dataset for studying the contents of the economics literature. The extraction layer also changes what search can mean: keyword search depends on exact wording, embeddings recover semantic similarity, and LLM preprocessing maps varied paper language onto controlled labels such as method family, dataset type, citation role, or limitation type. The broader contribution is to turn a repeated manual task—reading papers to code methods, datasets, findings, limitations, citations, and replication information—into reusable research infrastructure. For agents, the same layer augments retrieval with domain-specific evidence and provides guardrails by comparing proposed claims, methods, or checks against documented practice in the literature.

The paper makes three contributions. First, it proposes and implements an economics-specific structure for machine-readable papers: seven linked information types, source locations, controlled-vocabulary labels, and typed links among extracted objects. Second, it turns full-text papers

²Technically, this coding guide is a controlled vocabulary and ontology used with LLM extraction. It behaves less like a fixed dictionary of trigger phrases and more like a controlled vocabulary with thesaurus-like behavior: categories and canonical labels set boundaries, while the model can recognise paraphrases, abbreviations, and field-specific variants in context. The source and contract metadata record the prompt bundle, taxonomy, schema, model binding, run configuration, and source location, so a later error can be traced to the relevant extraction component and repaired locally before revalidation. On controlled vocabularies as semantic infrastructure for reconciling synonyms and variants, see Talisman (2025); on workflow ontology design and governance, see Talisman (2026).

into a cross-paper dataset and measurement layer. This is complementary to nearby economics text work. Goldsmith-Pinkham (2026) measures broad method trends with full-text keyword search; Garg and Fetzer (2026) extract a targeted claim graph of standardized concept nodes, claim edges, and evidentiary basis; and Briggs et al. (2026) code null-result reporting. This paper extracts core paper elements and preserves the relationships among them, so the same preprocessing can support generalisable analysis across several units: papers, paper objects, method–data pairs, finding–method links, limitation–open-question links, citation contexts, and corpus-level variables. Once findings, methods, datasets, disagreements, stated limitations, open questions, information gaps, and citation contexts are recorded as typed objects, they can be searched as text, aggregated as variables, or assembled into knowledge-graph inputs. Third, the paper applies the method to the 2025–2026 NBER corpus, reporting representation-coverage descriptives, a preliminary validation review, and four task setups—living systematic review, automated diagnostics, agentic reproduction, and AI peer review—designed for future benchmarking rather than benchmarked here.

The 2025–2026 NBER extraction is the first corpus run. Its descriptives focus on whether the representation is populated across papers, not on substantive economic findings. A first single-reviewer validation review in Appendix G diagnoses the current extraction quality for the main non-ML paper-object layers in a 100-paper sample after adding matched intermediate records; raising recall, automating promotion into final tables, and validating spans, links, and retrieval are the engineering priorities before larger-corpus release. The paper proceeds as follows. Section 2 motivates the layer. Section 3 defines the extraction method and paper objects. Section 4 describes the access modes: structured filters, keyword search, semantic search, and graph or edge traversal. Section 5 reports representation coverage in the 2025–2026 NBER run, and Appendix G reports the preliminary validation review. Section 6 sets out four illustrative downstream task setups for future benchmarking. Section 7 states the current scope, cost, and scaling path. Section 8 concludes.

2 Why AI Research Tools Need Machine-Readable Papers

Recent AI research tools show a common input problem. Across augmented and automated systems, papers are used to retrieve related work, extract claims and equations, check methods, attempt reproduction, generate code, and produce or evaluate manuscripts (Zheng et al., 2025). This paper adds an upstream layer for economics: source-grounded paper records that agents and researchers can query, compare, or check against. Figure 1 organises four task families where that layer is useful. The same representation supports researcher-facing augmentation—search, checking, comparison, and synthesis—and more autonomous workflows that carry tasks across the research cycle, from information gathering through hypothesis development, experimentation, analysis, and iteration (Lu et al., 2026).

Two adjacent approaches motivate but do not replace this layer. Retrieval-augmented scientific QA systems such as PaperQA retrieve full-text articles, select relevant passages, and generate cited answers (Lála et al., 2023). Scholarly knowledge-graph systems such as ORKG show how research contributions can be made machine-actionable through structured graph records (Jaradeh et al., 2019). The gap here is economics-specific preprocessing: full working-paper PDFs are converted once into reusable records whose methods, data, findings, citations, limitations, open questions, and replication metadata remain linked to their source evidence.

This distinction matters because several current tools can already read a PDF. A PDF reader can produce a task-specific analysis for one manuscript or one query. The object built here is different: a reusable evidence layer over many papers. It helps because (i) the same extracted record can answer many later questions without re-reading the PDF; (ii) answers can be com-

pared across papers because datasets, methods, findings, limitations, citation roles, and source locations use shared fields; (iii) a researcher can audit the exact passage, table, figure, equation, or citation context behind a claim; and (iv) the same records can be used through SQL tables, semantic search, and graph objects rather than only through an LLM transcript.

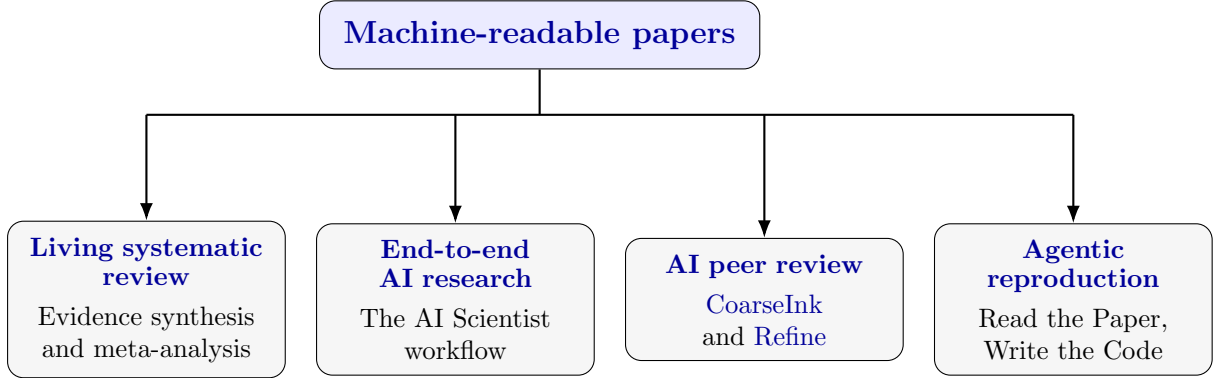


Figure 1: Machine-readable economics papers as reusable input for four AI research tasks (Kohler et al., 2026; Lu et al., 2026; Stocker et al., 2025; Zheng et al., 2025). For each task family, this paper contributes reusable structured paper evidence rather than a new end-user agent or one-off PDF reader: extracted data sources, methods, findings, limitations, open questions, citations, replication information, and source locations. Box titles link to representative source or tool pages.³

The four task families differ, but each needs papers in a structured form rather than only as PDFs or abstracts:

- (i) A living systematic review needs papers that can be screened, coded, updated, and re-analysed as the evidence base changes (Elliott et al., 2014; Iversen & Quintana, 2026).
- (ii) An end-to-end AI research system needs full-paper evidence when it checks novelty, chooses datasets, or chooses methods. In the public AI Scientist workflow, related work is retrieved through scholarly API results that expose metadata and abstracts, not a structured full-paper literature layer (Lu et al., 2026; Sakana AI, 2026).
- (iii) AI peer review needs both submitted-paper evidence and related-literature evidence: what the manuscript itself supports, what comparable papers usually report, and whether important prior work is being used as background, method precedent, comparison, or contradiction.
- (iv) Agentic reproduction needs the paper as a structured description of the study: data, sample, method, tables, equations, and the checks needed to decide whether a reimplementation failure reflects the agent, the extraction, missing data, or underspecification in the paper. It also needs related-paper context, especially when time has passed since publication, to identify later data treatments, validity checks, and robustness standards that go beyond simply rerunning the original code.

The peer-review case makes the distinction clearest. This paper does not claim that tools such as CoarseInk or Refine cannot analyse a submitted PDF. CoarseInk is publicly inspectable and reads the submitted file before running several review passes; Refine is a deployed review tool, but it is not currently open source, so its internal process is not audited here (CoarseInk, 2026; Refine Technologies, 2026). The contribution here is the evidence layer around the PDF: a

³The first box uses the established term *living systematic review*. It denotes an evidence synthesis that can be updated as new studies enter, including meta-analysis where appropriate, over structured full-paper records rather than only abstracts and bibliographic metadata.

checkable record of claims, tables, equations, references, assumptions, citation roles, related papers, and source locations that can ground, revise, or drop a review comment.

Agentic reproduction needs the original paper as an executable description of the study. *Read the Paper, Write the Code* shows that agents can recover published social-science results from paper-derived method and table descriptions plus the original data, without access to the authors’ code (Kohler et al., 2026). That benchmark addresses same-data reimplementations on papers that I4Replication had classified as fully reproducible; it is distinct from testing whether authors’ conclusions are robust to alternative specifications. A structured literature layer can help with the next step by selecting targeted alternative specifications, data treatments, and validity checks from nearby and later papers, while still leaving substantive judgment to the researcher (Dreber & Johannesson, 2025).

The implication is that parts of a paper can become reusable research components. Here, a component does not mean executable software; it means a source-linked record of a dataset, sample rule, constructed variable, method, table, equation, assumption, limitation, or robustness check that can be retrieved independently while remaining linked to the paper. Code is modular by design. Paper-derived records make the empirical components of a study more modular in a weaker but useful sense: they can be searched, compared, adapted, and checked across tasks without pretending that any component alone settles the substantive question. This is also a middle ground between abstract-only literature search and generating an empirical study from scratch. Systems such as The AI Scientist and APE-style economics pipelines need to choose ideas, datasets, methods, and checks; a structured paper layer can supply reusable components from existing papers that are candidates for adaptation to new research questions or datasets, while preserving the source evidence and uncertainty around each choice (Lu et al., 2026; Sakana AI, 2026; Social Catalyst Lab, 2026). Reproduction projects then become a natural testbed for these components, because they reveal whether the extracted records are specific enough to support reimplementations and whether failures come from the agent, the extraction, missing data, or underspecification in the original paper.

The same structure points to the next stage: literature-grounded robustness reproducibility. Recreate reproduction asks whether an agent can recover a published result from the paper and the same data. Robustness reproducibility asks whether the conclusion survives alternative plausible analytical decisions on the same data (Dreber & Johannesson, 2025). The structured corpus helps define those alternatives from the literature itself: papers already nearby in the literature and, as the corpus is updated, papers published later that use the same dataset, outcome, method family, or identification strategy; validity checks commonly reported for that design; author-stated limitations; and citation contexts where later papers compare, qualify, or contradict the original estimate. This matters most for older papers, where reimplementations is only the first question and later disciplinary practice may imply additional checks before treating the result as current evidence. This follows the logic of specification-curve analysis: the alternatives should be theoretically justified, statistically valid, and non-redundant, not arbitrary perturbations (Simonsohn et al., 2020).

Appendix B expands these task families into robustness, source-tracing, and reuse tables. The main text keeps the focus on the input problem: current tools would benefit from structured paper records with source links before they are used for high-stakes research workflows.

The limit of metadata- and abstract-centric retrieval. Current end-to-end research agents often access prior literature through scholarly metadata APIs such as Semantic Scholar or OpenAlex. In the public AI Scientist implementation, novelty checking and citation insertion query these APIs and show the model titles, authors, venues, years, citation counts, and abstracts where available (Lu et al., 2026; Sakana AI, 2026). Abstracts describe findings but omit the methods, validity tests, data structures, and assumptions that determine whether a result transfers to a new setting. An abstract that mentions “difference-in-differences” does not reveal

whether the parallel-trends test was run, reported, or passed — information the agent needs to judge reusability. Abstract retrieval also biases discovery toward direct thematic matches, suppressing cross-field connections that only become visible when methods and data structures are compared across literatures. Structured extraction addresses both gaps: extracted fields are queryable even when they describe what a paper did *not* do, and matching on methods or data can connect papers that share a design even when they use different vocabulary.

From literature review to living systematic review. The synthesis standard for evidence-based research is not a narrative literature review but a systematic review: pre-specified eligibility criteria, structured search and screening, study appraisal, and data extraction (Elliott et al., 2014; Iversen & Quintana, 2026). A meta-analysis is the quantitative component when comparable effects can be re-analysed; it is not synonymous with the review itself. Its value is to reconcile apparently contradictory findings and quantify “what is known and what is not yet known” (Gurevitch et al., 2018). Static reviews rapidly age: search dates lag publication, author-maintained review files go stale, and new studies can change pooled estimates or the interpretation of a field. *Living* systematic reviews address this by keeping eligibility, extraction, appraisal, and, where relevant, meta-analytic estimates current (Elliott et al., 2014; Iversen & Quintana, 2026).

Systematic Online Living Evidence Summaries extend this logic by framing living evidence products as continuously updated, curated, interrogatable evidence datasets enabled by automation (Hair et al., 2023). This paper borrows that database logic but changes the structure being maintained: rather than only keeping study-level evidence for a review, it decomposes each economics paper into source-linked data, methods, findings, limitations, open questions, citation evidence, and replication information.

The hard part is maintenance. Living reviews require transparent update rules, guardrails against reviewer bias, mechanisms for incorporating revised papers, and reusable outputs. Elliott et al. (2014) already pointed to semi-automation through text mining, data repositories, linked data, and semantic technologies. Later automation reviews show that much of the work has focused on search, screening, and primary-study selection, while extraction and knowledge representation remain harder; where extraction is automated, ontologies are often the mechanism for representing concepts and relationships (de la Torre-López et al., 2023; Van Dinter et al., 2021). Structured full-text extraction supplies the missing input layer. A living review can add new papers and revised working papers, then triage whether they add a new estimate, dataset, identification strategy, validity test, limitation, contradiction, or open question. Routine additions can be integrated into the review database; papers that materially change estimates, introduce new data/method combinations, or qualify a central claim can be flagged for closer human review. The database and graph outputs are useful research products in their own right: SQL tables let reviewers filter, aggregate, and re-analyse extracted information, while RDF/OWL-style graph representations expose shared concepts and relationships through common semantic conventions.⁴ That makes the processed literature reusable by other researchers, not just by the original extraction pipeline.

Citation roles as a quality signal for primary studies. Felizardo et al. (2012) use citation networks to help validate the selection of primary studies in systematic reviews. Citation role adds the semantic layer that a network edge lacks. Rather than recording only whether paper A cites paper B, the extraction records the citation function: background context, foundational

⁴This follows the living-review argument in Elliott et al. (2014): structured research outputs, linked data, and semantic technologies can make systematic-review data reusable beyond the original review and support a broader evidence ecosystem. RDF represents knowledge as subject–predicate–object triples, OWL 2 provides ontology vocabulary and semantics for classes, properties, individuals, and their relationships, and SHACL validates RDF data graphs against shape constraints (Berners-Lee et al., 2001; Cyganiak et al., 2014; Knublauch & Kontokostas, 2017; W3C OWL Working Group, 2012). Here, RDF/OWL-style refers to exporting extracted paper items and relationships in interoperable semantic-web forms, not to using RDF or OWL as the extraction model.

theory, methodological precedent, an effect-size benchmark, or a target of critique (Anderson & Lemken, 2023; Jurgens et al., 2018). This should not replace formal quality appraisal by review authors. It is a complementary signal from future citing authors: when later papers use a study as identification precedent, calibration input, or a result to contradict, they provide a different signal than when they cite it only as background. That distinction can guide snowballing and prioritisation in living reviews, especially when new papers arrive faster than human reviewers can read every PDF.

3 Extraction Method and Paper Objects

This section defines the paper representation used to make the PDF machine-readable in the database-and-query sense described above. The unit is a full working-paper PDF, including appendices contained inside that PDF. Each paper goes through five main stages, with enrichment as a 4.7+ substage before contract gating: reading the PDF and identifying sections and tables; classifying the paper type (empirical, structural, or theoretical); extracting the seven kinds of paper information; linking related items, such as findings to methods and limitations to open questions; and contract gating.⁵ The base path averages under two minutes per paper; enrichment adds roughly 30–80 seconds when triggered. For tables, figures, and equations, the system reads the downloaded working-paper PDF, including appendix tables, figures, and equations that are inside that PDF. Separate online appendices are not yet downloaded and processed as independent documents.

The output is deliberately redundant across access modes. The same extracted paper information is written as source-linked JSON, SQL tables, semantic-search views, and graph objects, so non-LLM tools can filter, aggregate, cluster, link, and traverse the records before any LLM is asked to interpret a retrieved passage. In database form, the records support retrieval, filtering, joins, and source inspection. In dataset form, the same records become observations and variables for analysing papers across the corpus. The indexing step is mechanical rather than interpretive: it takes the contract-gated paper information, source passages, table/figure/equation records, and document sections and creates paper profiles, extracted-item views, individual-field views, nearby-sentence views, broader context windows, relationship views, and claim-evidence bundles. These views are stored in DuckDB. When semantic search is needed, the text is embedded with `text-embedding-3-large`. Appendix H shows the database format and the search views in detail. Figure 2 summarizes the resulting paper representation.

⁵SHACL is used as a contract-checking layer, not as an extraction model. The shapes encode required fields, allowed relationship types, cardinality expectations, and controlled-vocabulary constraints. This catches malformed records before downstream use; it does not certify substantive extraction accuracy.

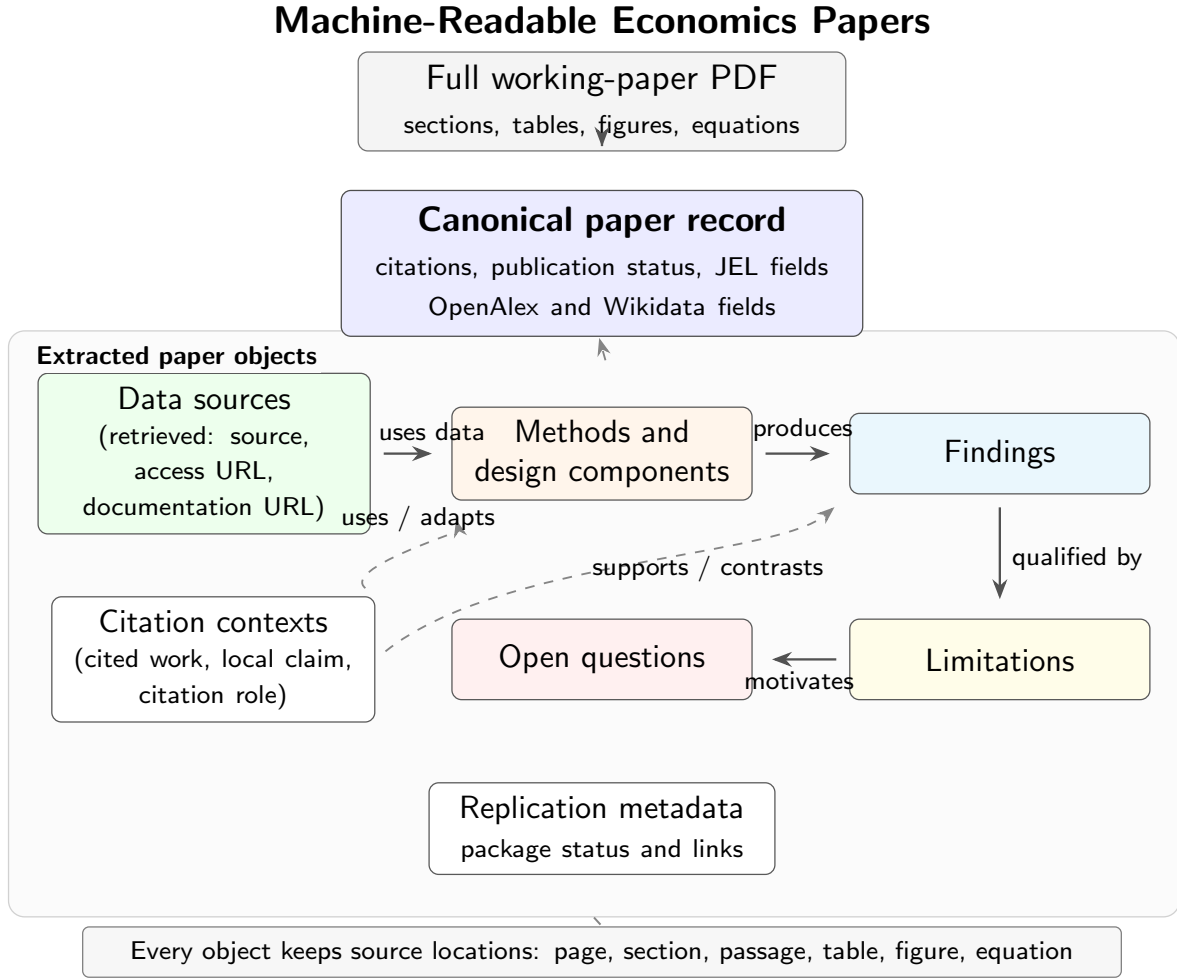


Figure 2: Machine-readable economics paper representation. A full working-paper PDF is converted into a canonical paper record and extracted paper objects. Solid arrows show the main inferential chain from data sources to methods, findings, limitations, and open questions. Dashed arrows show citation-context links and source-location provenance; the full relationship vocabulary is reported in Appendix E.

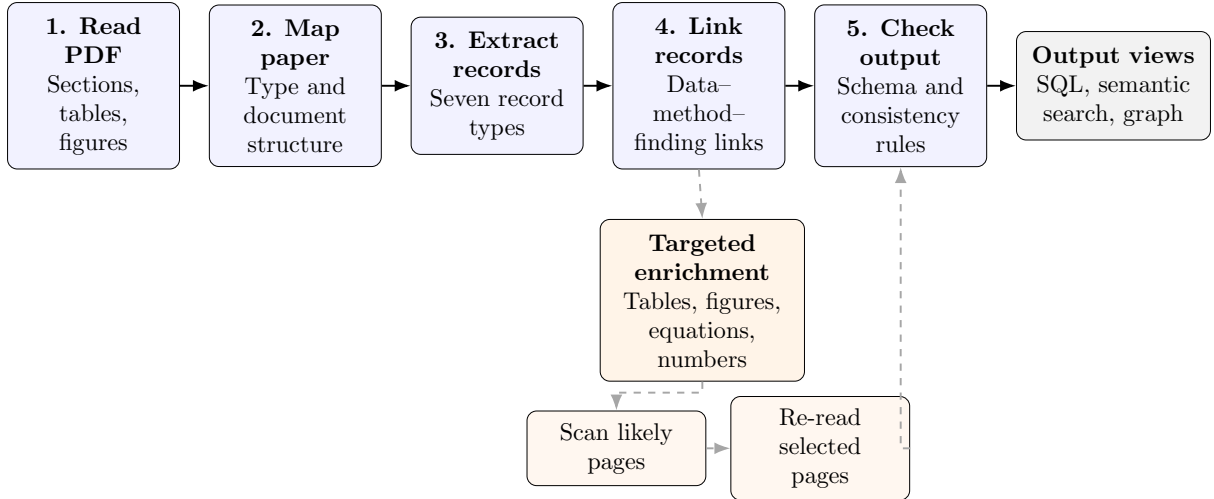


Figure 3: Extraction pipeline for turning one working-paper PDF into searchable paper records. The numbered boxes show the five interpretive stages: read the PDF and detect sections, tables, and figures; map the paper type and document structure; extract the seven record types; link related records such as data, methods, findings, citations, limitations, and replication fields; and check schema and consistency rules. The dashed branch is a targeted enrichment pass for tables, figures, equations, and numeric results: likely pages are scanned and selected pages are re-read before final checks. The gray box writes the checked record to SQL, semantic-search, and graph views without changing the extracted content.

Three properties of the multi-stage design matter for extraction quality. First, each stage has a narrower job: a method stage focuses on methods, while a finding stage focuses on findings. Second, important fields are checked more than once. Methods extracted in Stage 3 are reviewed when links are created in Stage 4, and precision fields such as standard errors and p -values are checked again during enrichment. Third, every extracted item is linked to the page, figure, table, equation, or passage where it appears. If an error reaches downstream use, the user can inspect the source, identify whether the failure came from source parsing, prompt logic, taxonomy mapping, linking, or enrichment, and then update or flag the affected component without re-running the full extraction. This isolation does not remove the need for validation after a fix, but it limits the surface area over which unrelated parts of the pipeline can regress. Full stage-by-stage prompts are in Appendix S.

The seven extracted information types, **data sources**, **methods**, **findings**, **limitations**, **open questions**, **citation evidence**, and **replication metadata**, capture the core inferential chain of an economics paper, from the data that identifies variation, through the methods that exploit it, to the findings that result, the limitations that qualify them, and the open questions they raise (Morgan, 2012). Labels use controlled vocabularies rather than free text wherever possible. Methods are coded at three levels: a broad purpose, displayed in the current run as measurement, causal, descriptive, structural parameter, structural, decomposition, or theoretical; one of 91 method-family labels, and, where needed, embedded method components. Algorithmic work can therefore appear either as the method itself—for example language models, causal forests, double machine learning, or prediction models—or as a component inside another design, such as record linkage inside an event study or a numerical solver inside a structural counterfactual. Shocks are recorded as variation sources inside causal identification designs; counterfactuals are recorded as structural or simulation method families. The remaining controlled vocabularies cover data-collection method categories and citation-function labels (Jurgens et al., 2018). Full controlled-vocabulary tables appear in Appendix R.

- **Data sources:** dataset name and provider, with web evidence where it can be resolved;⁶ collection method (8-category taxonomy); geographic scope; sample size; access status (public / restricted / proprietary); identification role.
- **Methods:** empirical, structural, descriptive, algorithmic, and measurement approaches used in the paper; method or design family (91-type taxonomy, see Appendix R); validity tests (test name, reported statistic, p -value, result) (Brodeur et al., 2020); how the identifying assumption was defended (statistical test / verbal defense / indirect evidence / not addressed); whether the strategy was deployed, mentioned as an alternative, or proposed for future work.
- **Findings:** reported results, including effect sizes where available, with standard error; null result (affirmatively recorded, not inferred from absence of text); whether the proposed causal mechanism was tested; whether the mediating variable was observed in the data.
- **Limitations:** author-stated constraints, including constraint type from a controlled vocabulary; severity; links to the open questions this limitation blocks.
- **Open questions:** author-stated follow-up questions, including data requirements; question type (mechanism, extension, external validity, policy, methodology, identification).
- **Citation evidence:** references that play a visible role in the citing paper, labelled by function from a taxonomy based on Jurgens et al. (2018) with economics-specific extensions (data or measurement, method justification, compares findings, background, etc.).⁷
- **Replication metadata:** paper-level replication-package status; package or repository URL where resolved; source of the resolution (paper text, OpenAlex, or directed search); and verification evidence for the matched package.

Extracted items are linked to show which data source supports which method, which method produces which finding, and which limitation motivates which open question. Appendix E lists the relationship types. The ten primary citation roles used in the main tables are literature survey, prior evidence, theoretical foundation, method justification, data or measurement, contextual motivation, contribution framing, estimate comparison, identification strategy, and contradiction. Coverage for the 1,608 selected processed NBER working-paper PDFs is reported once, in Section 5: the headline counts are 9,880 methods and design components, 10,498 findings, 7,659 data sources, 6,337 limitations, 4,364 open questions, 1,597 paper-level replication-package status records, and 120,064 role-coded citation links. Citation-role links are reported separately because their natural unit is the local citation context rather than the paper-level information list.

The paper counts differ across layers because the extraction target differs. The method layer records design components used in a paper’s own analysis; the 70 papers without a method record are almost all descriptive review, synthesis, agenda, or commentary-style papers (69 descriptive and one theoretical in the current JSON audit). The finding layer is broader, so many of those papers still contribute extracted claims or summary findings. Conversely, seven papers have no finding record under the current finding prompt; these are mostly method, database, measurement, or overview papers where the useful content is captured in methods, data sources, limitations, or open questions rather than as a discrete result statement.

⁶Dataset resolution uses web search, institutional pages, DOI metadata, OpenAlex, Crossref, and Wikidata where available. Ambiguous cases are marked for review rather than treated as fully resolved.

⁷Cited-work keys are resolved to DOIs via OpenAlex and Crossref where possible. The citing documents are the selected NBER working-paper PDFs; cited works may be working papers, published articles, books, reports, datasets, or other reference types. Unresolved references fall back to author-year identifiers.

4 How the Layer Is Queried and Used

The layer separates the access mode from the research task. Four access modes are used throughout the paper. **Structured filters** query fields such as method family, dataset, paper type, JEL code, source location, or citation role. **Keyword and BM25-style search** find exact terms and phrase variants in extracted records and source passages. **Semantic search** retrieves nearby passages, tables, and records when authors use different wording for similar concepts. **Graph or edge traversal** follows recorded links among paper objects, such as a method producing a finding, a data source supporting a method, or a limitation motivating an open question. Appendix H gives the implementation details, including the DuckDB views, embedding model, and rank-merging procedure.

These modes can be combined. A researcher designing a new study can filter to comparable IV papers, search for validity-test language, retrieve semantically similar source passages, and follow links from methods to findings, data sources, and reported limitations. At the paper-set level, database queries return counts and comparisons such as citation-role shares, algorithm-task frequencies, and data-access patterns. At the paper level, search can combine structured fields, page-level table/figure/equation evidence, and surrounding text. At the passage level, links trace a finding to the method that produced it, the data source it used, and the limitations the authors acknowledged.

This is the practical value of the graph representation. A search result can start from one known item and recover the related parts of the paper without rebuilding context from raw text. In the reproduction trace in Section 6.3, a 2025 NBER shock-design method links to its administrative data, source figure, event-time construction, and related robustness cluster. In the peer-review trace in Section 6.4, a broad comment is routed through focal-paper claims, methods, tables, limitations, citations, and comparator papers before becoming a source-checkable revision request. The same pattern can start from a dataset, method, table, limitation, open question, or review comment.

At corpus scale, each paper contributes information to the same structure, so the extraction supports direct comparison across papers: which identification strategies are shared across literatures, which datasets appear across JEL fields, which limitation types cluster across otherwise disconnected papers, which effect sizes are consistent or contradictory for the same treatment–outcome pair. These are systematic review and meta-analysis tasks (Gurevitch et al., 2018) that previously required manual evidence extraction. The structured paper layer makes them queries. The same objects also provide the inputs for a knowledge graph: documented facts, disagreements, limitations, and missing-information claims can be connected to the datasets, methods, citation contexts, and source passages that support or qualify them.

Table 1 gives concrete examples from the current paper. This is the practical sense in which the representation goes beyond a small hand-coded sample: the same source-linked record structure is used for corpus coverage, dataset navigation, within-paper link-following, living-review search, reproduction diagnostics, and peer-review enrichment. Some examples are database queries, some are semantic searches over passages and tables, and some require following graph edges. In each case, the result returns to extracted paper evidence rather than a free-form model answer.

The current edge layer is deliberately limited. It supports within-paper traversal among extracted objects, local citation links by citation role, and light cross-paper context through cluster labels, shared fields, and JEL codes. Full cross-paper citation networks and citation-role-weighted traversal are future work, because they require more complete cited-work resolution and additional validation of citation roles.

Embeddings are used to find candidate evidence, not to generate facts. A question, or a set of

Table 1: Examples of structured-paper records used in the paper

Use	Concrete example in this paper	What the records link or check	Where to inspect
Corpus coverage	The 1,608-paper NBER build records methods, findings, data sources, citations, limitations, open questions, and replication metadata.	Common paper IDs, item IDs, layer denominators, source locations, and processing metadata.	Section 5; processing ledger .
Dataset navigation	ACS, CPS, CRSP, Compustat, WDI, FRED, PWT, PSID, Comtrade, and NLSY entries distinguish product, provider, access status, documentation, and source role.	Data-source names, provider URLs, documentation links, access restrictions, product variants, and related table references.	Appendix I; dataset overview .
Within-paper link-following	The reproduction trace starts from a 2025 shock-design method and follows data, event-time, figure, and robustness links.	Method–data, method–artifact, finding, limitation, and related-cluster edges.	Section 6.3; retrieval trace .
Living-review search	The AI-economics pilot combines keyword search, semantic search, citation-role snowballing, and a 50-entry screening list.	Research questions, open questions, AI/ML method roles, citation functions, datasets, and source-linked paper groups.	Section 6.1; workflow table .
Reproduction and diagnostics	The reproduction trace links a focal shock design to its data, method, figure, and related-paper robustness cluster; the diagnostics trace checks generated ACS/CPS measurement claims.	Sample, data, method, table or figure anchors, and related-paper robustness or measurement precedents.	Sections 6.2 and 6.3.
Peer-review enrichment	The AI peer-review example turns broad comments on a GitHub Copilot field-experiment paper into source-checkable revision requests.	SQL filters, semantic matches, and graph edges connecting focal-paper claims, tables, methods, outcomes, limitations, citations, and comparator papers.	Section 6.4; Appendix Q; side-by-side edits .

subquestions, is embedded and compared with indexed paper text. Database fields still provide filters, joins, source locations, and the option to search only selected rows or columns. Keyword or BM25-style matching can narrow the candidate set before embedding search, or rerank embedding results that contain required terms. This lets a semantic query find nearby evidence even when authors use different phrasing, while the returned object remains a database row with paper ID, extracted-item ID, source passage, and linked table, figure, or equation evidence. Appendix H gives the search views, the simplified DuckDB format, and a worked link-following trace.

Strategy diffusion and technology adoption. Each identification strategy is classified as actually deployed, mentioned as an alternative, or only proposed for future work. The transition from proposed in one paper to deployed in a later paper is a technology adoption signal that exists nowhere in the text of any single paper and cannot be made reusable and auditable at corpus scale from raw text alone.

A simple dataset query can be made equally concrete: “What datasets do related education papers use to study student achievement and school accountability?” The main text leaves the long dataset answer to Appendix I, where the table has room to show product mode, access conditions, paper context, variables, transformations, and documentation links without interrupting the central argument.

A worked trace in Appendix [H](#) shows how a broad task can be routed from a paper-set pattern to a current-corpus paper and then to the method, data, figure or table anchors, and related-paper robustness context needed for the next check. The point is not fluent prose; it is that each step remains tied to extracted fields, paper links, and source evidence.

5 What the 2025–2026 Extraction Contains

This section treats the extracted layer as a dataset: papers are observations, and extracted objects become variables or linked records describing data use, methods, findings, citation roles, replication metadata, and stated uncertainty. The descriptives document coverage of the representation rather than headline claims about economics. They summarize coverage, data sources, methods, findings, replication-package metadata, citation roles, algorithmic-method use, and author-stated frontier signals.

5.1 Coverage of the Structured Paper Files

The first descriptive question is coverage: what information is available before any researcher or agent asks a substantive question. Table 2 uses the selected 2025–2026 processed papers as the denominator. It reports extracted items, papers with at least one item, and the per-paper density among nonempty papers. These counts describe representation coverage rather than substantive economic findings: they show how much of each paper has been converted into reusable research information.

Table 2: Main structured paper layers in the selected 2025–2026 NBER extraction ($N = 1,608$; the Papers column reports papers with at least one item of that type, and item counts use the unit named in the Counted item column)

Information type	Counted item	Extracted items	Papers	Mean	Median
Methods and design components	method or design component	9,880	1,538	6.4	7
Findings	finding statement	10,498	1,601	6.6	7
Data sources	data source	7,659	1,396	5.5	5
Limitations	author-stated limitation	6,337	1,492	4.2	4
Open questions	author-stated open question	4,364	1,439	3.0	3
Citation links	role-coded citation link	120,064	1,572	76.4	66

Note: The denominator is 1,608 selected processed papers from the 2025–2026 NBER corpus. No-item counts relative to this set are 70 for methods, 7 for findings, 212 for data sources, 116 for limitations, 169 for open questions, and 36 for nonblank citation-role links. Citation links are counted at the link level and are therefore reported separately from other paper-level information. Papers can contribute zero items in one layer while still contributing items elsewhere: papers without method records are mostly descriptive review, synthesis, agenda, or commentary-style papers; the few papers without finding records are mainly method, database, measurement, or overview papers.

Replication-package metadata is paper-level rather than item-level, so it is reported separately from the per-paper density table. The resolver records a package status for 1,597 of the 1,608 selected papers, and 132 papers have a resolved package or repository URL. Appendix Table 26 reports resolver details.

Layer denominators differ because several reviewed layers were rerun or checked separately; Appendix Table 25 gives the consolidated processing ledger.⁸

5.2 Data Sources

Data-source extraction records the empirical inputs that papers use, not just the variables named in a table. The reviewed data-source type layer covers 1,391 papers with data-source

⁸The base selected processed-paper set contains 1,608 papers. Citation roles cover 1,572 papers with at least one nonblank role-coded link. The reviewed data-source-type layer covers 1,391 papers, while the main extraction contains data sources in 1,396 papers. Replication-package metadata has status records for 1,597 papers. The algorithmic and ML method layer was rerun on 1,603 papers. The frontier-signal cluster figures use routed open-question and limitation statements from 1,381 papers; Appendix L reports the embedded, assigned, and displayed-cluster counts.

labels. Administrative records dominate the sample, but the paper set also contains extensive survey, proprietary, document, digital-trace, experimental, geospatial, and author-constructed sources. That mix matters because access conditions, linkage steps, measurement roles, and replication feasibility differ sharply across source types.

Table 3: Most common data-source types and data structures in the reviewed data-source layer

Panel A. Data-source types			
Data-source type	Papers	Share	Common measurement roles
Administrative records	1,101	79.2%	not specified (1049); outcome (898)
Survey data	557	40.0%	outcome (425); not specified (245)
Proprietary/commercial records	499	35.9%	outcome (324); not specified (265)
Text/document corpus	415	29.8%	not specified (272); treatment/exposure (134)
Digital trace/platform data	369	26.5%	not specified (164); outcome (135)
Census or population register	153	11.0%	not specified (71); outcome (64)
Synthetic/generated data	107	7.7%	not specified (47); calibration (39)
Experimental data	106	7.6%	outcome (83); treatment/exposure (23)
Panel B. Data structures			
Data structure	Papers	Share	Interpretation
Unbalanced panel	1,050	75.5%	Most common reusable format for linked administrative, firm, household, or regional panels.
Cross-section	767	55.1%	One-period or pooled snapshots used for descriptive, survey, or benchmark designs.
Time series	620	44.6%	Macroeconomic, market, policy, and repeated aggregate settings.
Repeated cross-section	318	22.9%	Repeated survey or administrative snapshots without persistent unit linkage.
Spatial data	128	9.2%	Geographic units, exposure assignment, or map-based measurement.
Balanced panel	92	6.6%	Repeated observations with complete panel structure.
Hierarchical data	74	5.3%	Nested individuals, schools, firms, places, or markets.
Document/text data	31	2.2%	Text corpora and document-derived measurement layers.

Note: Shares in this table use the reviewed 1,391-paper data-source type layer. Shares do not sum to 100% because a paper can use several source types or structures. The field-composition figures and full type table are kept in Appendix N.

5.3 Methods and Design Components

The method layer treats a method as a paper-specific design component: an estimator, identification strategy, measurement procedure, simulation, model, data-construction step, or diagnostic that the paper actually uses. This is broader than one item per identification strategy, because economics papers often use several methods and design components together. Table 4 shows both the high-level purpose of these method items and the most common method families. The key descriptive point is that the paper set is not only about causal identification; measurement and data construction are the largest part of the structured method layer.

These are component-level method records, not one primary research design per paper. The denominator therefore mixes primary designs, supporting estimators, data-construction steps, diagnostics, and robustness components. The table should be read as coverage of the structured method layer, not as a claim about the share of economics papers whose main design is measurement, causal, or structural.

The same method records also retain diagnostic fields useful for targeted design review: standard-error treatment is captured for 8,832 method items (89.4%), fixed-effect structure for 8,656 (87.6%), identifying or key assumptions for 6,693 (67.7%), specification variants for 4,363 (44.2%), and validity defenses and acknowledged threats for 2,300 each (23.3%). These fields let a reviewer filter a method family by reported assumptions, standard-error treatment, or validity defenses before returning to the source passage.

Table 4: Component-level methods and design records in the selected 2025–2026 extraction

Panel A. Method purpose			
Purpose	Method items	Share	Interpretation
Measurement	5,501	55.7%	Constructing variables, links, classifications, indices, or descriptive empirical inputs.
Causal	2,334	23.6%	Design components where authors claim causal identification or estimate causal effects.
Descriptive	877	8.9%	Summary, descriptive, accounting, or non-causal empirical patterns.
Structural Parameter	501	5.1%	Estimating structural parameters or model-implied primitives.
Structural	278	2.8%	Structural model solution, calibration, or simulation components.
Decomposition	260	2.6%	Variance, contribution, shift-share, Oaxaca, or accounting decompositions.
Theoretical	124	1.3%	Formal, conceptual, or model-building components used in the paper.
Panel B. Most common method families			
Method family	Method items	Share	Typical role in the paper set
Ols Regression	1,139	11.5%	Baseline regression and measurement/association work.
Index Construction	674	6.8%	Composite measures, scores, bundles, or constructed empirical inputs.
Event Study	565	5.7%	Dynamic treatment, policy, or event-window estimates.
Fixed Effects	520	5.3%	Panel or high-dimensional controls used inside empirical specifications.
Record Linkage	483	4.9%	Matching people, firms, places, products, texts, or administrative sources.
Specification Test	440	4.5%	Validity, sensitivity, diagnostic, or robustness checks.
Reduced Form Ols	415	4.2%	Reduced-form empirical models and first-stage style specifications.
Difference In Differences	414	4.2%	Treatment-control designs using timing or policy variation.
Descriptive	392	4.0%	Descriptive statistics, accounting, maps, tabulations, or plots.
Instrumental Variables	323	3.3%	IV/2SLS and related exclusion-restriction designs.
Counterfactual Simulation	308	3.1%	Counterfactual, policy, or model-based simulated scenarios.
Structural Estimation	256	2.6%	Estimated structural models and parameterized mechanisms.

Note: The denominator is 9,880 component-level method items in 1,538 papers. A method item is a paper-level design component, so a single paper can contribute multiple methods and multiple method purposes. Field labels are normalized for display from the extracted controlled vocabulary.

5.4 Findings and Source Evidence

The finding layer connects each empirical result to the rest of the paper information. Findings link to methods, data sources, tables, sections, and source passages where available. This matters because an agent should not only retrieve a claim; it should know which method produced it, which table or paragraph supports it, and whether a numeric estimate, significance statement, sample description, or data link is present.

Table 5: Finding-item types and source evidence in the selected 2025–2026 extraction

Panel A. Finding families			
Finding family	Findings	Share	Interpretation
Associational	2,689	25.6%	Non-design empirical associations and descriptive regressions.
Heterogeneity	2,370	22.6%	Subgroup, treatment-effect heterogeneity, mechanisms, or differential effects.
Null	2,002	19.1%	Explicit null, insignificant, or no-detectable-effect findings.
Causal	1,796	17.1%	Findings attached to causal designs or causal effect claims.
Structural	1,055	10.0%	Structural estimates, model-implied moments, calibrations, or scenarios.
Missing	586	5.6%	Findings with family not normalized in the source file.
Panel B. Source evidence and quantitative fields			
Field present in finding item	Findings	Share	Why it matters
Source location present	9,928	94.6%	Allows the result to be checked against the paper text.
Source location matched to full text	9,515	90.6%	Indicates the source location was matched back to parsed full text.
Point estimate recorded	7,947	75.7%	Supports quantitative comparison and meta-analysis candidates.
Table reference recorded	7,074	67.4%	Links a result to a table or regression output.
Linked data source	8,362	79.7%	Connects findings to empirical inputs.
Sample description recorded	7,017	66.8%	Supports eligibility and comparability checks.
Significance indicator recorded	6,467	61.6%	Captures author-reported statistical interpretation.
Significance level recorded	6,420	61.2%	Preserves reported threshold or notation.
Observation count recorded	2,396	22.8%	Supports scale, precision, and reproducibility checks.
p-value recorded	1,972	18.8%	Records exact inferential evidence when stated.
Standard error recorded	1,151	11.0%	Supports direct precision calculations when stated.
Confidence interval recorded	83	0.8%	Supports interval overlap and meta-analytic screening.

Note: The denominator is 10,498 finding items in 1,601 papers. A finding can be useful even when some quantitative fields are not present, but field gaps should be interpreted through source review rather than treated as validated absence.

5.5 Citation Roles and Functional Grounding

The extracted citation graph is functionally differentiated rather than purely bibliographic. Across 120,064 role-coded citation links in the 2025–2026 citation-role layer, literature-survey links are the largest category in both years: 40.4% of 2025 links and 38.1% of 2026 links. That is the lower-specificity background baseline against which more evaluative forms of citation should be read. Method-justification, identification-strategy, estimate-comparison, and contradiction links together account for 13.0% of all role-coded links, identifying cases where citing authors use prior work as method infrastructure, identification precedent, estimate benchmark, or object of explicit disagreement. The value is comparative rather than definitive: two cited papers can receive the same amount of attention but very different profiles of substantive uptake.

The current citation-role claim is strongest at the local-context level: how a processed paper uses a cited work in a specific passage. Cited-work profiles and cross-paper citation networks require bibliographic resolution, which remains preliminary in this release.

The citing side is the selected NBER working-paper PDFs. The cited side is broader: references can point to NBER working papers, published articles, books, reports, datasets, or other sources;

where DOI resolution identifies a published version, the cited-work key can point to that version. The unit in the counts is a local citation context, not a unique cited work. Each local citation link receives one primary role so the role distribution is additive,⁹ but the same cited work can receive different roles elsewhere in the same paper or across other citing papers. For that reason, the main text reports the role distribution of local citation contexts, not a resolved cited-work network. Appendix P reports the cited-work resolver audit and treats cited-work profiles as exploratory.

Table 6: Citation-link-level role distribution in the 2025–2026 citation-role layer

Role	2025 citations	2025 share	2025 papers	2026 citations	2026 share	2026 papers
Literature survey	35,840	40.4%	1,106	11,966	38.1%	401
Prior evidence	14,494	16.4%	1,022	5,430	17.3%	378
Theoretical foundation	10,340	11.7%	1,019	3,941	12.5%	384
Method justification	7,422	8.4%	977	2,553	8.1%	342
Data or measurement	7,027	7.9%	1,028	2,483	7.9%	381
Contextual motivation	6,886	7.8%	966	2,559	8.1%	360
Contribution framing	2,664	3.0%	759	891	2.8%	271
Estimate comparison	1,958	2.2%	639	762	2.4%	243
Identification strategy	1,468	1.7%	539	663	2.1%	203
Contradicts	529	0.6%	279	188	0.6%	103

Note: Shares are within role-coded citations for each year. Paper counts report the number of papers with at least one citation in the role. The 2025 layer contains 88,628 role-coded citations; the 2026 layer contains 31,436.

A few reviewed examples make the distinction concrete. A literature-survey citation places a work in a field; a method-justification citation uses it as procedural precedent; an estimate-comparison citation benchmarks magnitudes; and a contradiction citation marks explicit disagreement. Appendix O reports one reviewed example for each role.

5.6 Algorithmic and Machine-Learning Use in the 2025–2026 Paper Set

The 2025–2026 algorithm-method layer changes how machine-learning use is counted. By the broader definition used here, algorithmic and machine-learning use is common: 592 of 1,603 papers in that layer contain at least one extracted algorithmic task or machine-learning use. This is broader than counting papers with an explicit machine-learning method label. Algorithms can be standalone methods, but they can also be components inside causal, structural, measurement, or data-construction designs. The relevant descriptive unit is therefore not a binary “ML mention.” For most economics papers, algorithms enter as *input tasks*: linking records, coding documents, imputing or predicting variables, assigning locations, compressing high-dimensional inputs, or estimating heterogeneity inside an otherwise standard causal or descriptive design. The pattern matches the use documented by Ash and Hansen (2023), where text algorithms serve as the measurement step rather than the final estimand.

This complements, rather than replaces, keyword-based trend measurement. Goldsmith-Pinkham (2026) shows that full-text keyword matching is a low-cost way to track broad credibility-revolution trends, with benchmark comparisons showing that keyword methods can perform well for many broad method categories. That is the right tool for broad method mentions and time trends. The structured extraction here is useful for a different question: where the algorithm enters the research process, what empirical object it creates, and what paper evidence supports that classification. Hybrid search can then combine keyword filters with structured fields and embeddings, so a reader or agent can move from a broad method term to the nearby method context, data-construction step, validation table, or figure evidence (Edge

⁹Future versions will store raw citation-count fields separately from role-coded local-context counts, and will allow secondary role labels where a single local citation context serves multiple functions. The current tables keep one primary role per local context so shares remain additive.

et al., 2024).

Table 7: Paper-level algorithmic use by contextual family, 2025–2026 NBER sample

Contextual use family	Papers	Share of algorithmic papers
Linked panels and matched units	304	51.4%
Text-coded treatments, mechanisms, and documents	238	40.2%
Spatial, visual, and OCR measurement	130	22.0%
Predicted exposures, risks, and imputed states	99	16.7%
Latent types and community structure	31	5.2%
Heterogeneity and policy learning	24	4.1%
Engineered measures and derived inputs	12	2.0%

Note: The denominator is 592 papers with at least one extracted algorithmic or machine-learning method. Shares do not sum to 100% because papers can use algorithms for more than one empirical purpose.

The table makes the main pattern visible: the dominant algorithmic use is data construction—record linkage, document coding, spatial assignment, OCR, and prediction-imputation—rather than primary estimation. Text-as-data tasks (Ash and Hansen (2023)) alone appear in 40% of algorithmic papers. Detailed JEL distributions, researcher-facing coding fields, exact algorithm families, validation signals, and method-citation groups are reported in Appendix J.

5.7 Frontier Signals: Open Questions, Limitations, and Sources of Variation

The extraction makes it possible to read the corpus not only for findings, but for authors’ own descriptions of what remains unresolved. The analysis aggregates two stated objects: *open questions*, which say what future work still needs to learn, and *limitations*, which say why the current evidence does not fully settle the issue. This is not a topic model over abstracts or a citation community map. The units are paper-level statements about uncertainty, data needs, external validity, measurement, and identification.

Keeping the two objects separate is useful. Open-question clusters provide signals about the substantive frontier: care channels, belief formation, employer adjustment, transportability of credit and health-policy findings, school mechanisms, climate and energy policy, AI and innovation, and related agendas. Limitation clusters describe the binding obstacles: missing pay margins, local institutions, place-bound green-policy evidence, system-specific health payments, selected platform samples, weak statistical power, imperfect links, and the absence of credible shocks. Together they show both what is not known and why it remains hard to know.

These are signals, not a complete map of the frontier. Authors may understate limitations or open questions when they are already pursuing follow-up work, when they do not want to weaken the paper’s contribution, or when uncertainty is strategically framed. Aggregation is still useful because repeated constraints across unrelated papers can reveal field-level bottlenecks that any one author may not emphasize. Future work can combine these author-stated statements with forward citation contexts, where later researchers describe, qualify, or criticize earlier work. That external signal is also strategic: later authors may overstate prior limitations to position their own contribution. The useful object is therefore a source-linked set of signals, not any single limitation statement treated as definitive.

5.7.1 Stated Open Questions and Limitations

Figures 4 and 5 show the 15 largest researcher-facing frontier-signal groups for open questions and limitations. These 15 groups are drawn from 85 open-question clusters and 129 limitation clusters. Bar length shows each group’s share within the displayed 15-group set; the numbers correspond to the researcher-facing labels in Appendix K. The input contains extracted

open-question and limitation statements from 1,381 2025–2026 NBER papers. The bars are shares within the displayed largest assigned clusters, not shares of all extracted open-question or limitation records. Appendix L defines the clustering denominators, embedded-record counts, assigned-record counts, and cluster-display choices.

Open-question frontier ladder: largest 15 researcher-facing groups

Selected from 85 assigned clusters. Bar length shows each group's share within the 15 displayed groups. Labels stay outside the bars; numbers match the map and lookup table.

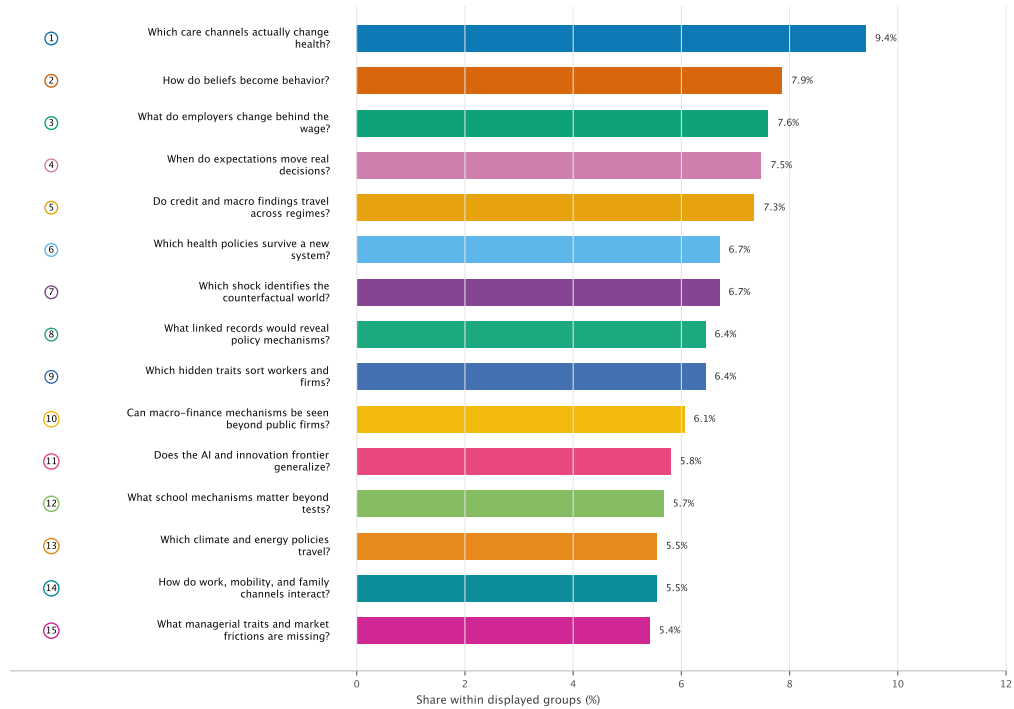


Figure 4: Relative prominence of the 15 largest open-question frontier-signal groups in the 2025–2026 NBER extraction. The displayed groups are the largest researcher-facing groups drawn from 85 HDBSCAN open-question clusters over statements from 1,381 papers. Bar length shows each group's share within the displayed 15-group set, not its share of all extracted open-question records. Codex (GPT-5.5, extra-high reasoning) reviewed all input text for each displayed cluster before assigning the researcher-facing labels. Numbers and labels correspond to Appendix Table 50; Appendix L reports the full clustering denominators and technical choices.

Limitation frontier ladder: largest 15 researcher-facing groups

Selected from 129 assigned clusters. Bar length shows each group's share within the 15 displayed groups. Labels stay outside the bars; numbers match the map and lookup table.

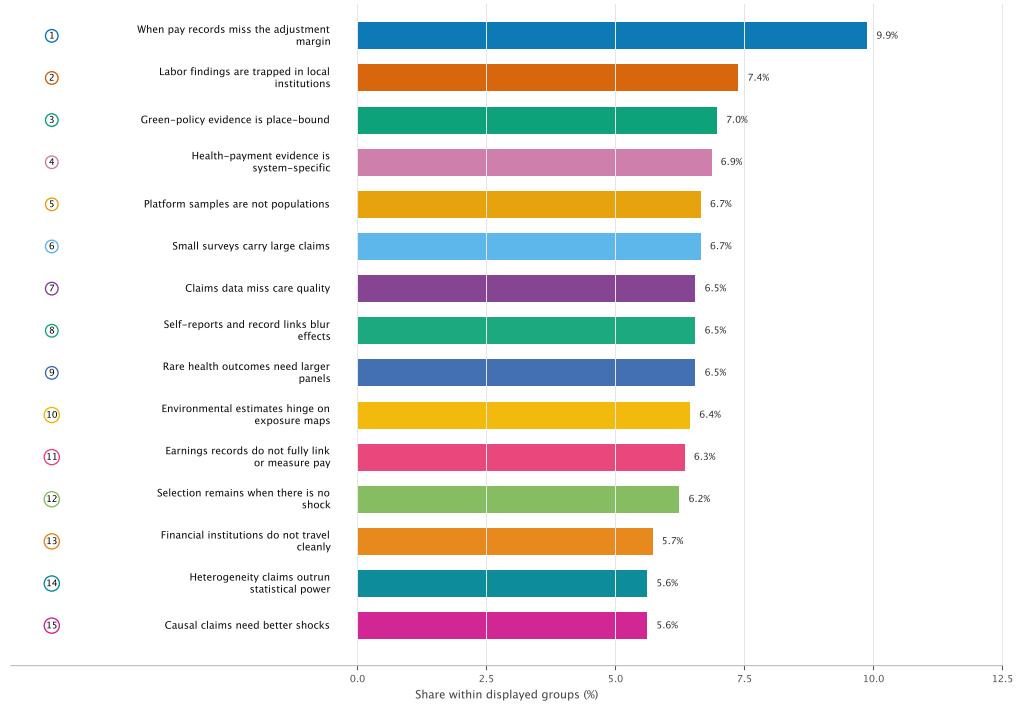


Figure 5: Relative prominence of the 15 largest limitation frontier-signal groups in the 2025–2026 NBER extraction. The displayed groups are the largest researcher-facing groups drawn from 129 HDBSCAN limitation clusters over statements from 1,381 papers. Bar length shows each group's share within the displayed 15-group set, not its share of all extracted limitation records. Codex (GPT-5.5, extra-high reasoning) reviewed all input text for each displayed cluster before assigning the researcher-facing labels. Numbers and labels correspond to Appendix Table 51; Appendix L reports the full clustering denominators and technical choices.

Table 8: Reading paired frontier-signal clusters

Open-question signal	Limitation pattern	Reading
Care channels and health	Claims data miss care quality	Health papers often identify utilization or spending effects more easily than care processes, patient experience, or clinical mechanisms.
Beliefs, attention, and behavior	Beliefs are hard to observe repeatedly	Papers can measure stated expectations, but linking belief updating to later choices requires repeated belief and action data that are rarely available together.
Employer adjustment margins	Pay records miss adjustment margins	Labor papers often observe wages or employment, while hours, benefits, job content, and internal employer responses remain less visible.
Credit and macro transportability	Financial institutions do not travel cleanly	Findings tied to particular credit systems, regulatory regimes, or balance-sheet definitions may not generalize across settings.
Linked records and mechanisms	Agency, identifier, and privacy barriers	Many frontier questions require linking administrative systems that were not designed to be joined.
Climate and energy policy	Exposure maps and local infrastructure	Climate and energy estimates depend heavily on geography, exposure measurement, infrastructure, and local policy environments.
School mechanisms beyond tests	Education records emphasize achievement	Education studies often observe test scores better than classroom mechanisms, peer interactions, or non-test outcomes.
AI, innovation, and technology adoption	Rapidly changing technologies	The technology frontier moves faster than many long panels and administrative data systems, making generalization difficult.

Note: The table gives a reader-facing interpretation of selected large open-question and limitation clusters. The full 15-cluster lookup tables and clustering denominators are reported in Appendices [K](#) and [L](#).

A second clustering exercise applies the same logic to method-level shock and natural-experiment design records. This is useful because the controlled method taxonomy identifies broad design families, while the cluster labels show recurring sources of identifying variation inside those families: information disclosures, eligibility margins, public-service rollouts, policy surprises, place-based exposure, credit conditions, and incentive offers. These clusters are especially relevant for literature-grounded robustness and agentic reproduction, where the question is not only whether a paper uses a difference-in-differences, IV, or event-study design, but what kind of variation the design relies on.

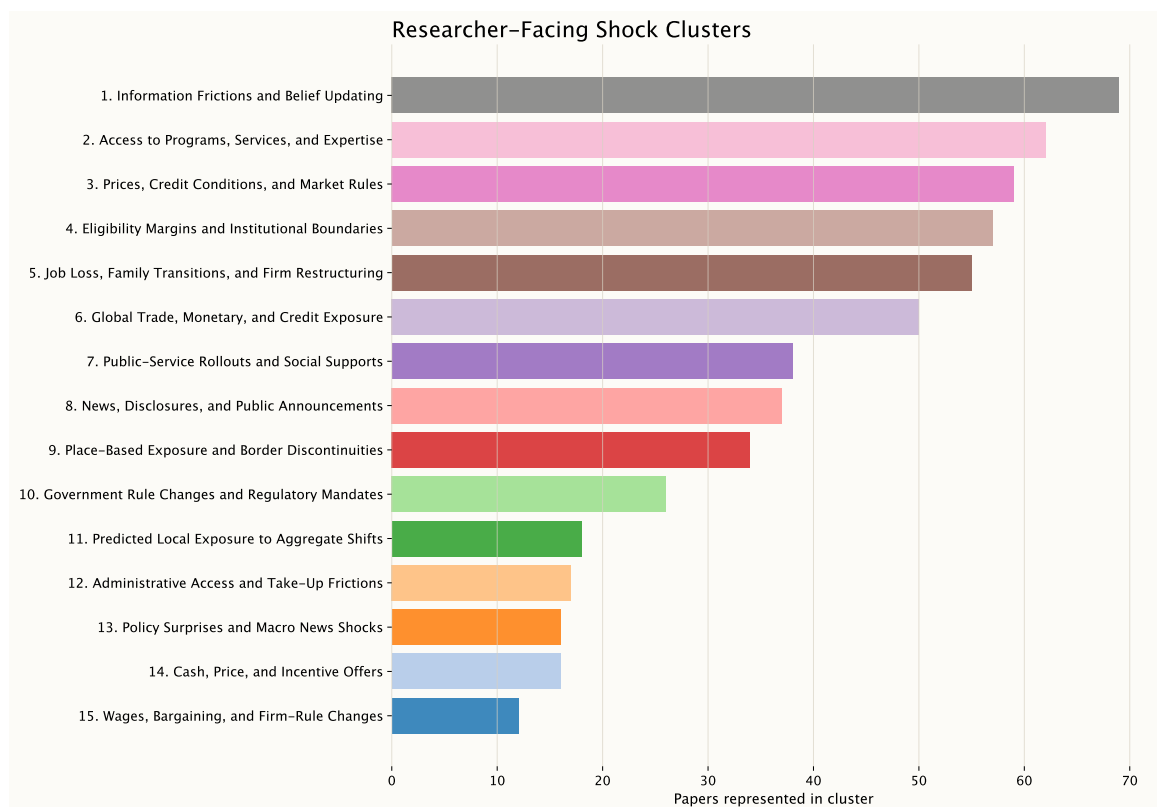


Figure 6: Relative prominence of the largest shock and natural-experiment design clusters in the 2025–2026 NBER extraction. The displayed groups are the 15 largest researcher-facing clusters by paper count among assigned method-level shock and design records. Bar length shows the number of papers represented in each displayed cluster, not a share of all methods or all papers. Appendix M reports the clustering denominators, embedding map, and representative source-linked examples.

6 Illustrative Task Setups

The descriptive tables show what becomes visible once papers are available in a structured form. Appendix G reports a preliminary validation review, and broader accuracy checks and task-level benchmarks remain in progress. The next four examples are task setups rather than benchmarks of complete downstream systems. Their purpose is to make the use cases concrete and to define experiments that can be run in future versions. The hypothesis is that the source-linked layer improves retrieval, review, diagnosis, or reproduction decisions; the examples below show how such tests would be structured rather than estimating the effect of the layer on task performance.

Table 9: Illustrative task setups and future benchmark designs

Task family	Current illustration	Future comparator	Outcome to measure
Living systematic review	AI-related economics search, citation follow-up, and 50-paper starting list	Keyword-only or abstract-only search followed by reviewer screening	Relevant papers found, important citations recovered, reviewer time, and agreement with expert inclusion decisions
Automated research diagnostics	Checks for generated economics papers against related data, method, and limitation records	Generated-paper review using only the manuscript PDF and general web or scholarly search	Unsupported novelty claims, weak data or method choices detected, and false alarms requiring reviewer correction
Agentic reproduction	Reproduction-failure diagnosis and literature-grounded robustness menu	Reimplementation from the focal paper alone	Correct attribution of failures, matched tables/equations, and quality of robustness checks selected from related papers
AI peer review	Evidence-grounded rules for keeping, revising, or dropping review comments	PDF-only review plus title/abstract literature search	Comment accuracy, source support, missing-literature detection, and rate of unsupported comments removed

Note: The current paper reports task setups and compact illustrative traces. It does not estimate the effect of the layer on these downstream outcomes.

The examples are organized around the same simple pattern: (i) ask a research-task question; (ii) retrieve structured records and their source locations; and (iii) decide whether the evidence supports an action, such as adding a paper to a review queue, rejecting a weak generated design, planning a robustness check, or keeping, revising, or dropping a review comment.

6.1 Living Evidence Map: AI-Related Economics

A living systematic review repeatedly adds new and revised papers, screens them, records the key study information, and updates the synthesis when the new evidence can change the answer (Elliott et al., 2014; Iversen & Quintana, 2026). The AI-related economics example should be read as a living-review starting search, not as a completed systematic review. Its purpose is to show what changes when papers have already been converted into source-linked records: the reviewer can search research questions and open questions, follow citation contexts by function, separate core evidence from methods context, and decide which papers need human reading next.

The first pass combined two discovery routes. A keyword search for AI terms found 91 candidate papers. A meaning-based search over extracted research questions and open questions found 329

candidate papers, and the combined starting set contained 358 papers. The initial appendix list retained 50 paper or cited-work entries for reviewer screening. A deeper follow-up then processed 58 first-stage NBER papers from the top review queue and extracted 4,392 citation-role assignments. That distinction matters: the 50-entry list is the compact review queue, while the 58-paper extraction is the richer evidence map used to classify subtopics, citation roles, datasets, constructed variables, limitations, and open questions.

Table 10: AI living-review pilot as an evidence-map workflow

Step	Structured layer used	Example output	Reviewer action
Search	Keyword and semantic search over extracted research and open questions	91 keyword candidates, 329 semantic candidates, and 358 combined candidates	Define the first screening universe.
Citation snowballing	Citation contexts labelled by function	Data, method, benchmark, identification, and contradiction references separated from background citations	Download and inspect references that do review work.
Paper-level coding	Extracted methods, findings, data, limitations, open questions, and ML methods	58 processed first-stage papers grouped by AI object, AI measurement, theory, finance, methods, and adjacent-comparator roles	Decide which papers are core evidence, method anchors, or context only.
Variable audit	Constructed-variable and ML-in-data-pipeline fields	Logs, surveys, embeddings, classifiers, policy texts, patent labels, image inputs, and model outputs used to construct variables	Code raw input, construction method, ML role, validation, and downstream use.
Evidence synthesis	Findings, limitations, open questions, citation roles, and artifact QA	Adoption, workplace effects, exposure construction, AI-measurement validity, policy and innovation measurement, finance, and long-run theory groups	Build subtopic memos and screen which branches are ready for later meta-analysis.

The most important result of this exercise is not a ranked bibliography but a cleaner review architecture. The AI-related set is not one pool of comparable effect estimates. It separates into AI use and adoption; exposure and task construction; workplace, productivity, and inequality evidence; AI-as-measurement and text-as-data methods; validity, leakage, and model-evaluation papers; finance and macro applications; innovation and policy measurement; long-run theory; and adjacent comparator papers. These groups use different evidence: some estimate short-run causal effects of AI tools, some describe adoption or expectations, some build AI-derived variables, some provide theory, and some are useful mainly as method or citation anchors. Pooling them mechanically would erase the distinction the extraction is meant to preserve.

The citation-role layer also changes which references are worth snowballing. Raw citation frequency is noisy, but the cleaned contexts reveal a small trunk of references that do repeated review work: they benchmark short-run productivity, define adoption and exposure measures, discipline organizational and macro interpretation, or warn about measurement validity.

Table 11: AI economics subtopics and citations to follow

AI economics area	NBER papers to inspect	Key outside citations to follow	Synthesis point
Adoption and diffusion	w34255 , w34995 , w34984 , w34836	Bick–Blandin–Deming; Bonney et al.; McElheran et al.	Worker use, firm use, executive expectations, and official adoption measures are related but not interchangeable.
Exposure and labor demand	w33509 , w33536 , w33451 , w34386	Eloundou et al.; Webb; Acemoglu et al.; Acemoglu–Autor	The exposure object is usually constructed before effects are estimated, so task and occupation mappings need validation.
Workplace productivity and inequality	w34851 , w33795 , w33641 , w34660	Noy–Zhang; Dell’Acqua et al.; Brynjolfsson–Li–Raymond	The strongest causal evidence is task-bounded and short-run; it should be screened separately from long-run labor-market claims.
AI-as-measurement and validity	w33344 , w34834 , w34714 , w33598	Ludwig–Mullainathan; Gentzkow–Kelly–Taddy; Hassan et al.; Glasserman–Lin	AI-generated labels, scores, and hypotheses are research inputs whose leakage, prompt sensitivity, and validation must be coded.
Policy, innovation, and official measurement	w34330 , w34947 , w33821 , w35022	Babina et al.; Giczy–Pairolero–Toole; Dell; policy-text and patent-data anchors	Many papers are building the policy or innovation dataset itself; the measurement model is part of the evidence.
Finance, model behavior, and long-run theory	w34054 , w34965 , w34256 , w34859	Gu–Kelly–Xiu; Acemoglu; Brynjolfsson–Rock–Syverson; Garicano; Trammell–Korinek	These papers mainly provide prediction, mechanism, organizational, or macro interpretation rather than directly comparable AI-effect estimates.

Two examples show the added depth. In the adoption branch, [w34255](#), [w34995](#), and [w34836](#) combine usage logs, worker surveys, firm surveys, and occupation or task links, so “AI adoption” becomes a set of constructed measures rather than a single variable. In the measurement-validity branch, [w34834](#), [w34714](#), [w33344](#), and [w33598](#) show why AI-generated variables need validation, leakage checks, prompt and model sensitivity analysis, and source-linked audit trails before they can support economic claims.

The open-question layer adds a second triage signal. Across the 58 processed papers, the system extracted 177 open-question records and grouped them into 12 reader-facing subtopics. These records do not merely say that more research is needed; they identify which branches first need better measurement, which are closer to effect-size screening, and which should be used as mechanism or validity evidence rather than pooled as AI-effect estimates.

Table 12: Open research questions by AI economics area

AI economics area	Open-question records	Unresolved empirical issue	Implication for synthesis
Adoption and exposure	28	Usage, exposure, and firm forecasts are not yet consistently linked to realized productivity, wages, employment, output, or reorganization.	Treat as measurement-first until usage, task, worker, and firm records can be linked to outcomes.
Workplace, sector, and content effects	28	Short-run task gains, creator effects, healthcare support, and trust responses may not scale to teams, markets, wages, or welfare.	Candidate effect-size branch, but only after outcomes, horizons, and adoption mediators are harmonized.
AI-as-measurement, policy, and innovation	50	LLM labels, similarity scores, policy-text measures, official-statistical constructs, and patent classifiers need validation and leakage checks.	Code the AI-generated variable role before using the result as economic evidence.
Model behavior, finance, and ML methods	37	Benchmark and prediction gains may depend on training data, model versions, prompts, market frictions, or implementation constraints.	Use mainly as mechanism, validity, and method evidence unless comparable deployed outcomes are coded.
Theory and adjacent comparators	34	Long-run mechanisms, organizational complements, skill change, and text-measurement analogies guide interpretation but are not comparable AI-effect estimates.	Keep as context and quality screens; do not pool with empirical AI-treatment estimates.

The most useful extracted tables and figures are the examples that make the evidence structure concrete for an economics reader. They are not all publication-ready without checking the page image, but they show what a reader should inspect first when browsing the AI evidence map.

Table 13: Concrete paper evidence by AI economics area

AI economics area	Papers and evidence to inspect first	Synthesis takeaway
Adoption measurement	w34255 message-classification pipeline; w34995 worker-country adoption figure; w34836 firm adoption and expectations table	Adoption is built from logs, surveys, occupation links, and firm forecasts, not one common variable.
Workplace effects	w33795 Copilot event-study figures; w33641 solution-quality by treatment group; w34851 education-gap experiment tables	Current causal evidence is strongest for bounded tasks, time allocation, and team output, with short horizons.
Exposure measures and labor outcomes	w33509 exposure-construction diagram; w33536 work/leisure time-use figures; w34386 historical exposure plots	Labor-market claims depend on how tasks, patents, occupations, and applications are linked.
LLM measurement validity	w34834 GPT measurement-cost and method tables; w33344 prompt/model sensitivity figures; w33598 leakage and Books3 pipeline figures	LLM measurement is powerful, but validity, leakage, and sensitivity are part of the empirical design.
Policy, innovation, and official measurement	w34615 policy-text validation tables; w35022 AI-patent classifier tables; w34947 regulatory deep-learning validation figures	The artifact to teach is often the measurement pipeline before the downstream result.
Finance, macro, and long-run mechanisms	w34636 visual-sentiment time series; w35016 Fed text-signal figures; w34859 task-chaining diagrams	These branches teach signal extraction and mechanism structure; they should not be pooled with worker-adoption effects.

The later taxonomy and audit add one more guardrail. The review should not move from these tables directly to pooled estimates. The paper-level reread and graph and evidence audit make the next step a readiness screen: decide what is core evidence, what is measurement infrastructure, what is theory or adjacent context, and which variables were constructed with ML before any economics result is interpreted.

Table 14: Screening rules before any AI meta-analysis

Screening decision	Evidence from the later audit	Coding fields needed	Implication for pooling
Paper-level unit	The 58 papers split into AI effects/adoption, AI/ML measurement infrastructure, theory, finance, methods, model evaluation, and adjacent comparators.	Primary and secondary review role, adjacent/core flag, design type, and evidence-quality tier.	Pool papers only within compatible roles and designs.
AI object versus AI method	46 of 58 papers have some ML/LLM or data-pipeline involvement, but many use AI to construct measures rather than study AI adoption.	ML placement as treatment, exposure, outcome, covariate, prediction, validation, or theory input.	AI-built variables need validation before their downstream findings can be compared.
Direct counts versus semantic buckets	Structured paper fields are strongest for counts; semantic topic buckets remain useful for discovery but can over-classify sparse snippets.	Coverage notes, result family, data role, measurement task, limitation type, and open-question type.	Counts should support triage and missingness checks, not final effect claims by themselves.
Citation function before frequency	Raw citation rankings contain parser fragments, while cleaned contexts identify benchmark, measurement, theory, identification, and validity anchors.	Canonical citation string, role, mechanism bucket, DOI/download status, and whether the citation is background or review-critical.	Snowball functional anchors first rather than every frequently mentioned reference.
Effect-size eligibility	Workplace productivity, selected creative/content papers, and some finance applications are closest to effect-size screening.	Outcome, task, horizon, comparator, heterogeneity margin, adoption mediator, and validity risk.	Adoption, exposure, measurement, and theory branches should usually remain narrative or mechanism evidence until harmonized.

This is the sense in which the output supports a living systematic review rather than a one-time search result. New NBER papers, revised working papers, or newly added background papers can be compared against the same review roles, citation functions, constructed-variable fields, and open-question categories. The review can then update which subtopics are active, which datasets and methods are being reused, which citations are functioning as benchmarks or measurement anchors, which studies are candidates for later meta-analysis screening, and which limitations or open questions are repeatedly stated by authors.

The example remains a starting point because it begins from the 2025–2026 NBER extraction and selected cited papers. A comprehensive systematic review would add recent working papers and published articles from outside this slice, specify eligibility and screening rules, appraise study quality, and code comparable effect estimates only where the designs and outcomes justify pooling. Appendix C gives the compact role-coded list retained from this starting search.

6.2 Automated Economics Research: Diagnosing Weak Generated Papers

AI systems that draft economics papers need more than broad web search once they move from a research question to an empirical study. Public information for APE indicates a large AI-generated-paper effort, while its detailed methodology page remains a work in progress. This case study is therefore not an audit of APE’s internal process. It shows where a structured collection of economics papers can help a researcher check a generated paper before treating it as serious evidence: whether the dataset has been prepared in the usual way, whether a constructed or coded variable has been validated, whether the proposed causal design resembles accepted designs in related papers, and whether related papers already state important limitations or contradictory findings.

End-to-end agent-generated papers also create a new reproducibility requirement. If a system generates a paper from a fixed literature corpus, data input, prompt bundle, and model setting,

the contribution should be checkable by rerunning the same inputs or by asking independent agents to recover the same core components: research question, dataset, sample, method, diagnostics, and main claims (Lu et al., 2026; Sakana AI, 2026; Social Catalyst Lab, 2026). The expectation is not identical prose. It is that the substantive paper objects and the evidence for them remain stable enough to audit. A structured paper layer gives that audit a reusable comparison set, rather than leaving the generated draft as a one-off transcript.

A compact trace illustrates the check. Suppose a generated paper proposes to measure local labor-market exposure using ACS or CPS occupation shares. A dataset query returns the ACS and CPS navigation records in Appendix I: ACS entries distinguish PUMS/IPUMS microdata, published geography tables, stitched Census–ACS panels, and denominator roles, while CPS entries separate Basic Monthly, ASEC/March, ORG, supplements, and linked rotation-panel uses. The action is not to accept or reject the generated paper automatically. It is to require the draft to name the product, geography, period, and construction step it uses; otherwise the comment is that the measurement is underspecified rather than merely unconventional.

6.3 AI-Assisted Reproduction: From Reimplementation to Robustness

Kohler et al. (2026) show that agents can sometimes reproduce social-science results from the paper and original data even without the authors’ replication code. That result makes the paper itself central to reproduction. The structured version of the paper adds two things. First, it helps the agent check the exact sample, data, method, table, equation, and result it is trying to reproduce. If reproduction fails, that structure helps distinguish an agent coding error from an extraction error, missing data or code, or underspecification in the paper. Second, it lets the researcher look across related papers for robustness checks that are standard for that dataset, outcome, method, or source of variation. This related-paper context becomes more important as time passes after publication: later papers may establish data treatments, validity checks, or robustness expectations that go beyond rerunning the original code. If reproduction succeeds, the next checks can therefore be drawn from actual practice in the literature rather than arbitrary specification changes.

A compact trace uses the shock-design records. For [w34188](#), the layer links the focal method to a firm-level GAP difference-in-differences design, the Uruguayan employer–employee administrative records, the causal-effect section (pp. 26–27), and the closest visual anchor, Figure B.1 (p. 10). A reproduction agent can use those records as the first checklist: construct the GAP measure, match the sample and event time, reproduce the linked figure or table, and only then move to related-paper robustness checks from the same wages, bargaining, and firm-rule-change cluster in Appendix M.

6.4 AI Peer Review: Grounding Comments in Paper and Literature Evidence

AI review tools use papers in two distinct ways: they read the submitted manuscript to check internal support, and they search related literature to check positioning. As Section 2 explains, the contribution here is not another PDF-reading review tool but the reusable evidence layer around the PDF. A comment can be tied to the exact claim, table, equation, assumption, reported diagnostic evidence, relevant citation role, or prior paper that supports the comment; related-paper records can also show whether a missing citation is merely background or a substantive method, data, comparison, or contradiction signal.

A recent AI-productivity manuscript gives a more concrete version of the workflow. Starting from a review of Cui et al.’s field experiments on GitHub Copilot ([10.2139/ssrn.4945566](#)), the first step is not to ask a model whether the referee comment sounds plausible. It is to anchor

the comment in the focal-paper graph: treatment, adoption, method, finding, table, figure, limitation, and citation nodes, linked by directed edges. The second step is a hybrid SQL, semantic, and graph query. SQL filters restrict the search to structured paper fields such as AI topic, method family, outcome type, citation role, table or figure links, and DOI. Semantic retrieval then finds adjacent language where papers discuss the same issue without sharing the same keywords. Graph traversal follows directed edges among claims, methods, findings, tables, figures, limitations, and citations to separate benchmark evidence from method precedent and measurement context. In this trace, the AI-related evidence map covered 128 unique AI papers and a narrow semantic index of 3,816 paper snippets, while the broader NBER index supplied method analogies from the full working-paper corpus. The review output changed from generic comments—“usage is vague,” “weights matter,” “spillovers may exist,” and “pull requests may not equal tasks”—into source-checkable revision requests: name the site-specific complier populations, define the weighted-IV estimand, add a coworker or team-exposure diagnostic, and describe the main outcome as pull-request or workflow throughput unless task scope and quality are validated. Appendix Q documents the sub-questions, SQL-style filters, semantic queries, and graph traversals behind this example, including the side-by-side comparison between free-tier review points and corpus-enriched edit suggestions in Table 78.

Table 15: Hybrid SQL, semantic, and graph enrichment of an AI peer-review comment

Review step	Hybrid query layer used	Query action	Resulting revision guidance
Focal-paper grounding	Graph edges from adoption and weighted-IV methods to Table 3, Table 9, Figure 11, and the quality-measure table	Traverse from the review comment to the method, finding, table, figure, and caveat nodes inside the focal paper	Keep the estimand and weighting comments, but state them as LATE and adoption-gap-weighted-estimand issues rather than loose wording concerns.
Adoption and encouragement context	SQL filters for AI-adoption and employer-tool records, plus topic and citation-role edges to Mind the Gap and Still Waters, Rapid Currents	Search for papers where encouragement, training, tool provision, and employer initiatives predict AI use	Ask for a site-by-site table separating access, encouragement, registration or first use, assignment level, and likely complier group.
Team and spillover context	SQL filters for workplace experiments, plus method and outcome edges to Shifting Work Patterns and The Cybernetic Teammate	Follow edges to teammate treatment, coworker exposure, and team-versus-individual production evidence	Convert the interference concern into a concrete diagnostic: treated share of team, repository, reporting line, or code-review network.
Outcome-measurement context	Structured outcome filters, semantic matches on productivity language, and artifact/finding edges to Generative AI at Work , Jagged Frontier , and Noy-Zhang	Compare pull requests, commits, builds, task completion, human or expert quality scores, customer outcomes, and correctness measures	Revise “completed tasks” to “workflow throughput” unless the paper validates pull-request scope, complexity, and quality across treatment status and sites.
Methods-positioning context	Method-family SQL filters, semantic retrieval, and graph edges to econometric anchors, including Building Bridges and Harvesting Differences-in-Differences	Find language on instrument-specific treatment effects and heterogeneous weighting	Present pooling as a precision-weighted summary across implementations, not as one common productivity parameter for high-skilled work.

The additional process detail worth retaining is the hybrid-query decision rule, not the retrieval transcript. A comment is kept when the focal-paper graph shows an unsupported or overstated claim and related-paper edges show the relevant comparison. It is revised when the broad concern is right but the graph points to a narrower interpretation, as when “usage” becomes encouragement-induced adoption or “completed tasks” becomes pull-request throughput. It is split when one comment bundles separable objects, such as treatment content, weighted-IV timing, team spillovers, and outcome measurement. It is dropped when the source-linked methods, tables, figures, limitations, and citation roles already supply the requested support.

This makes the unit of AI-assisted review a claim connected to source anchors and comparison edges, rather than a free-form judgment.

A compact trace starts from a review comment such as “the event-study estimator should be justified against recent staggered-adoption practice.” The hybrid query filters to method-family records, retrieves semantically similar estimator language, and then follows citation-role edges to local contexts where papers use prior work as method justification or identification strategy rather than background. Appendix O shows examples: [w34389](#) uses Cengiz et al. (2019) for a stacked event-study approach, while [w33746](#) uses de Chaisemartin and d’Haultfoeuille (2024) as a dynamic difference-in-differences estimator. The review action is therefore to keep the comment if the focal paper lacks a comparable justification, revise it if the issue is only citation positioning, or drop it if the source-linked methods and citations already supply the needed support.

7 Practical Scope, Cost, and Scaling

Scope. The main manuscript uses the 2025–2026 extraction layers. The selected processed-paper set contains 1,608 papers: 1,179 from the 2025 batch and 429 from the 2026 batch.¹⁰ Layer-specific denominators are reported in Section 5 because the citation-role, data-source, algorithm-method, and frontier-signal analyses were rerun or reviewed separately.

Cost and local processing. The primary extraction model is GPT-5-mini running parallel extraction stages. In the current run, the fully linked extraction costs about US\$0.30 per processed paper, including extraction of tables, figures, equations, source locations, and typed links. The timing profile makes repeated corpus builds feasible, while manual review can be reserved for validation, ambiguous records, and high-stakes downstream uses. A practical design goal is to reduce repeated dependence on hosted model calls: once papers have been converted into source-linked records, researchers can query, filter, embed, and audit those records without re-sending the full PDF text for each downstream question. For licensed, confidential, or institutionally restricted papers, the initial extraction would need to substitute a self-hosted or locally approved model for GPT-5-mini; after conversion, querying and auditing can proceed without repeatedly sending full PDF text to a hosted model.¹¹

Validation status. Appendix G reports a preliminary single-reviewer validation review of extraction quality in a 100-paper sample. Validation remains in progress: future versions will add additional reviewers, layer-specific accuracy checks, external benchmark comparisons where overlap exists, and task-level experiments for search, review, reproduction, and diagnostics. Once extraction accuracy is stable, those experiments will test whether APE-style generated-paper diagnostics and agentic reproduction systems such as *Read the Paper*, *Write the Code* improve when they can retrieve reusable paper components rather than only the focal PDF, abstracts, or raw search results.

Repairability. The same source and contract metadata used for validation also make the pipeline easier to improve. A false method label, missed data source, or incorrect link can be traced to the source passage and to the stage, prompt bundle, taxonomy version, model binding, and run configuration that produced it. In many cases the remedy is therefore local: revise a taxonomy boundary, adjust a prompt instruction, rerun an enrichment pass, or repair a linking rule, then revalidate the affected layer. This component isolation is part of the research design. It makes extraction errors diagnosable and correctable without treating the whole corpus as an opaque model output, while still requiring regression checks before corrected records are released.

Scaling. Scaling to the full NBER working-paper archive, and then to the broader economics literature, remains a future implementation target rather than a result claimed here. The contribution is that the method makes literature-scale measurement plausible at full-paper depth. At that scale, source-linked labels would resolve to corpus-wide quantitative patterns: not “labor and environmental papers share a data constraint” but “what share of papers requiring linked administrative records flag the same restricted-access limitation, and across how many JEL categories does it appear.”

Uses. The same extraction can support agent grounding, meta-analysis, research design assistance, knowledge graphs, and graph-based search (Cognee, 2025; Edge et al., 2024; Stocker

¹⁰The current coverage file reports 1,289 downloaded 2025 NBER PDFs, of which 1,179 are in the selected processed set used here; the remaining 110 downloaded 2025 PDFs are not yet included in this build.

¹¹Generic graph-retrieval systems such as GraphRAG and cognee are complementary rather than substitutes (Cognee, 2025; Edge et al., 2024): they provide graph and vector-retrieval tools, including clustering or community detection, but still require upstream source-linked records to represent economics paper objects. A hosted API model is a language model accessed programmatically through an application programming interface, usually billed by input and output tokens rather than through a fixed chat subscription.

et al., 2025), without re-reading the PDF for each task. Future releases can expose the layer as downloadable tables, graph objects, and, where copyright and access rules permit, a browsable or API-backed database for direct researcher use.

8 Conclusion

This paper shows that economics working papers can be converted from PDF prose into source-linked research objects. The 2025–2026 NBER extraction demonstrates the method at corpus scale: seven linked information types, source locations, controlled labels, and relationship links make papers searchable, comparable, and auditable across database, semantic-search, and graph interfaces.

The first descriptives show what this representation makes visible: citation links can be decomposed by function, algorithmic and machine-learning use can be located in the research workflow, and author-stated open questions can be separated from the limitations that make them difficult. The broader contribution is an auditable evidence layer that is both a database for source-grounded inspection and a dataset for cross-paper measurement. Because records carry source locations and contract fingerprints, the layer is also repairable: misclassifications can be traced, corrected at the relevant component, and checked without rebuilding every unrelated part of the extraction. The downstream task value remains a testable hypothesis: future versions will add formal accuracy validation, expand the corpus, prepare public-facing access where rights permit, and run experiments on living-review, automated-diagnostics, reproduction, and peer-review tasks.

Guide to the Appendices

The appendices provide the audit trail for the extraction layer: prior-work comparisons, task details, pipeline metadata, schema definitions, corpus denominators, validation, retrieval examples, descriptive lookup tables, citation-role checks, taxonomies, and prompts. Table 16 summarizes the main appendix materials and provides direct references to key sections, tables, and figures. The bibliography follows this guide, and the appendices begin immediately after the bibliography.

Table 16: Guide to appendix materials

Topic	Contents	Key references
Prior systems and contribution	Positions the contribution against keyword search, economics claim graphs, scholarly knowledge graphs, and paper-search systems.	<i>See:</i> section ; comparison table .
Downstream research tasks	Task setups for living review, automated diagnostics, agentic reproduction, and AI peer review.	<i>See:</i> task details ; robustness table ; task-use table ; AI starting-list appendix ; starting-list table .
Pipeline and reproducibility	How PDFs become records, with run metadata for audit, reproduction, and error repair.	<i>See:</i> section ; pipeline stages ; reproducibility metadata .
Schema and links	Typed relationships among data sources, methods, findings, limitations, open questions, citations, and replication metadata.	<i>See:</i> section ; edge-type table .
Corpus accounting	Processing ledger and denominator notes for paper-level, item-level, and link-level tables.	<i>See:</i> section ; processing ledger ; replication metadata coverage .
Validation	Preliminary 100-paper single-reviewer review, split between main and supplementary objects.	<i>See:</i> section ; main-object validation ; supplementary-object validation .
Retrieval and multihop search	Database, embedding, and source-linked retrieval examples.	<i>See:</i> retrieval detail ; query lanes ; DuckDB format ; retrieval trace .
Dataset navigation	Dataset-family navigation tables and aggregate dataset-type figures.	<i>See:</i> top dataset families ; dataset overview ; aggregate dataset types ; type-count figure ; JEL-composition figure .
Algorithmic and ML use	Machine-learning, algorithmic, prediction, text-analysis, and computational use by field and empirical role.	<i>See:</i> section ; JEL counts ; algorithm-role table .
Frontier and cluster evidence	Open-question, limitation, shock, and natural-experiment cluster labels, lookup tables, and maps.	<i>See:</i> lookup tables ; open-question labels ; limitation labels ; frontier detail ; open-question map ; limitation map ; shock clusters ; shock map .
Citation evidence	Citation roles, resolver limits, cited-work profiles, frontier-setting citations, and context topic coding.	<i>See:</i> role examples ; example table ; citation appendix ; resolver audit ; cited-work profiles .
AI peer-review query trace	Documents the sub-questions and hybrid SQL, semantic, and graph queries behind the AI peer-review case study.	<i>See:</i> query trace ; side-by-side edits ; sub-question table ; returned-evidence table .
Taxonomies and prompts	Controlled vocabularies and prompt text for method, finding, data-source, citation, theory, linking, and enrichment passes.	<i>See:</i> taxonomies ; method families ; citation roles ; prompts ; common method fields ; artifact enrichment .

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A Prior Extraction Systems

Prior systems either work at one article-level classification layer, extract one high-value object, or require manual curation. Some are economics-specific, especially Goldsmith-Pinkham (2026) and Garg and Fetzer (2026); others cover related social-science or general scholarly corpora. The comparison in Table 17 is therefore about depth of machine-readable structure, not about whether those systems are useful for their own tasks. The distinction matters because an ontology used with an LLM is closer to a controlled vocabulary than to a fixed dictionary: it gives the model field-specific categories and boundaries, but lets it recognise paraphrases and related formulations in context, with the source passage retained for checking (Talisman, 2025). For Goldsmith-Pinkham (2026), the comparison is specifically to the search-category design in Appendix A, Table 4: most outcomes are paper-level indicators for whether at least one trigger phrase appears, while the Figure and Table categories use average word counts per paper.

Table 17: Depth comparison: prior extraction systems versus this paper

System	Current scope	Main machine-readable unit	Field depth	Relation depth
Goldsmith-Pinkham (2026)	Approximately 44,000 papers: 31,500 NBER working papers from 1982–2025 and 12,300 top economics and finance journal articles from 2011–2024	Paper-level search-category indicators from Appendix A, Table 4	Trigger phrases for data, method, empirical-practice, figure, table, and econometrics categories; unless noted otherwise, category outcomes are fractions of papers with at least one phrase match	No within-paper item graph; phrase categories are not source-spanned paper objects, and Figure/Table categories are average word counts per paper
Lee et al. (2025)	10,490 context-annotated causal triplets from 2,595 economics and finance studies	Causal triplet benchmark	Cause, effect, context, and identification metadata	Single cause-effect relation; no multi-entity paper graph
Torreblanca et al. (2026)	91,632 political-science articles from 156 journals, 2003–2023	Article-level design, credibility-practice, and citation-pattern labels	Research-design and practice tags	No relationship links among extracted paper items
Briggs et al. (2026)	101,475 political-science articles from 156 peer-reviewed journals, 2010–2024	Article-level null-result and highlighted-claim dataset	Null, non-null, and precise-null claim coding; statistical-method use; open-science, subfield, geography, and time metadata	Validated article-level metadata for meta-science; no multi-entity within-paper graph
Xu et al. (2025)	AI research-paper limitation benchmark with synthetic and human-written subsets	Limitation statement	Limitation text, limitation type, and retrieval-grounding context	No persistent typed graph
Jaradeh et al. (2019)	General scholarly contributions, curated through crowdsourcing plus automated support	Research contribution record	Variable contribution properties	Curated scholarly graph objects
Garg and Fetzner (2026)	44,852 NBER and CEPR working papers, 1980–2023; first-30-page extraction	Standardized concept nodes and claim edges	Research question, methods, key claims, relation type, evidentiary basis, and JEL concepts	Evidence-annotated causal and non-causal claim graph
This paper	1,608 NBER working papers in the 2025–2026 extraction layers	Seven linked kinds of paper information	15–25 structured fields per kind, with page/section evidence	14 directed relationship types across extracted items

Note: Prior-system scopes are taken from the current source pages or papers available at the time of this build. *Relationship links* are directed relationships connecting extracted items across types: method *produces* finding; data source *calibrates-with* method; finding *qualified-by* limitation; limitation *motivates* open question; and the further types listed in Appendix E: robustness checks, data-use links, table/figure evidence, method lineage, supporting and contradicting findings, mechanism links, data-blocked open questions, citation evidence, and shared-method paper links.

B Agentic Research Task Details

The main text summarizes the agentic and researcher-facing workflows that motivate a machine-readable paper layer. The tables here give the supporting detail: how the paper set can help define robustness checks, how different task families use papers, and how source links turn updates and failures into diagnosable events.

Table 18: Literature-grounded checks for robustness reproducibility

Check family	Same-data robustness check	Corpus evidence used
Data construction	Alternative sample filters, panel balance rules, missing-data treatment, winsorization, survey weights, deflators, merge rules, and linkage thresholds.	Dataset records, variable-construction snippets, data-source evidence, and author-stated data limitations.
Method specification	Alternative controls, fixed effects, clustering levels, functional forms, lags/leads, transformations, and model-family variants.	Method-family labels, table/equation evidence, reported diagnostics, and nearby-paper method profiles.
Identification	DiD pre-trend and event-window checks; IV first-stage and weak-IV tests; RD bandwidth, polynomial, and donut choices; synthetic-control donor and predictor choices; RCT attrition and compliance checks.	Identification strategy records, validity-test fields, reported diagnostics, and citation precedents.
Literature contradiction	Re-run choices used by papers that compare, qualify, or contradict the original estimate.	Citation-role records for estimate comparison, identification precedent, method justification, and contradiction.
Author-stated limitations	Same-data tests motivated by the original paper's own limits, such as sample coverage, measurement error, unobserved mechanisms, or alternative channels.	Limitation and open-question records with source passages.
Reporting	Compare sign, significance, effect-size ratio, confidence-interval overlap, the share of checks supporting the original claim, and dispersion across checks.	Finding records linked to table cells, equations, and every robustness output.

Note: The system does not decide which robustness checks are substantively correct; it assembles a source-linked menu of defensible checks for researcher review.

Table 19: Why agentic research tasks use papers, and what the structured layer adds

Task family	Why papers matter	How papers are used	What this paper adds
Living systematic review	Papers are the evidence units: eligibility, design, estimates, uncertainty, limitations, and later updates must be tracked as new work appears.	Reviews search, screen, extract, appraise, and re-analyse studies; meta-analysis re-analyses comparable effect estimates, and living reviews repeat the process as evidence changes.	Queryable full-text fields for methods, outcomes, standard errors, limitations, citation roles, and provenance, so updates and citation-led snowballing do not start from a blank PDF.
End-to-end AI research	Agents need prior work to choose questions, avoid duplication, select methods, design experiments, and interpret results.	The AI Scientist uses scholarly API searches for novelty checks and citation writing; retrieved related work is passed to the model as metadata and abstracts, not full-paper records.	Structured economics context: what nearby papers actually did, which data and methods they used, what validity checks they reported, and where their stated limits are.
AI peer review	Reviewing depends on checking whether manuscript claims are supported by the submitted paper, its tables, equations, citations, assumptions, and prior literature.	CoarseInk’s open-source implementation extracts the submitted file to markdown, parses sections, runs section/proof/cross-section review agents, searches related literature from the title and abstract, and verifies comments against the extracted full text. Refine is not currently open source, so its internal pipeline is not audited here.	A checkable record of claims, tables, equations, references, assumptions, and source locations, plus corpus context for whether a claim or method is positioned accurately.
Agentic reproduction	Reproduction needs the exact method, data, sample, table structure, equations, and replication metadata; robustness work also needs defensible alternative choices, especially when later papers have changed the expected checks.	Read the Paper, Write the Code extracts method descriptions and blinded table templates from the full paper, then agents reimplement with original data while the authors’ code, reported results, and paper PDF are withheld from them.	Focal-paper inputs for recreate reproduction, plus nearby and later-paper evidence on common specifications, data treatments, and validity checks for targeted robustness exercises.

Table 20: Provenance as update and diagnostic infrastructure across research tasks

Task family	Highest-impact use	What provenance helps triage or design	Reusable output
Living systematic review	Ingest new papers and revisions, then identify which updates can change the review.	New results, datasets, identification strategies, validity tests, limitations, citation contradictions, and revised estimates.	Living evidence database: review records, study records, datasets, methods, estimates, and update log.
End-to-end AI research	Prevent false novelty claims and weak method selection.	Prior uses of a dataset or method, nearby identification strategies, unresolved limitations, and contradictory findings.	Grounded research brief with source-linked prior work and method/data options.
AI peer review	Ground every criticism in submitted-paper text or corpus evidence.	Unsupported claims, diagnostic questions, inconsistent tables, overstated novelty, and ignored contradictory evidence.	Evidence-anchored review comments.
Agentic reproduction	Move from recreate reproduction to designed robustness reproducibility.	Same-data alternative specifications and validity checks derived from related and later literature rather than arbitrary perturbations.	Robustness dashboard or specification curve with provenance for every check.
Citation-role quality signal	Separate attention from substantive uptake.	Background citation volume versus explicit method use, estimate comparison, identification precedent, and contradiction.	Cited-work profile showing how later authors use a paper.
Dataset/method discovery	Make reuse defensible.	Dataset construction, access restrictions, variables, shocks, instruments, counterfactual methods, and comparable studies.	Dataset and method registry reusable by researchers.

C AI Living-Review Starting List

The AI-related economics case study is a starting search, not a completed systematic review. The table below records the papers and citation anchors retained for the next review round. Each row begins with its review role, separating core AI-effect evidence from adoption measurement, exposure construction, AI-as-measurement methods, validity checks, theory, finance and market applications, and external benchmark citations. A comprehensive review would expand this list with recent working papers and published articles outside the 2025–2026 NBER extraction, then apply explicit eligibility, screening, appraisal, and coding rules.

Table 21: AI-related economics starting list from the living-review case study

No.	Paper or reference	Review role and next use
<i>Direct NBER papers retained for screening</i>		
1	How People Use ChatGPT (w34255)	<i>Adoption measurement.</i> Core example for consumer use, work/non-work classification, task use, message logs, and occupation links.
2	Artificial Intelligence and the Labor Market (w33509)	<i>Exposure and labor demand.</i> Core labor-market paper for firm AI adoption, task exposure, labor demand, and measurement construction.
3	Still Waters, Rapid Currents: Early Labor Market Transformation under Generative AI (w33777)	<i>Labor-market effects.</i> Early labor-market evidence on adoption, employer initiatives, reported benefits, mobility, and earnings.
4	Does Generative AI Narrow Education-Based Productivity Gaps? Evidence from a Randomized Experiment (w34851)	<i>Causal productivity evidence.</i> Candidate effect-size paper for task performance and education-based heterogeneity.
5	Shifting Work Patterns with Generative AI (w33795)	<i>Workplace effects.</i> Core workplace experiment for time allocation, Outlook/Teams/Word use, take-up, and spillovers.
6	The Cybernetic Teammate: A Field Experiment on Generative AI Reshaping Teamwork and Expertise (w33641)	<i>Teams and organizations.</i> Core field-experiment paper for teamwork, expertise, quality, and task-bounded productivity.
7	Large Language Models: An Applied Econometric Framework (w33344)	<i>AI-as-measurement validity.</i> Method and validity anchor for LLM-generated measurements, leakage, validation samples, and inference sensitivity.
8	Making AI Count: The Next Measurement Frontier (w34330)	<i>Official measurement.</i> Measurement-agenda paper for official statistics, welfare measurement, free AI services, and adoption tracking.
9	A Research Agenda for the Economics of Transformative AI (w34256)	<i>Theory and agenda.</i> Agenda paper for frontier questions, long-run mechanisms, and review scope.
10	Artificial Intelligence, Productivity, and the Workforce: Evidence from Corporate Executives (w34984)	<i>Firm adoption.</i> Firm and workforce evidence on executive expectations, adoption, productivity, and employment margins.
11	Behavioral Economics of AI: LLM Biases and Corrections (w34745)	<i>AI behavior and bias.</i> Model-behavior evidence for LLM biases, correction prompts, and decision settings.
12	General Social Agents (w34937)	<i>Agent behavior.</i> Agent paper for social-agent capabilities, behavioral simulation, and external-validity questions.
13	Measuring Human Leadership Skills with AI Agents (w33662)	<i>AI measurement proxy.</i> AI-agent measurement paper for leadership assessment and validation against human-team outcomes.
14	Artificial Writing and Automated Detection (w34223)	<i>Detection and writing.</i> Validity and policy paper for generated writing, detector performance, and stage-of-use measurement.
15	Out of the Black Box: Uncertainty Quantification for LLMs via Conditional Probabilities (w34965)	<i>Uncertainty and reliability.</i> Methods paper for LLM uncertainty, inner confidence, look-ahead bias, and reliability checks.
16	AI and the Extended Workday: Productivity, Contracting Efficiency, and Distribution of Rents (w33536)	<i>Workplace and rents.</i> Workplace paper for exposure, hours, leisure, contracting, monitoring, and distributional effects.
17	AI and Women’s Employment in Europe (w33451)	<i>Gender and labor exposure.</i> Labor paper for gender, exposure, employment shares, and cross-country external validity.
18	Firm Data on AI (w34836)	<i>Firm data infrastructure.</i> Firm-level measurement paper for adoption surveys, realized impacts, expected impacts, and official data infrastructure.
19	Mind the Gap: AI Adoption in Europe and the U.S. (w34995)	<i>Adoption comparison.</i> Cross-country adoption paper for worker use, time savings, adoption gaps, and productivity comparisons.

No.	Paper or reference	Review role and next use
20	Natural Language Processing and Innovation Research (w33821)	<i>Innovation methods.</i> Methods paper for text-based innovation measures and AI-adjacent research methods.
21	Does Generative AI Crowd Out Human Creators? Evidence from Pixiv (w34733)	<i>Creative markets.</i> Platform-shock paper for creator entry, uploads, attention, and substitution in creative markets.
22	GenAI Misinformation, Trust, and News Consumption: Evidence from a Field Experiment (w34100)	<i>Information and trust.</i> Information experiment for misinformation, trust, media consumption, and belief outcomes.
23	Does LLM Assistance Improve Healthcare Delivery? An Evaluation Using On-site Physicians and Laboratory Tests (w34660)	<i>Sector evidence.</i> Healthcare evidence for LLM assistance, care-plan quality, safety, lab tests, and selected-review caveats.
24	Deep Learning Projects Jurisdiction of New and Proposed Clean Water Act Regulation (w34947)	<i>Policy measurement.</i> Computer-vision and regulatory-measurement paper for policy classification and validation.
25	AI-Powered Trading, Algorithmic Collusion, and Price Efficiency (w34054)	<i>Finance and markets.</i> Finance paper for algorithmic trading, price efficiency, collusion, and simulation-based mechanisms.
<i>External citation anchors retained for snowballing and interpretation</i>		
26	Noy and Zhang (2023), Experimental evidence on the productivity effects of generative artificial intelligence	<i>Productivity benchmark.</i> Frequently used benchmark for bounded-task productivity and heterogeneity.
27	Acemoglu and Restrepo (2018), Modeling Automation	<i>Automation theory.</i> Theory anchor for task replacement, reinstatement, and productivity channels.
28	Acemoglu and Restrepo (2020), Unpacking Skill Bias: Automation and New Tasks	<i>New-tasks theory.</i> New-tasks framework for interpreting labor-market adjustment and skill bias.
29	Eloundou et al. (2023/2024), GPTs are GPTs: Labor market impact potential of LLMs	<i>Exposure measurement.</i> Core task-exposure measure for occupational comparisons and labor-market relevance.
30	Brynjolfsson et al. (2018), What can machine learning do? Workforce implications	<i>ML exposure measure.</i> Machine-learning exposure benchmark for comparing task-suitability approaches.
31	Acemoglu et al. (2022), Artificial Intelligence and Jobs: Evidence from Online Vacancies	<i>Jobs benchmark.</i> Empirical jobs benchmark using vacancy data, skills, and labor-demand measures.
32	Acemoglu and Autor (2011), Skills, Tasks and Technologies	<i>Task framework.</i> Task framework linking technology exposure to employment, wages, and earnings.
33	Webb (2020), The Impact of Artificial Intelligence on the Labor Market	<i>AI exposure measure.</i> AI exposure measure for labor-market comparison and measurement choices.
34	Babina et al. (2024), Artificial Intelligence, Firm Growth, and Product Innovation	<i>Firm and innovation benchmark.</i> Firm benchmark linking AI investment, growth, innovation, and measurement design.
35	Korinek (2023), Generative AI for Economic Research	<i>Research-workflow methods.</i> Methods and use-case survey for economics research workflows.
36	Ludwig and Mullainathan (2024), Machine Learning as a Tool for Hypothesis Generation	<i>Hypothesis-generation method.</i> Methods paper for hypothesis generation, validation, and research design.
37	Vaswani et al. (2017), Attention Is All You Need	<i>Technical foundation.</i> Transformer foundation for interpreting LLM-based methods and models.
38	Radford et al. (2019), Language Models are Unsupervised Multitask Learners	<i>Technical foundation.</i> GPT-style technical background and language-model capability anchor.
39	Humlum and Vestergaard (2024), The Adoption of ChatGPT	<i>Adoption benchmark.</i> Adoption benchmark for who uses ChatGPT, why, and how adoption diffuses.
40	Humlum and Vestergaard (2025), Large Language Models, Small Labor Market Effects	<i>Labor-market benchmark.</i> Labor-market benchmark for realized employment effects and short-run muted outcomes.
41	Tomlinson et al. (2025), Working with AI	<i>Occupational measurement.</i> Occupational measurement paper for generative-AI exposure and task implications.
42	Carnehl and Schneider (2025), A Quest for Knowledge	<i>Knowledge framework.</i> Decision and knowledge framework for AI-assisted information search and belief formation.
43	Eisfeldt et al. (2023), Generative AI and Firm Values	<i>Firm-value benchmark.</i> Firm-value benchmark for market expectations about generative AI.
44	Brynjolfsson, Li, and Raymond (2025), Generative AI at Work	<i>Workplace benchmark.</i> Workplace productivity benchmark for customer support and AI assistance.
45	Dell (2024), Deep Learning for Economists	<i>Deep-learning methods.</i> Methods paper for deep-learning applications in economics.
46	Gentzkow, Kelly, and Taddy (2019), Text as Data	<i>Text-as-data foundation.</i> Methods foundation for text measurement, validation, and interpretation.
47	Hassan et al. (2019), Firm-Level Political Risk	<i>Text-measurement benchmark.</i> Text-measurement benchmark for validating firm-level text signals.
48	Glasserman and Lin (2023), GPT sentiment analysis and look-ahead bias	<i>Finance validity benchmark.</i> Finance benchmark for bias checks in language-model trading signals.

No.	Paper or reference	Review role and next use
49	Marcus (2020), The next decade in AI	<i>Robustness perspective.</i> Robustness and limitation perspective for current AI systems.
50	Trammell and Korinek (2023), Economic Growth under Transformative AI	<i>Transformative-AI theory.</i> Long-run theory for growth, policy, and transformative-AI scenarios.
<i>Additional trunk anchors surfaced by the follow-up citation-role review</i>		
51	Dell'Acqua et al. (2023), Navigating the Jagged Technological Frontier	<i>Knowledge-work productivity benchmark.</i> Field-experiment benchmark for task-specific gains, frontier/caveat framing, and professional-work interpretation.
52	Bick, Blandin, and Deming (2024), The Rapid Adoption of Generative AI	<i>Worker adoption benchmark.</i> Adoption-speed anchor for worker use, demographic gaps, and diffusion comparisons.
53	Bonney et al. (2024), Tracking Firm Use of AI in Real Time	<i>Firm adoption benchmark.</i> BTOS-based firm-use anchor for comparing worker reports, firm reports, and official adoption measures.
54	McElheran et al. (2023), AI Adoption in America: Who, What, and Where	<i>Organizational adoption anchor.</i> Firm-level anchor for management practices, organizational complements, and adoption heterogeneity.
55	Brynjolfsson, Rock, and Syverson (2021), The Productivity J-Curve	<i>Complementarity theory.</i> Explains why AI adoption and productivity may separate because complementary investments arrive slowly.
56	Garicano (2000), Hierarchies and the Organization of Knowledge in Production	<i>Organization theory.</i> Knowledge hierarchy anchor for AI as a coordination, expertise, and organizational-redesign technology.
57	Acemoglu (2025), The Simple Macroeconomics of AI	<i>Macro AI benchmark.</i> Aggregate-growth and labor-share framing for interpreting why task gains may imply modest macro effects.

D Extraction Pipeline

Table 22: Pipeline stages, model tiers, and approximate time per paper (single cloud instance)

Stage	Name	Primary function	Model	Avg. time
1	Parse	PDF $\rightarrow$ structured text, section index, artifact registry (tables, figures, equations as first-class objects)	Docling (local)	~ 10 s
2	Blueprint	Paper-type classification (empirical / structural / theoretical); identification strategy and section identification	GPT-5-mini	~ 15 s
3	Extract	Parallel entity extraction across seven types: data sources, methods, findings, limitations, open questions, citation evidence, replication metadata	GPT-5-mini (4 parallel)	~ 45 s
4	Synthesize	Cross-entity linking, rigor classification, causal DAG construction; only stage requiring sequential cross-entity reasoning	GPT-5-mini	~ 30 s
4.7+	Enrich	Artifact backfill via two-pass vision LLM (scan pass at 144 DPI; extract pass at 302 DPI); precision enrichment (standard errors, p -values); table structure parsing (cluster level, FE levels, SE type)	GPT-5-nano (scan) GPT-5-mini (extract)	~ 80 s
5	Contract gate	Three-profile deterministic gate: core profile (entity lists, unique IDs, empirical papers must have methods); extension profile (advisory warnings); contract integrity (schema, prompt, and taxonomy hash consistency; SHA-256 payload fingerprint)	Rule-based	~ 5 s

The base pipeline path averages under two minutes per paper. The enrichment substages (4.7–4.78) add approximately 30–80 s when triggered, depending on table density, and are parallelised within each paper. Papers passing the core contract profile are cleared for downstream use; failures are held for review with logged exclusion reasons. Contract fingerprinting enables exact reproducibility audits.

Table 23: Reproducibility metadata stored with final extraction packets

Field	Purpose
<code>schema_version</code>	Names the final-extraction schema version, currently <code>final_extraction.v1</code> .
<code>schema_hash</code>	SHA-256 hash of the schema file used by the contract gate.
<code>prompt_bundle_hash</code>	SHA-256 hash of the prompt bundle used by the extraction run.
<code>taxonomy_hash</code>	SHA-256 hash of the controlled-vocabulary bundle.
<code>run_config_hash</code>	SHA-256 hash of run configuration metadata.
<code>models</code>	Stage-level model identifiers, including blueprint, extraction, data-source, findings, and synthesis models.
<code>fingerprint</code>	SHA-256 payload fingerprint for the final extraction packet.

Note: These fields support contract-level reproducibility checks and error diagnosis. They do not by themselves establish substantive extraction accuracy; they identify which source, prompt bundle, taxonomy version, schema, model binding, or run configuration should be inspected when a record is misclassified or missed.

E Schema and Relationship Detail

Table 24: Typed directed relationships between extracted entities

Relationship	Direction	Meaning
<i>produces</i>	method → finding	Method generates this result
<i>calibrates_with</i>	data source → method	Dataset provides identifying variation
<i>qualified_by</i>	finding → limitation	Limitation qualifies the finding’s scope
<i>motivates</i>	limitation → open question	Constraint blocks resolution of the question
<i>robustness_check_of</i>	finding → finding	Sensitivity analysis on the target result
<i>uses_data</i>	method → data source	Method draws on this dataset
<i>evidenced_by</i>	finding → table/figure	Artifact containing the reported result
<i>builds_on</i>	method → method	Method extends or uses a parent method
<i>supports</i>	finding → finding	Cross-paper corroboration
<i>contradicts</i>	finding → finding	Cross-paper conflict
<i>has_mechanism</i>	finding → mediator	Proposed causal channel
<i>blocked_by</i>	open question → limitation	Data gap preventing resolution
<i>cites_for</i>	citation evidence → entity	Citation supports or identifies an entity
<i>shared_method</i>	paper → paper	Cross-paper edge: same method family and design

F Corpus Selection and Layer Denominators

Corpus. The main manuscript uses the 2025–2026 NBER extraction layers. The citation-role layer contains 1,608 selected processed papers: 1,179 papers from the 2025 batch and 429 papers from the 2026 batch. The data-source and algorithm-method layers have separate reviewed denominators, so their tables report their own paper counts rather than inheriting the citation-role denominator. The processed object is the NBER working-paper PDF, not a later journal version; published versions appear only as cited works when a reference resolves to them. Each paper is tracked from candidate PDF through parsed, extracted, and contract-gated stages; exclusions are logged in run metadata with reason codes.

Table 25: Processing ledger and layer denominators for the 2025–2026 build

Layer	Unit	Coverage	Main use	Denominator rule
Selected processed corpus	Paper	1,608 papers	Main corpus scope	Selected 2025–2026 processed-paper set.
Methods	Method or design component	9,880 items in 1,538 papers	Method descriptives and method–finding links	Component-level method records; papers can contribute multiple primary, supporting, construction, diagnostic, or robustness items.
Findings	Finding statement	10,498 items in 1,601 papers	Finding descriptives and source-evidence links	Finding-item denominator; field-presence rates are shares of extracted finding items.
Data sources	Data-source record	7,659 sources; reviewed type labels in 1,391 papers	Data-source type and structure descriptives	Main extraction contains data sources in 1,396 papers; type tables use the reviewed 1,391-paper subset.
Citation roles	Local citation context	120,064 links in 1,572 papers	Citation-role distribution and local citation use	Citation-link-level denominator; not a unique cited-work or resolved-network count.
Algorithmic and ML use	Paper and algorithmic-use record	592 algorithmic papers in a 1,603-paper rerun	Algorithmic-use descriptives	Separately rerun algorithm-method layer; shares are within papers with at least one algorithmic or ML use.
Frontier clusters	Routed statement / assigned cluster / displayed subset	Statements from 1,381 papers	Open-question, limitation, and shock-cluster figures	Cluster figures use routed statements and displayed largest-cluster subsets; Appendix L reports assigned and displayed denominators.
Replication metadata	Paper-level status record	1,597 status records; 132 resolved URLs	Replication-package coverage	Resolver records whether a package URL is resolved or not found, plus resolution source when available.

Note: The table separates paper-level, item-level, citation-link-level, cluster-level, and resolver-level units so the descriptive tables are not read as a single unified denominator.

Table 26: Replication-package metadata coverage and resolver details

Measure	Count	Interpretation
Selected processed papers	1,608	Main 2025–2026 processed-paper denominator.
Paper-level package status present	1,597	The resolver records either a resolved package or a not-found status for the paper.
Resolved package or repository URL	132	Papers with a matched package URL from paper text, OpenAlex, or directed search.
Status not populated	11	Selected papers without a paper-level package-status record in the current build.
Data-source records with a non-null source-level replication flag	622	Data-source-level flag, not part of the main coverage denominator; spans 177 papers.
Data-source records marked available through a replication package	246	Subset of source-level flags where the data source is marked available through package evidence.

Note: The main text uses the paper-level coverage figures. Source-level flags are reported here as resolver details because their denominator is data-source records rather than papers.

G Validation Review

This appendix reports a first single-reviewer validation review for the 2025–2026 extraction. The review covers 100 papers. For each paper, the reviewer read the source text and coded source-supported gold labels for the main paper-object layers: methods and design components, datasets, findings, limitations, open questions, and machine-learning or algorithmic use. The comparison is object-level. A matched extracted object is a true positive, an unmatched extracted object is a false positive, and an unmatched reviewer-coded object is a false negative. Because there is no finite universe of possible non-objects in an open-ended paper, object-level true negatives are not defined; precision, recall, and F1 are the relevant metrics. True negatives are defined only for closed paper-level checks, such as whether a paper contains any machine-learning or algorithmic use. Because this is a single-reviewer audit, the current release does not report inter-rater agreement; future validation will add independent reviewers and adjudication.

The review reports two groups of objects. **Main paper objects** are the records a researcher would usually expect to find in a first-pass search: primary methods and designs, named datasets, headline findings, core limitations and open questions, explicit machine-learning or algorithmic uses, and paper-level machine-learning or algorithmic presence. **Supplementary and extension objects** are lower-level records that add depth: robustness checks, heterogeneity analyses, diagnostics, constructed or auxiliary data, table-specific findings, model extensions, broad non-machine-learning algorithms, solvers, linkage rules, and related implementation details.

The tables below report the current extraction after adding matched intermediate records. This is a diagnostic of the paper information already surfaced by the pipeline when recall-stage candidates are promoted correctly to final records. It is not a claim that the current fully automated final tables already achieve these scores without promotion improvements.

This review validates whether source-supported paper objects are surfaced by the pipeline and can be matched to reviewer-coded gold items. It does not yet validate four separate production-readiness targets: fully automated final-table accuracy without manual promotion, source-span correctness, typed-link correctness, or retrieval effectiveness. Those are reported as next validation tasks rather than as completed claims in this release.

Table 27: Validation results for main paper objects after adding matched intermediate records

Object type	Gold items	Precision	Recall	F1
Methods and designs	798	0.95	0.64	0.76
Datasets	632	0.78	0.52	0.62
Main findings	790	0.96	0.46	0.62
Limitations	669	0.82	0.40	0.54
Open questions	394	0.85	0.54	0.66
Machine-learning or algorithmic presence	58	0.84	0.47	0.60
Machine-learning or algorithmic objects	69	0.70	0.30	0.42

Note: The review covers 100 papers. The machine-learning or algorithmic presence row is a paper-level binary task. The remaining rows are object-level comparisons. Precision, recall, and F1 are computed after adding source-supported intermediate records that match reviewer-coded gold items.

Overall, the validation review supports the current release as a practical source-linked discovery layer for first-pass search and audit, while separating that claim from full production validation. The diagnostic scores are strongest on the main paper objects needed for hybrid retrieval, within-paper link-following, and researcher inspection: methods, datasets, findings, open questions, and paper-level machine-learning or algorithmic use. In this first release, link-following mainly means tracing within-paper links between paper objects, following local citation links by citation role, and using cluster labels or JEL codes to gather related context across papers. Future versions will add citation networks and citation-role-weighted citation networks. For the core filterable

Table 28: Validation results for supplementary and extension objects after adding matched intermediate records

Object type	Gold items	Precision	Recall	F1
Methods and designs	381	0.93	0.57	0.71
Datasets	154	0.60	0.19	0.29
Findings	509	0.83	0.53	0.65
Limitations	157	0.90	0.29	0.44
Open questions	145	0.92	0.32	0.48
Machine-learning or algorithmic objects	299	0.74	0.06	0.11

Note: Supplementary and extension objects include lower-level design details, robustness checks, heterogeneity analyses, diagnostics, constructed or auxiliary data, table-specific findings, model extensions, and broad computational procedures.

objects, precision is high, ranging from 0.78 to 0.96 after adding matched intermediate records. The dominant remaining gap is recall, not precision, so the immediate validation task is to improve record capture and promotion rather than to rebuild the search layer. Some relevant objects are already present in intermediate extraction records but are not yet promoted to the final tables; others, especially supplementary checks, detailed data construction, limitations, and broad computational procedures, require improved extraction prompts and routing. These refinements define the work needed before expanding to a substantially larger corpus.

The lowest score is supplementary machine-learning or algorithmic object recall, which identifies broad computational procedures and lower-level algorithmic components as the clearest target for the next extraction pass.

The unrecovered method objects show where recall needs the most work. Misses are usually not the headline empirical design. They are lower-level design and implementation components: constructed measures, indices, treatments, and outcomes; record linkage, geocoding, and harmonization; sample and panel construction rules; robustness, validation, and diagnostic checks; heterogeneity and subgroup analyses; calibration, simulation, counterfactual, and decomposition methods; and theoretical or model components. Examples include point-in-time macro-vintage alignment, forward-contract return construction, mover-type outcome construction, placebo discontinuity checks, alternative bandwidth and kernel checks, establishment-worker linkage, star-researcher move definitions, welfare-cost model components, and dynamic-program or solver details. These are exactly the objects needed for deeper reproduction and robustness work, so they define the next extraction-improvement target rather than undermining the first-pass search use case.

H Retrieval Detail: DuckDB and Embeddings

The main text summarizes the retrieval architecture. The retrieval layer is deliberately simple. The central DuckDB object is a `chunks` table: each searchable unit stores text, relational keys, metadata, and, when embedded, a 3072-dimensional vector. Side tables attach page spans, table/figure/equation artifacts, claim-evidence links, and bundle membership. This creates a dual-store representation inside DuckDB: the same paper unit is available as a typed relational row for filters and joins and as a stored embedding for semantic search. At query time, the query text is embedded and compared with stored chunk embeddings; the returned object is still the underlying row with its paper, entity, span, and artifact links. For compound questions, the same mechanism can embed narrower subqueries and combine them with SQL predicates over `entity_type`, `chunk_lane`, `field_path`, paper metadata, or artifact type. Lexical scoring, including BM25-style keyword ranking, can be used as a pre-filter over candidate rows or as a reranker after vector retrieval; the current implementation uses SQL filters, lane-specific vector search, lexical-overlap candidate selection for research-synthesis queries, and reciprocal-rank fusion across lanes. Semantic search therefore returns ordinary database rows, not model-generated memories. Table 29 spells out the retrieval units; Table 30 gives a compact schema view; Table 31 gives a current-corpus retrieval trace.

Table 31: Current-corpus retrieval trace over organized paper knowledge

Hop	Retrieved fields	What the researcher learns
Task routing	Paper metadata; method families; shock-cluster labels; source-linked method records	The reproduction example first narrows to a 2025 NBER paper with a firm-level shock design, then asks which data, event-time construction, and artifact evidence must be checked before robustness work begins.
Paper design	Methods; data sources; sample and timing fields	The focal record links w34188 to a firm-level GAP difference-in-differences design, Uruguayan employer–employee administrative records, and the event-time construction used in the causal-effect section.
Page-level evidence	Figure, table, and source-span anchors	The same query resolves to the causal-effect discussion on pages 26–27 and the closest visual anchor, Figure B.1 on page 10, rather than leaving the agent to infer which part of the paper to reproduce.
Related-paper follow-up	Shock-cluster labels; methods; limitations; robustness cues	After the focal figure or table is reproduced, the next checks can be drawn from related wages, bargaining, and firm-rule-change papers in the same shock-design cluster.

Table 29: Discrete chunk lanes over the machine-readable paper record

DuckDB lane	Embedded unit	Example text embedded	Returned context	Typical use	Model
paper_profile	Paper-level summary and coverage record	Paper type, year, topics, research questions, chunk coverage	Paper ID, title, year, coverage metadata	Shortlist papers before entity-level drilldown	OpenAI large embedding
structured_entity	Whole method, finding, data-source, limitation, open-question, or citation record	Method summary plus treatment, outcome, validity-test fields	Entity ID, linked methods, findings, data, artifacts	Compare design objects across papers	Same
field_text	One typed field from an entity record	A single pre-trend result, data requirement, or access-status field	Entity ID, field path, linked paper context	Search exact substantive fields semantically	Same
anchor_span	Resolved local source span around an entity	Sentence or paragraph naming a first-stage test, dataset use, or citation role	Page, section, entity ID, parent paper	Verify a specific claim or diagnostic	Same
structured_artifact	Table, figure, or equation artifact chunk	Table caption, headers, notes, figure caption, equation text	Page number and artifact ID	Retrieve tables, figures, equations, and appendix artifacts	Same
fulltext_context	Section-aware window of surrounding paper text	800-token window from methods, data, results, discussion, or conclusion	Section ID, page span, neighboring overlap	Recover broader context when entity fields are too narrow	Same
entity_edge	Text representation of a typed graph edge	data source --[used_by]--> method plus rationale	Source and target entity IDs	Traverse paper-internal relationships after retrieval	Same
bundle	Claim-evidence bundle	Finding statement with linked method, table, data, and caveat evidence	Bundle ID, members, artifact support	Build auditable multihop evidence chains	Same

Note: Embedded lanes use OpenAI **text-embedding-3-large** embeddings with 3072 dimensions. Structured chunks target 400 tokens. Full-context windows use 800 target tokens with 120-token overlap and are capped per paper. Appendix artifacts inside the downloaded PDF are indexed as table, figure, or equation chunks; separate online appendices are outside the current ingestion set.

Table 30: Simplified DuckDB data format linking relational and semantic retrieval

DuckDB object	One row represents	Key relational fields	Text/vector field	Example row	How it is used
chunks	One searchable unit from the validated paper record	chunk_id , paper_id , entity_type , entity_id , chunk_lane , field_path	text ; embedding FLOAT[3072]	structured_entity row for one method, or field_text row for one data-access field	Dual store: vector search plus SQL joins over the same row.
chunk_spans	Source location for a chunk	chunk_id , paper_id , section_id , page_numbers , char_start , char_end	Resolved source text, no vector required	Anchor span for a method sentence on pages 6–7	Lets the reader verify a retrieved row in the PDF.
Artifact entities	One extracted table, figure, or equation	artifact_id , paper_id , artifact_type , label , page_numbers	caption_or_text ; artifacts also appear as embedded chunks	Equation, Figure 2, or Appendix Table A1	Returns page-level evidence rather than only prose summaries.
Claim-evidence edges	One link from a finding to an artifact	claim_id , artifact_id , relation_type , confidence	No vector required	Finding evidenced by Table 4	Connects extracted claims to their table, figure, or equation support.
Bundle tables	One claim-evidence bundle and its member entities	bundle_id , finding_id , member entity IDs, roles, artifact support counts	Bundle text is embedded through the chunks table	Finding plus method, data source, artifact, limitation, and unresolved-method member rows	Supports multihop evidence chains without rebuilding context from raw text.
Paper metadata	One paper-level metadata record	paper_id , title, year, DOI/OpenAlex identifiers, journal fields	No vector required	Publication year and title for a candidate NBER paper	Supports cohort filters, joins, and external metadata checks.

Note: The same **chunks** rows support SQL and semantic retrieval: relational columns constrain the search, while **embedding** stores the OpenAI **text-embedding-3-large** vector used by DuckDB cosine-similarity search. The query or subquery is embedded at retrieval time and compared against these stored chunk vectors; it is not a separate source of evidence. SQL predicates and keyword/BM25-style lexical matching can select the rows or columns searched, or rerank semantic candidates that contain required terms. The embedding is a retrieval index, not a generated claim. Returned answers remain tied to source rows, spans, artifacts, and entity links.

I Dataset Navigation Appendix: Top Ten Dataset Families

This appendix turns the dataset-discovery result into a researcher-facing navigation layer. A dataset entry is useful only if it tells a reader what empirical object was used, how to access it, what product mode or release was involved, what the paper did with it, and where the evidence can be checked in the paper. The compact index below is intentionally short; the full report for each dataset follows in separate subsections so that caveats and product-mode decisions are readable rather than forced into narrow table columns. All paper counts in this appendix refer to the 2025–2026 dataset-use review sample, not to the full NBER corpus. *Mention papers* are papers where the extraction names the dataset. *Linked-evidence papers* are papers where the dataset use is connected to a method, construction, finding, or paper artifact; these are the most useful counts for page-level navigation.

Table 32: Metadata required for researcher-facing dataset navigation

Researcher question	Required metadata fields	Why this matters
What dataset is this, exactly?	Canonical name; accepted aliases; provider; product mode; release, vintage, version, or supplement; documentation URL.	Generic labels such as ACS, CPS, CRSP, WDI, or FRED are not reproducible objects. The appendix must distinguish microdata from aggregate tables, annual from quarterly products, original source from distributor, and raw fields from constructed bundles.
Can I access and cite it?	Access tier; license or subscription status; application or restricted-data requirement; DOI or registry identifier when available; citation instruction.	A public aggregate table, an IPUMS extract, an RDC file, and a proprietary WRDS product imply different replication and reuse paths.
What did the paper use it for?	Role in paper; measurement task; identification role; method links; finding links; whether the dataset supplies treatment, outcome, mechanism, control, benchmark, calibration target, denominator, or sampling frame.	The same source can be the focal outcome in one paper and only a denominator or control in another. Dataset-discovery tools should not promote denominator-only uses as dataset-centered findings.
What variables or transformations were used?	Native variables; constructed variables; harmonized extract layer; unit of observation; geography; time period; frequency; weights; linkage keys; concordances.	Researchers often need the variable construction, not merely the dataset name. This is especially important for CPS ORG wages, PWT PPP-vintage GDP, UN Comtrade HS/SITC flows, or CRSP/Compustat linked panels.
Where do I verify it in the paper?	Paper URL; PDF page URL when available; section title; table, figure, equation, appendix, and data-availability statement; linked entity IDs.	Page-level navigation is the practical value-add: the reader can jump from a dataset entry to the exact table, appendix, or construction paragraph that verifies it.
What caveat travels with the claim?	Product ambiguity; access caveat; constructed-bundle caveat; source-versus-distributor caveat; sparse artifact support; missing code-side evidence.	The appendix should preserve uncertainty. A claim should be demoted when the final JSON or artifact path does not identify the product mode, variable construction, or paper role.

Table 34: Page-level evidence index to populate for each dataset entry

Field in appendix table	Example value or format	Researcher use
Dataset and product mode	CPS ASEC/March; CRSP monthly returns; PWT 10.1 GDP/PPP; UN Comtrade HS6 imports.	Makes the entry reusable rather than a loose dataset mention.
Paper and quick-read link	w33323 ; w34093 ; w35004 .	Lets the reader open the paper directly from the appendix.
Paper page and section	w33323 PDF, p. 17; Data; Empirical Strategy; Appendix Data Construction.	Lets the reader jump to the exact paper page once page anchors are available.
Evidence locator	Table, figure, equation, appendix table, data-availability statement, replication-file mention, or codebook citation.	Distinguishes empirical result tables from background literature mentions and appendix-only data construction.
Linked paper objects	Dataset entry; method entry; finding entry; constructed variable; limitation; open question.	Shows how the dataset connects to the paper’s design, result, caveat, and follow-up question.
Verification status	Final extraction checked; page or table link checked; text-only lead; code-side not reviewed; product ambiguous; access unresolved.	Prevents overclaiming. Text-only leads should be usable for follow-up reading, not treated as fully verified evidence.

I.1 Full Dataset Reports

I.1.1 American Community Survey (ACS)

Documentation. [U.S. Census Bureau ACS](#). **Coverage.** Rank 1 in the 2025–2026 sample; 127 mention papers; 113 linked-evidence papers. Access is mostly public, with restricted/internal and missing-access cases preserved as caveats.

Empirical role. ACS is not one empirical object in the corpus. It appears as public-use person/household microdata, ACS PUMS or IPUMS extracts, 1-year and 5-year published estimates, tract/block-group/county/ZCTA tables, and Census–ACS stitched repeated cross-sections. Microdata support labor, income, demographic, insurance, disability, family, and household-resource outcomes; published tables support neighborhood, tract, county, and denominator controls; stitched Census–ACS objects support long-run demographic and labor-market panels.

Credible variables and constructions. Employment, labor-force status, wages and earnings, insurance coverage, disability, education, race/ethnicity, nativity, family and household structure, income, housing costs, rents, values, population counts, household counts, and tract/county demographic covariates. Constructed objects include ACS occupation shares, ACS-based non-citizen estimates, ACS/Census harmonized panels, and small-area denominators.

Product-mode decision. Promote ACS claims only after distinguishing public microdata, published geography tables, Census–ACS stitched objects, constructed bundles, denominator/-calibration/imputation roles, and restricted/internal access. Generic “American Community Survey” entries remain leads until unit, product, geography, period, and role are identified.

Representative evidence leads. [w33323](#) for IPUMS/ACS microdata and occupation-share construction; [w33328](#) for ACS 5-year household denominators; [w33329](#) for Census/ACS questionnaire items and IPUMS ACS microdata. Page-level entries should point to the data-construction paragraph, table or figure using the derived measure, and the linked data-source entity.

Curation rule. Promote ACS-centered relationships only when ACS supplies the interpreted response, mechanism, or central constructed object. Demote cases where ACS supplies only tract context, county demographics, denominators, controls, imputation donors, or one component of an author-built panel.

I.1.2 Compustat

Documentation. [S&P Global fundamental financial data](#). **Coverage.** Rank 2 in the 2025–2026 sample; 119 mention papers; 84 linked-evidence papers. Linked-evidence papers are focal and used in the paper. Product mode is mostly direct source use, with constructed bundles and derived-measure-only cases separated. Access is mainly proprietary or restricted, with missing-access cases retained.

Empirical role. Compustat is both a focal estimation input and a measurement input. It measures public-firm financial outcomes, firm characteristics, controls, treatment proxies, mechanisms, calibration targets, and sampling frames for public-firm panels. It frequently appears with CRSP, I/B/E/S, ExecuComp, BoardEx, DealScan, Worldscope, patents, or SEC filings.

Credible variables and constructions. Assets, sales, book equity, market capitalization, Tobin’s q , leverage, secured and unsecured debt, R&D, capital expenditures, cash flow, profitability, ROA, employee counts, industry codes, headquarters or listing identifiers, firm age, revenue growth, inventory-to-COGS, markups, liquidity, payout, debt issuance, and patent-linked firm counts.

Product-mode decision. Keep North America annual fundamentals, quarterly fundamentals, Compustat Global, CRSP/Compustat or CCM market-linked bundles, adjacent vendor/research-link bundles, and generic under-specified Compustat entries separate. Frequency, geography, and linkage role are substantive metadata.

Representative evidence leads. [w33463](#), [w33929](#), and [w34882](#) are useful page-level leads for linked firm-security panels, security/fundamental panels, and accounting-return merges. Navigation entries should show whether the paper used accounting fundamentals, the CCM link, a linked-vendor bundle, or market-return evidence supplied by another source.

Curation rule. Demote market-return, security-price, patent, payout, and vendor-linked claims when Compustat is only an identifier, public-firm frame, bridge, control vector, or accounting-side component. Do not make CRSP return designs look Compustat-centered.

I.1.3 Current Population Survey (CPS)

Documentation. [BLS CPS documentation](#). **Coverage.** Rank 3 in the 2025–2026 sample; 106 mention papers; 71 linked-evidence papers. Linked-evidence papers are all focal and used in the paper. Evidence is mainly direct use, with calibration-target and appendix-data-note cases preserved.

Empirical role. CPS supplies labor-market, income, demographic, family, health-insurance, program-participation, supplement, flow, calibration, and official benchmark series. It is primarily an outcome and measurement input, but it also appears as covariates, robustness evidence, sampling metadata, and calibration targets.

Credible variables and constructions. Employment status, unemployment, labor-force flows, job-finding rates, wages, hourly wages, weekly earnings, annual hours, occupation and industry shares, education, marital and cohabitation matrices, fertility and children, health insurance, income, poverty, taxes and transfers, household weights, licensing status, telework, tobacco-use variables, top-code flags, and imputation indicators.

Product-mode decision. Separate Basic Monthly CPS, ASEC/March, ORG/MORG earnings files, topical supplements, official aggregates, harmonized IPUMS or CEPR extracts, linked rotation panels and flows, and acronym collisions such as Chicago Public Schools.

Representative evidence leads. [w33323](#) for monthly CPS microdata and occupation shares; [w33354](#) for March/IPUMS family and cohabitation content; [w34981](#) for Basic Monthly, ASEC, and ORG combined use. Page-level entries should distinguish monthly status, annual income/-

family content, earnings-rotation files, supplements, and constructed flows.

Curation rule. Promote CPS when it directly supplies the labor, earnings, family, income, insurance, or supplement variable. Demote when CPS is only a calibration target, official aggregate, simulation input, ATUS sampling frame, or acronym collision.

I.1.4 CRSP

Documentation. [CRSP U.S. stock databases](#). **Coverage.** Rank 4 in the 2025–2026 sample; 94 mention papers; 66 linked-evidence papers. Linked-evidence papers are focal and used in the paper; evidence is mostly direct use, with calibration-target cases separated. Access is mainly proprietary or restricted.

Empirical role. CRSP supplies stock returns, prices, market equity, identifiers, delisting outcomes, benchmark index/factor inputs, mutual-fund flows and holdings, Treasury security measures, risk-free rates, and security-fundamental linkages. It is often the measured response in event studies and asset-pricing designs, but it is also a linkage backbone in CRSP/Compustat panels.

Credible variables and constructions. Daily return, monthly return, abnormal return, cumulative abnormal return, total return, ex-dividend return, value-weighted market return, market cap, price, shares outstanding, PERMNO, PERMCO, delisting code and return, volatility, beta, portfolio or factor return, Treasury market value, Treasury basis, risk-free or T-bill rate, mutual fund AUM, holdings, flows, cash ratios, and ownership-buffer measures.

Product-mode decision. Separate stock-security files, daily returns and prices, monthly returns and delisting files, CRSP/Compustat linked panels, index/factor/portfolio series, mutual-fund products, Treasury/risk-free-rate products, and author-created return objects.

Representative evidence leads. [w33477](#) for daily stock data and value-weighted index use; [w33535](#) for monthly U.S. stock database use; [w35007](#) for CRSP mutual-fund and stock database links. Page-level entries should say whether CRSP supplies the interpreted return object or only identifiers and merge keys.

Curation rule. Promote CRSP-centered evidence when CRSP directly measures returns, prices, delistings, fund holdings, fund flows, Treasury series, or identifiers used in the finding. Demote author-created returns, model outputs, Compustat accounting outcomes, TAQ/WRDS quote objects, Bloomberg/FRED bundles, and broad financial-shock labels unless CRSP is the measured object.

I.1.5 World Development Indicators (WDI)

Documentation. [World Bank World Development Indicators](#). **Coverage.** Rank 5 in the 2025–2026 sample; 63 mention papers; 37 linked-evidence papers. Linked-evidence papers include focal and cited-literature uses, used in the paper and mentioned-not-exploited uses, direct uses, calibration targets, and appendix-data-note evidence.

Empirical role. WDI supplies public country-year indicators for GDP, GDP per capita, PPP/base-year GDP, population, labor force, unemployment, urban share, fiscal and debt series, trade and exports, inflation and exchange-rate series, private credit, market capitalization, infrastructure, internet users, and benchmark macro controls. It is commonly a covariate, denominator, calibration target, benchmark, or merge component.

Credible variables and constructions. GDP current USD, GDP per capita, GDP per capita PPP, constant-base-year GDP, GNI per capita Atlas, population, labor force, unemployment, fertility, life expectancy, urban population share, exports of goods and services as percent of

GDP, trade over GDP, FDI, inflation CPI, exchange rate, tax revenue, military expenditure, private credit, deposits, market capitalization, internet users, and infrastructure metrics.

Product-mode decision. Separate national-accounts denominators, vintage/base-year GDP, demographic/labor/health indicators, fiscal and public-finance indicators, external-sector and price series, financial-development indicators, infrastructure and technology indicators, and constructed or linked-source bundles.

Representative evidence leads. [w34093](#) for GDP and vintage sensitivity; [w34389](#) for private credit and deposits; [w34730](#) for internet users and technology indicators. Page-level entries should expose indicator family, base year or vintage, source lineage, and whether WDI is focal or supporting.

Curation rule. Promote WDI only when it directly measures the reported outcome, denominator, indicator, or control family. Demote World Bank non-WDI sources, author-created country-year panels, fiscal cyclical coefficients, migration exposures, conflict panels, market-cap node sizes, and relationships where WDI is only a denominator, benchmark, control, or calibration target.

I.1.6 Federal Reserve Economic Data (FRED)

Documentation. [Federal Reserve Economic Data](#), [Federal Reserve Bank of St. Louis](#). **Coverage.** Rank 6 in the 2025–2026 sample; 43 mention papers; 31 linked-evidence papers. Linked-evidence papers are focal and used in the paper. Evidence is mostly direct use, with calibration-target cases separated. Access is public when resolved.

Empirical role. FRED supplies public time-series inputs for Treasury and TIPS yields, fed funds and policy rates, EFFR–IOR–reserve demand, inflation and price indexes, GDP and investment benchmarks, unemployment controls, VIX and credit spreads, curated FRED-MD/FRED-QD panels, and source-agency or vendor series distributed through FRED. It is often a covariate source, outcome series, benchmark, deflator, calibration target, or component in a constructed macro panel.

Credible variables and constructions. Federal funds rate, fed funds target rate, effective federal funds rate, EFFR minus IOR, IOR, ON RRP spread, reserves, monetary base, GS2, DGS3MO, constant-maturity Treasury yields, TIPS yields, breakeven inflation, PCEPI, PCE inflation, GDP deflator, GDPC1, PNFI, PCECC96, private fixed investment, industrial production, county unemployment, VIX, Moody’s BAA/AAA yields, commercial paper rates, risk-free rates, mortgage rates, FRED-MD macro factors, and FRED-QD standardized series.

Product-mode decision. Separate Treasury yield and term-structure series, monetary-policy rates and reserves, inflation and price-index series, curated FRED-MD/FRED-QD panels, real-activity and national-accounts series, labor or local-area series, financial-market spreads and indexes, and distributor/source-agency bundles.

Representative evidence leads. [w34243](#) for Treasury and TIPS yields; [w34972](#) for EFFR, IOR, ON RRP, and reserve-demand constructions; [w33701](#) for FRED-MD vintage and transformations. Page-level entries should show the FRED series identifier when available, original source agency, vintage, and transformation.

Curation rule. Promote FRED when it directly supplies the rate, spread, yield, inflation, reserve, real-activity, or macro series. Demote recursive shocks, high-rate indicators, rate-cycle indexes, event-window statistics, source-agency cases where FRED is only the access portal, and non-FRED vendor series.

I.1.7 Penn World Table (PWT)

Documentation. [Groningen Growth and Development Centre PWT](#). **Coverage.** Rank 7 in the 2025–2026 sample; 33 mention papers; 17 linked-evidence papers. Linked-evidence papers include focal and cited-literature uses, mostly used in the paper and direct-use, with calibration targets separated.

Empirical role. PWT supplies public country-year national-accounts, PPP, growth-accounting, and benchmark variables. It is used as an outcome series in some macro panels, but more often as a covariate, denominator, benchmark, deflator, calibration target, model input, or component in a constructed panel.

Credible variables and constructions. Real GDP per capita, GDP per capita PPP, output-side real GDP at chained PPPs, log GDP per capita, real GDP per capita growth, population, working population, GDP shares, trade openness, merchandise exports to GDP, capital stock, RRNA, capital-output ratios, TFP, human capital index, labor share, employment, labor costs and wages, consumption, depreciation, price levels, deflators, and investment shares.

Product-mode decision. Separate versioned release or PPP-vintage mode, GDP/PPP/national-accounts mode, growth-accounting inputs, calibration or structural benchmark use, trade-openness/deflator/GDP-share mode, and constructed, simulated, or linked-source bundles. PWT version and PPP vintage are substantive metadata.

Representative evidence leads. [w34142](#) for Feenstra et al. PWT release evidence; [w34718](#) for ETD/GGDC/PWT/WDI/IPUMS merged panels; [w33364](#) for output per capita and national-accounts scaling. Page-level entries should show release, GDP concept, PPP or price concept, and whether PWT is outcome, control, benchmark, or calibration target.

Curation rule. Promote PWT only when final JSON identifies the PWT variable or release and the role in the paper. Demote simulated outcomes, regional panels, harmonized macro series, structural wedges, highway-cost constructions, and relationships where PWT is only a scaling factor, deflator, benchmark, or model calibration input.

I.1.8 Panel Study of Income Dynamics (PSID)

Documentation. [PSID Online](#). **Coverage.** Rank 8 in the 2025–2026 sample; 32 mention papers; 18 linked-evidence papers. Linked-evidence papers are all focal and used in the paper. Evidence is mostly direct use, with one calibration-target case. Access is public, restricted, public/restricted, or missing depending on file and linkage.

Empirical role. PSID supplies longitudinal household, individual, parent-child, child-development, wealth, earnings, labor, consumption, health, and family-structure data. It is often a focal estimation input, but it also appears as robustness evidence, calibration input, sampling frame, mechanism evidence, or imputation source.

Credible variables and constructions. Family income, adult-equivalent family income, parental income, child adult income, income class, race, education, parental education, family structure, marriage and divorce, labor earnings, hourly wages, wage residuals, hours, employment, housework hours, consumption, net worth, housing wealth, property value, home equity, stock share, child outcomes, cognitive and noncognitive skill indexes, video hours, health, QALY, 401k balances, child birth years, and FSP-linked labor-supply outcomes.

Product-mode decision. Separate core family-individual files, wave extracts, income/earnings/labor files, Child Development Supplement and time diaries, wealth/portfolio/housing/retirement files, genealogical or intergenerational links, calibration/simulation/imputation uses, and restricted or mixed-access modes.

Representative evidence leads. [w33465](#) for individual and family files; [w34466](#) for Child

Development Supplement and time-diary content; [w34053](#) for recent-wave PSID wealth and household evidence. Page-level entries should expose file, wave, unit of observation, weights, and public/restricted status.

Curation rule. Promote PSID when the final JSON identifies the file, wave, module, unit, and measured variable. Demote when PSID is only cited background, a calibration moment, imputation source, robustness check, or one component in a structural simulation.

I.1.9 UN Comtrade

Documentation. [UN Comtrade](#). **Coverage.** Rank 9 in the 2025–2026 sample; 32 mention papers; 23 linked-evidence papers. Linked-evidence papers are all focal, used in the paper, and direct-use. Product mode is mostly direct source use, with constructed-bundle cases separated. Access is public when resolved.

Empirical role. Comtrade supplies bilateral trade values, quantities, product-country-year flows, HS/SITC commodity extracts, import/export shares, unit values, sectoral exports, benchmark trade totals, and inputs to trade-exposure or policy-shock constructions. It is a focal input for outcomes, treatments, mechanisms, instruments, structural calibration, or validation depending on the paper.

Credible variables and constructions. Reporter-partner trade values, import/export flows, HS6 and HS4 product panels, SITC frontier-technology volumes, country-product-year values and quantities, export-weighted unit values, advanced-technology export share, PV/wind annual trade values, world seaborne crude exports, country-industry export values and growth, import shares, RCA, market HHI, import penetration, WID/WES shifts, bilateral trade shares, tariff-inclusive spending, sector-country export shares, and intermediate-trade indexes.

Product-mode decision. Separate reporter-partner import/export panels, commodity-classification or concordance mode, value/quantity/price/unit-value mode, harmonized BACI/WITS/services bundles, constructed trade-exposure or policy-shock mode, benchmark/validation/calibration mode, and sector, industry, or firm-linked bundles.

Representative evidence leads. [w34003](#) for reporter-partner-product flows; [w35004](#) for trade-flow and exposure construction; [w34160](#) for sector or firm-linked trade instruments. Page-level entries should expose reporter, partner, direction, product classification, value versus quantity, frequency, and concordance.

Curation rule. Promote Comtrade when it directly measures the trade flow, value, quantity, unit-value input, or explicitly identified exposure component. Demote political, wage, revenue, GDP, firm-growth, or markup relationships when Comtrade only supplies an exposure, denominator, validation benchmark, concordance input, or linked-source component.

I.1.10 National Longitudinal Survey of Youth (NLSY)

Documentation. [NLSY documentation](#). **Coverage.** Rank 10 in the 2025–2026 sample; 27 mention papers; 7 linked-evidence papers. The linked-evidence layer is narrower than 2025–2026 mention coverage, so method and construction links are high-quality leads rather than full-corpus summaries. Linked-evidence papers are focal, used in the paper, and direct-use.

Empirical role. NLSY supplies longitudinal person-level outcomes, test scores, education, employment, wages, assets, debt, mobility, health, child development, crime, job satisfaction, employee ownership, work-from-home items, and restricted-geography linkages. It is often a focal estimation or measurement input, but also appears as robustness evidence, calibration or imputation input, sampling frame, or cited/reused sample.

Credible variables and constructions. NLSY79/NLSY97 cohort indicators, AFQT/ASVAB

and SAT/GPA/transcript measures, wages and earnings, hourly wages, job satisfaction, ESOP indicator, WFH hours/dummy/scaled measures, education and degree timing, college enrollment and loans, net assets, work status, parent and child income percentiles, incarceration, child height-for-age, PIAT scores, BPI measures, smoking, county/metro location, migration, study time, performance pay, and sample weights.

Product-mode decision. Separate NLSY79, NLSY97, dual-cohort comparison mode, Children or Young Adults supplements, test-score/transcript/ability modules, restricted geography or linked-location use, labor/earnings/assets/debt/job-quality mode, and cited-sample, reused-extraction, or imputation mode.

Representative evidence leads. [w34221](#) for NLSY79 debt/assets and work states; [w34981](#) for NLSY97 job quality, ESOP, WFH, and wage profiles; [w33552](#) for restricted geography and ability or quality-of-life links. Page-level entries should expose cohort, module, wave, access status, and whether the reported variable is native or constructed.

Curation rule. Promote NLSY only when cohort, module or supplement, unit of observation, access tier, role in paper, and construction status are explicit. Demote structural states, lifetime projections, teleworkability or racial-capital indexes, external policy relationships, cited samples, and imputation pools unless the NLSY measurement path is direct.

Implementation note: The full reports are built from the completed dataset-use review reports and packet summaries. Paper-level links are reliable quick-read links. Page-specific PDF anchors should be populated from artifact links only after the final JSON evidence layer supplies stable page and artifact IDs.

Table 33: Top ten dataset families: compact navigation index, 2025–2026 sample

Rank	Dataset and documentation	Mention papers 2025–2026	Linked evidence papers 2025–2026	Main product modes to inspect	Report
1	American Community Survey (ACS)	127	113	PUMS/IPUMS microdata; published 1-year/5-year geography tables; Census–ACS stitched panels; restricted/internal ACS.	Section I.1.1
2	Compustat	119	84	North America annual fundamentals; quarterly fundamentals; Global; CRSP/Compustat links; linked-vendor panels.	Section I.1.2
3	Current Population Survey (CPS)	106	71	Basic Monthly; ASEC/March; ORG/MORG; topical supplements; official aggregates; linked rotation panels.	Section I.1.3
4	CRSP	94	66	Stock-security files; daily/monthly returns; delistings; mutual funds; Treasury products; CRSP/Compustat links.	Section I.1.4
5	World Development Indicators (WDI)	63	37	GDP and national accounts; base-year/vintage indicators; demographic, labor, fiscal, price, external-sector, financial-development, and infrastructure series.	Section I.1.5
6	Federal Reserve Economic Data (FRED)	43	31	Treasury and TIPS yields; monetary-policy rates; reserves; inflation and prices; FRED-MD/FRED-QD; real-activity and financial-spread series.	Section I.1.6
7	Penn World Table (PWT)	33	17	Versioned releases; PPP-vintage GDP; growth-accounting inputs; structural calibration benchmarks; trade openness and deflators.	Section I.1.7
8	Panel Study of Income Dynamics (PSID)	32	18	Core family/individual files; wave extracts; Child Development Supplement; wealth, labor, housing, retirement, and intergenerational modules.	Section I.1.8
9	UN Comtrade	32	23	Reporter-partner-product flows; HS/SITC classifications; value/quantity/unit values; BACI/WITS variants; trade-exposure constructions.	Section I.1.9
10	National Longitudinal Survey of Youth (NLSY)	27	7	NLSY79; NLSY97; Children/Young Adults; test-score and transcript modules; restricted geography; labor, assets, debt, and job-quality items.	Section I.1.10

Note: Counts are for the 2025–2026 dataset-use review sample. Mention-paper counts record papers where the dataset family appears in the extracted source list. Linked-evidence paper counts require the dataset use to be connected to a method, construction, finding, or paper artifact, and are therefore better evidence for page-level navigation.

Table 35: American Community Survey: dataset-specific navigation table, 2025–2026 sample

Field	Dataset-specific entry
Coverage	2025–2026 sample: 127 mention papers; 113 linked-evidence papers. Linked-evidence papers are mostly focal, direct-use, and source-level evidence.
Product modes	PUMS/IPUMS microdata; ACS 1-year and 5-year published tables; tract, block-group, county, and ZCTA geography tables; Census–ACS stitched panels; restricted/internal ACS; ACS-based constructed bundles.
Credible variables	Employment, labor-force status, wages, earnings, insurance, disability, education, race/ethnicity, nativity, family structure, income, housing costs, rents, values, population counts, household counts, and small-area demographics.
Evidence leads	w33323 for IPUMS/ACS occupation-share construction; w33328 for 5-year household denominators; w33329 for ACS/Census questionnaire items and IPUMS ACS microdata.
Researcher caution	Do not collapse microdata, published estimates, stitched Census–ACS products, imputation sources, and restricted files into a single ACS object.

Table 36: Compustat: dataset-specific navigation table, 2025–2026 sample

Field	Dataset-specific entry
Coverage	2025–2026 sample: 119 mention papers; 84 linked-evidence papers. Linked-evidence papers are focal and used in the paper, with proprietary and restricted access caveats.
Product modes	North America annual fundamentals; quarterly fundamentals; Compustat Global; CRSP/Compustat or CCM linked panels; linked vendor or research bundles; generic under-specified Compustat entries.
Credible variables	Assets, sales, book equity, Tobin’s q, leverage, debt, R&D, capital expenditures, cash flow, profitability, ROA, employees, industry codes, firm age, liquidity, payout, and debt issuance.
Evidence leads	w33463 , w33929 , and w34882 for firm-security links, security/fundamental panels, and accounting-return merges.
Researcher caution	Separate native accounting variables from market-return outcomes, CCM link-table evidence, linked vendor panels, and author-created firm-level constructs.

Table 37: Current Population Survey: dataset-specific navigation table, 2025–2026 sample

Field	Dataset-specific entry
Coverage	2025–2026 sample: 106 mention papers; 71 linked-evidence papers. Linked-evidence papers are all focal and used in the paper.
Product modes	Basic Monthly CPS; ASEC/March; ORG/MORG earnings files; topical supplements; official aggregates; harmonized IPUMS or CEPR extracts; linked rotation panels and flows; acronym-collision cases.
Credible variables	Employment, unemployment, labor-force flows, wages, hourly wages, weekly earnings, hours, occupation, industry, education, family, income, poverty, taxes and transfers, health insurance, licensing, telework, and tobacco use.
Evidence leads	w33323 for monthly CPS occupation shares; w33354 for March/IPUMS family content; w34981 for Basic Monthly, ASEC, and ORG combined use.
Researcher caution	Identify the CPS product before reuse; Basic Monthly labor status, ASEC income, ORG wages, topical supplements, and official aggregates answer different questions.

Table 38: CRSP: dataset-specific navigation table, 2025–2026 sample

Field	Dataset-specific entry
Coverage	2025–2026 sample: 94 mention papers; 66 linked-evidence papers. Linked-evidence papers are focal and used in the paper.
Product modes	Stock-security files; daily returns and prices; monthly returns and delistings; CRSP/Compustat linked panels; indexes, factors, and portfolio series; mutual funds; Treasury and risk-free-rate products.
Credible variables	Daily returns, monthly returns, abnormal returns, prices, market equity, market capitalization, PERMNO, PERMCO, delisting code and return, volatility, beta, index returns, fund AUM, holdings, flows, and Treasury measures.
Evidence leads	w33477 for daily stock data; w33535 for monthly U.S. stock database use; w35007 for CRSP mutual-fund and stock database links.
Researcher caution	State whether CRSP supplies the interpreted return object or only identifiers and merge keys in a CRSP/Compustat or vendor-linked panel.

Table 39: World Development Indicators: dataset-specific navigation table, 2025–2026 sample

Field	Dataset-specific entry
Coverage	2025–2026 sample: 63 mention papers; 37 linked-evidence papers. Linked-evidence papers include direct uses, calibration targets, appendix notes, and a small number of cited-literature or mentioned-not-exploited uses.
Product modes	National-accounts denominators; base-year and vintage GDP; demographic, labor, and health indicators; fiscal and public-finance indicators; external-sector and price series; financial-development indicators; infrastructure and technology indicators; linked-source bundles.
Credible variables	GDP, GDP per capita, PPP GDP, population, labor force, unemployment, fertility, life expectancy, urban share, exports/GDP, FDI, inflation, exchange rates, tax revenue, debt, private credit, deposits, market capitalization, and internet users.
Evidence leads	w34093 for GDP and vintage sensitivity; w34389 for private credit and deposit indicators; w34730 for internet and technology indicators.
Researcher caution	Record indicator family, source lineage, base year or vintage, and whether WDI is focal evidence, a denominator, a macro control, or one component of a constructed country-year panel.

Table 40: Federal Reserve Economic Data: dataset-specific navigation table, 2025–2026 sample

Field	Dataset-specific entry
Coverage	2025–2026 sample: 43 mention papers; 31 linked-evidence papers. Linked-evidence papers are focal and used in the paper, mostly as direct public time-series inputs.
Product modes	Treasury and TIPS yields; term-structure series; monetary-policy rates and reserves; inflation and price indexes; FRED-MD and FRED-QD panels; real-activity and national-accounts series; labor/local series; financial spreads and indexes; distributor/source-agency bundles.
Credible variables	Federal funds rate, EFFR, IOR, reserves, monetary base, Treasury yields, TIPS yields, breakeven inflation, PCEPI, GDP deflator, GDPC1, investment, industrial production, unemployment, VIX, Moody’s yields, commercial paper rates, and FRED-MD/FRED-QD factors.
Evidence leads	w34243 for Treasury and TIPS yields; w34972 for EFFR, IOR, ON RRP, and reserve-demand constructions; w33701 for FRED-MD vintage and transformations.
Researcher caution	FRED is often a distributor. Page-level entries should include the series identifier, original source agency, vintage, transformation, and whether the variable is raw or author-constructed.

Table 41: Penn World Table: dataset-specific navigation table, 2025–2026 sample

Field	Dataset-specific entry
Coverage	2025–2026 sample: 33 mention papers; 17 linked-evidence papers. Linked evidence includes direct uses and calibration targets.
Product modes	Versioned release or PPP vintage; GDP, PPP, and national-accounts series; growth-accounting inputs; calibration or structural benchmarks; trade openness, deflator, and GDP-share inputs; constructed or simulated linked-source bundles.
Credible variables	Real GDP per capita, output-side real GDP, PPP GDP, log GDP per capita, GDP growth, population, working population, capital stock, TFP, human capital, labor share, employment, labor costs, consumption, depreciation, price levels, deflators, and investment shares.
Evidence leads	w34142 for PWT release evidence; w34718 for ETD/GGDC/PWT/WDI/IPUMS merged panels; w33364 for output per capita and national-accounts scaling.
Researcher caution	PWT release and PPP vintage are substantive; generic PWT entries should be treated as leads until release, variable concept, price concept, and paper role are known.

Table 42: Panel Study of Income Dynamics: dataset-specific navigation table, 2025–2026 sample

Field	Dataset-specific entry
Coverage	2025–2026 sample: 32 mention papers; 18 linked-evidence papers. Linked-evidence papers are all focal and used in the paper, with public, restricted, mixed, and missing access statuses.
Product modes	Core family-individual files; wave extracts; income, earnings, and labor files; Child Development Supplement and time diaries; wealth, portfolio, housing, and retirement files; intergenerational links; calibration, simulation, or imputation roles; restricted or mixed access.
Credible variables	Family income, adult income, parental income, race, education, family structure, labor earnings, wages, hours, employment, consumption, net worth, housing wealth, property value, stock share, child outcomes, skills, health, QALY, 401k balances, and birth histories.
Evidence leads	w33465 for individual and family files; w34466 for Child Development Supplement and time diaries; w34053 for recent-wave wealth and household evidence.
Researcher caution	Record module, wave range, unit, weights, and public or restricted status before reusing a PSID entry. Constructed residuals, trajectories, and simulation inputs should not be treated as native PSID variables.

Table 43: UN Comtrade: dataset-specific navigation table, 2025–2026 sample

Field	Dataset-specific entry
Coverage	2025–2026 sample: 32 mention papers; 23 linked-evidence papers. Linked-evidence papers are all focal, used in the paper, and direct-use, with mostly public access.
Product modes	Reporter-partner import/export panels; HS/SITC commodity classifications; value, quantity, price, or unit-value mode; BACI, WITS, or services bundles; constructed trade-exposure or policy-shock mode; benchmark, validation, or calibration mode; sector, industry, or firm-linked bundles.
Credible variables	Trade values, quantities, HS6/HS4/SITC flows, import and export shares, unit values, sector exports, RCA, import penetration inputs, WID/WES shifts, bilateral trade shares, tariff-inclusive spending, and sector-country export shares.
Evidence leads	w34003 for reporter-partner-product flows; w35004 for trade-flow and exposure construction; w34160 for sector or firm-linked trade instruments.
Researcher caution	Expose reporter, partner, direction, product classification, value versus quantity, frequency, and concordance. Do not treat author-built exposures or concorded industry objects as raw Comtrade fields.

Table 44: National Longitudinal Survey of Youth: dataset-specific navigation table, 2025–2026 sample

Field	Dataset-specific entry
Coverage	2025–2026 sample: 27 mention papers; 7 linked-evidence papers. Linked evidence is narrower than 2025–2026 mention coverage, so it is best treated as high-quality lead evidence.
Product modes	NLSY79; NLSY97; dual-cohort comparisons; Children or Young Adults supplements; test-score, transcript, and ability modules; restricted geography or linked-location use; labor, earnings, assets, debt, and job-quality mode; cited-sample or imputation roles.
Credible variables	AFQT/ASVAB, SAT/GPA, transcripts, wages, earnings, job satisfaction, ESOP, work-from-home items, education, college loans, assets, work status, incarceration, PIAT, BPI, child height, smoking, migration, study time, performance pay, and sample weights.
Evidence leads	w34221 for NLSY79 debt/assets and work states; w34981 for NLSY97 job quality, ESOP, WFH, and wage profiles; w33552 for restricted geography and ability links.
Researcher caution	Generic NLSY is not enough. The entry needs cohort, module, wave, access tier, and whether the variable is a native item, restricted linkage, or author construction.

Table 45: Promotion rules for dataset-centered claims

Role classification	Promote as dataset-centered evidence when...	Demote to supporting or lead evidence when...
Focal estimation input	The final JSON links the dataset to a method or finding and the dataset supplies the interpreted treatment, outcome, mechanism, or key constructed variable.	The dataset appears only in a broad source bundle, bibliography, literature comparison, or unlabeled data appendix.
Outcome or mechanism measurement	The variables are native or transparently constructed from the dataset and the paper table or figure reports the linked result.	The outcome is measured in another source and this dataset supplies only demographics, controls, geography, or denominators.
Treatment or exposure construction	The construction path identifies source variables, unit, geography, timing, concordance, and linked method.	The treatment is an author-created shock, policy index, simulated state, or shift-share object where the dataset is one input among many.
Benchmark, denominator, calibration, or imputation	The entry is explicitly labeled as a benchmark, denominator, calibration target, or imputation source.	The report sentence is written as if the dataset estimates the causal relationship itself. These roles should be visible, but not overpromoted.
Restricted or proprietary access	The appendix reports access tier and likely replication path.	The entry is presented as if a public user can reproduce it from the provider URL alone.

J Algorithmic and Machine-Learning Detail Tables

This appendix reports the field distribution, coding fields, algorithm or task families, and method-citation groups behind the compact algorithm-method result in the main text. The main body keeps only the contextual-use table; the tables here are intended as audit and interpretation support.

Table 46: Primary JEL fields of papers with algorithmic methods, 2025–2026 NBER sample

JEL field	Papers	Share	JEL field	Papers	Share
J Labor/Demography	158	26.7%	E Macro/Monetary	46	7.8%
I Health/Education/Welfare	115	19.4%	N Economic History	45	7.6%
O Development/Innovation/Growth	105	17.7%	F International	44	7.4%
D Micro/Behavior	90	15.2%	Q Ag/Environment/Energy	42	7.1%
G Finance	88	14.9%	K Law	29	4.9%
L Industrial Organization	79	13.3%	M Business/Admin	29	4.9%
H Public	55	9.3%	P Systems	18	3.0%
C Methods/Econometrics	53	9.0%	Z Other Special Topics	12	2.0%
R Urban/Regional	51	8.6%	A General Economics	3	0.5%
			B History of Economic Thought	2	0.3%

Note: Counts use NBER JEL letters where available. Shares are relative to the 592 algorithmic papers and do not sum to 100% because papers can carry multiple JEL codes.

Table 47: Researcher-facing fields added for algorithmic and machine-learning methods

Field family	Fields retained for readers	Why it matters
Paper provenance	NBER paper URL; paper title; page, table, figure, or chunk reference where available	Lets a reader inspect the original evidence rather than relying on local file names or extraction paths.
Algorithm identity	Exact algorithm name where available; canonical algorithm or task family; contextual empirical object produced	Separates interpretable method names such as LASSO, GPT, OCR, and random forest from task families such as prediction, record linkage, and geocoding.
Research-stage role	Data assembly/linkage; treatment or exposure measurement; outcome measurement; control construction; main estimator; heterogeneity or targeting	Distinguishes data construction from identification. A text classifier that builds a policy measure is not the same object as a causal forest estimating treatment-effect heterogeneity.
Validation evidence	External or held-out validation; formal human agreement; manual audit; design or sensitivity check; no explicit validation	Shows whether text, OCR, and prediction outputs are treated as generated data and checked before entering regressions.
Empirical-method context	Co-occurring DiD/event study, IV, RCT, RD, structural, or descriptive panel design	Measures whether algorithms complement credibility-revolution designs or appear as standalone prediction and measurement exercises.
Author-cited methodology	Cited reference; citation role; manual citation type; empirical precedent flag	Separates true method foundations from empirical papers that set practical validation or measurement standards.

Table 48: Top algorithm or task families by empirical role, 2025–2026 NBER sample

Algorithm or task family	Main empirical role in the sample	Dominant validation signals
Record linkage / fuzzy matching (250 papers)	Data assembly and matched-entity construction: 245 papers use it for linkage; only a small number use the linked output directly for heterogeneity or controls.	No explicit validation (127); external or held-out validation (59); manual audit (42).
Prediction / scoring / imputation (137 papers)	Generated empirical inputs: outcomes (49), controls (55), treatments or exposures (21), and heterogeneity or targeting (14).	No explicit validation (66); external or held-out validation (46); manual audit (15).
Dictionary / keyword rules (112 papers)	Mostly linkage or coded document measures: linkage (71), outcomes (14), controls (27), and exposures (5).	No explicit validation (45); external or held-out validation (27); manual audit (21).
GPT / OpenAI LLM (64 papers)	Text-coded empirical variables: controls (36), outcomes (22), treatments or exposures (8), and linkage support (6).	Manual audit (24); external or held-out validation (16); formal human agreement (16).
Geocoding / spatial assignment (54 papers)	Data assembly and exposure construction: linkage/location assignment (36), exposures (7), outcomes (5), and controls (8).	No explicit validation (19); external or held-out validation (16); design or sensitivity check (10).
K-means / clustering (44 papers)	Latent groups and compressed features: controls (24), outcomes (11), treatments or exposures (10), and occasional linkage.	No explicit validation (17); external or held-out validation (12); manual audit (7).
NLP / supervised text classification (40 papers)	Coded text measures: outcomes (14), controls (12), linkage support (11), and limited treatment/exposure or heterogeneity use.	No explicit validation (20); external or held-out validation (6); manual audit (6).
LASSO / regularization (34 papers)	High-dimensional selection: controls (10), treatments or exposures (9), outcomes (8), heterogeneity (5), and main-estimator roles (3).	External or held-out validation (17); no explicit validation (14); manual audit (2).
Random forest (28 papers)	Prediction and heterogeneity: heterogeneity or targeting (12), controls (9), outcomes (7), and rare linkage support.	External or held-out validation (14); no explicit validation (9); design or sensitivity check (3).
OCR / digitization (20 papers)	Document digitization and data assembly: all 20 counted papers use OCR or digitization as a data-construction input.	No explicit validation (9); manual audit (5); external or held-out validation (3).

Note: Research-stage roles are not mutually exclusive because a paper can use one algorithmic output in multiple empirical roles. Some rows are exact method families, while others are task families where the papers do not name a single canonical algorithm; the table summarizes the stage-use profile in prose to avoid conflating data construction with identification.

The table makes two patterns visible. First, the largest algorithmic contribution is data construction, especially record linkage and geocoding, rather than primary estimation. Second, text and document methods are heterogeneous. GPT and LLM methods are unusually likely to report manual audits or formal human agreement, while OCR and supervised text classification still have many rows with no explicit validation record. This is the generated-data analogue of validity testing: once text or OCR output becomes a treatment, outcome, or control, the validation status of that generated input is part of the empirical design.

Table 49: Author-cited methodology foundations and empirical precedents

Manual citation group	Manual type	Papers	Reader interpretation
Text-as-data frameworks	Method overview	3	Justifies turning documents into empirical variables.
Record linkage and census matching	Linkage method	3	Supports linked-record and matched-panel construction.
LASSO and high-dimensional controls	Statistical foundation	12	Regularization or high-dimensional control selection inside causal designs.
Causal forests and policy learning	Statistical foundation	9	Heterogeneous treatment effects and policy targeting.
Debiased / double machine learning	Statistical foundation	4	Orthogonal scores and nuisance estimation for causal or structural parameters.
Transformer, BERT, and sentence embeddings	Model reference	9	Modern NLP encoders or sentence/document embeddings.
Word embeddings	Model reference	5	Vector-space text representations.
Other high-dimensional/statistical learning references	Related learning reference	5	Secondary technical context, not usually the clean canonical citation.
ML empirical application precedents	Empirical precedent	11	Applied papers that set practical standards for how generated measures are built, validated, or used in credible designs.

These fields support the more informative descriptive question: not whether a paper “uses ML,” but what empirical input the algorithm creates, where that input enters the research design, how it is validated, and which methodological or applied precedent the authors use to legitimate it.

K Frontier Plot Lookup Tables

Table 50: Open-question frontier groups: researcher-facing labels (Figure 4)

#	What this group asks	Why it remains open
1	Which care channels actually change health?	Claims and administrative data see utilization better than care processes or patient experience.
2	How do beliefs become behavior?	Beliefs and attention are hard to measure repeatedly; experiments rarely observe the full choice chain.
3	What do employers change behind the wage?	Worker records are more common than employer-side actions inside HR systems.
4	When do expectations move real decisions?	High-frequency belief data are rarely linked to subsequent choices.
5	Do credit and macro findings travel across regimes?	Comparable financial microdata and policy shocks are uneven across countries.
6	Which health policies survive a new system?	Payment rules, eligibility, and administrative records are system-specific.
7	Which shock identifies the counterfactual world?	Needed shocks are rare; model validation often requires outcomes not in the estimating sample.
8	What linked records would reveal policy mechanisms?	Relevant datasets sit behind different agencies, identifiers, and privacy rules.
9	Which hidden traits sort workers and firms?	Core traits are rarely measured directly; proxies are entangled with outcomes.
10	Can macro-finance mechanisms be seen beyond public firms?	Coverage and balance-sheet definitions differ sharply between public-firm and economy-wide data.
11	Does the AI and innovation frontier generalize?	Technology moves faster than long panels accumulate; evidence is concentrated in specific countries.
12	What school mechanisms matter beyond tests?	Administrative records track achievement better than classroom and peer-network mechanisms.
13	Which climate and energy policies travel?	Local geography, regulation, and infrastructure make direct transport difficult.
14	How do work, mobility, and family channels interact?	Linked household, employer, and location panels are still incomplete.
15	What managerial traits and market frictions are missing?	Managerial and operational variables are usually proprietary or indirectly inferred.

Table 51: Limitation frontier groups: researcher-facing labels (Figure 5)

#	Binding constraint	Why it persists
1	Pay records miss the adjustment margin	Wage and earnings records miss hours, benefits, job content, and within-job adjustments.
2	Labor findings are trapped in local institutions	Labor markets differ in unions, contracts, UI systems, and firm-worker matching rules.
3	Green-policy evidence is place-bound	Treatment effects depend on grid structure, local pollution transport, and existing technology.
4	Health-payment evidence is system-specific	Reimbursement, ownership, patient mix, and state rules define both the treatment and the counterfactual.
5	Platform samples are not populations	Opt-in users are selected on attention, wealth, digital access, and preferences.
6	Small surveys carry large claims	Low response rates and small cells create fragile estimates exactly where interpretation needs precision.
7	Claims data miss care quality	Claims are built for payment, not physiology or patient experience.
8	Self-reports and record links blur effects	Measurement error can be differential; linkage false positives are hard to audit.
9	Rare health outcomes need larger panels	Mortality and severe outcomes are rare, so subgroup estimates require very large linked panels.
10	Environmental estimates hinge on exposure maps	Exposure is spatially modeled without complete chemistry or defensive behavior.
11	Earnings records do not fully link or measure pay	Caps, missing identifiers, and absent pay-rate details leave key margins unobserved.
12	Selection remains when there is no shock	Without exogenous variation, richer controls cannot rule out unobserved selection.
13	Financial institutions do not travel cleanly	Market structure, supervision, and investor composition vary across settings.
14	Heterogeneity claims outrun statistical power	Samples split by subgroup or time leave standard errors dominating the signal.
15	Causal claims need better shocks	Observed exposure is often chosen, anticipated, or jointly determined with outcomes.

L Open-Question and Limitation Frontier Detail

This appendix reports the frontier maps and researcher-facing lookup tables behind the two prominence figures in the main text. The maps are visual indexes rather than the source of the interpretation: colored numbers identify the 15 largest researcher-facing groups, grey points show surrounding extracted statements, and the tables define the labels and composition of each group. The highlighted groups are selected from 85 open-question clusters and 129 limitation clusters. The inputs are extracted open-question and limitation records from 1,381 2025–2026 NBER papers; records were routed by extracted frontier or constraint family, embedded with OpenAI `text-embedding-3-large` embeddings, projected with UMAP for display, and clustered with HDBSCAN using the leaf-style frontier-clustering run. Codex (GPT-5.5, extra-high reasoning) reviewed all input text for each displayed cluster before assigning the researcher-facing labels. Table 52 defines the non-standard clustering quantities that are summarized more lightly in the main text.

Table 52: Frontier-cluster denominators and what each quantity measures

Quantity	Open questions	Limitations	What it measures
Papers represented	1,381	1,381	Papers contributing at least one routed open-question or limitation statement to the frontier-clustering input.
Extracted records routed to clustering	4,243	6,196	Author-stated records retained after routing to the open-question frontier family or limitation constraint family. This is a clustering-input count, not the full layer total.
Records with embeddings	3,632	5,575	Routed records with usable text embedded for semantic similarity calculations.
Assigned records	2,574	3,991	Embedded records assigned by HDBSCAN to a non-noise cluster. Records treated as noise are not part of any displayed cluster share.
HDBSCAN clusters	85	129	Non-noise clusters produced by the leaf-style clustering run after routing and embedding.
Displayed largest-15 records	776	962	Assigned records contained in the 15 largest clusters shown in the main ladder figures and highlighted in the appendix maps.
Bar-share denominator	776	962	The denominator for the main ladder bars: each bar is the cluster’s share within the displayed 15-cluster set, not a share of all extracted records.

Table 53: Interpretive summary of numbered frontier open-question clusters.

# Researcher-facing label	What this cluster summarizes	Why it remains open/hard
1 Which care channels actually change health?	Which margins of access, care quality, spending, pain, mobility, mental health, and provider behavior connect policy to outcomes.	Claims and administrative data see utilization better than care processes, symptoms, patient experience, and provider decision making.
2 How do beliefs become behavior?	When information, trust, norms, salience, and cognitive constraints actually change decisions rather than survey answers.	Beliefs and attention are hard to measure repeatedly in field settings, and experiments rarely observe the full downstream choice chain.
3 What do employers change behind the wage?	Whether firms adjust hiring, hours, job design, promotion, amenities, discrimination, or training when labor-market incentives move.	Worker records are more common than employer-side actions, and many firm responses are inside HR systems or informal managerial decisions.
4 When do expectations move real decisions?	Which expectation channels connect inflation, risk, policy news, and uncertainty to prices, investment, attention, and spending.	High-frequency belief data are rarely linked to subsequent choices, and attention is usually inferred rather than directly observed.

5 Do credit and macro findings travel across regimes?	Which banking, procurement, liquidity, and macro-finance mechanisms survive different countries, regulations, and market structures.	Comparable financial microdata and policy shocks are uneven across countries, and institutions mediate the same nominal shock differently.
6 Which health policies survive a new system?	Whether health-insurance, hospital, Medicaid, abortion, and provider-market estimates generalize to other states, cohorts, or health systems.	Payment rules, provider ownership, eligibility, and administrative records are system-specific, so replications do not map cleanly.
7 Which shock identifies the counterfactual world?	What exogenous variation can validate structural, welfare, and counterfactual claims in macro, climate, finance, and policy models.	The needed shocks are rare, endogenous exposure is common, and model validation often requires outcomes not observed in the estimating sample.
8 What linked records would reveal policy mechanisms?	How people, firms, agencies, and policies interact once individual, employer, administrative, and institutional records can be connected.	The relevant datasets sit behind different agencies, identifiers, privacy rules, and time coverage, leaving mechanisms split across files.
9 Which hidden traits sort workers and firms?	How latent productivity, preferences, attachment, managerial quality, discrimination, and amenities shape matches and earnings.	The core traits are rarely measured directly, and proxies are entangled with outcomes, institutions, and selection into jobs.
10 Can macro-finance mechanisms be seen beyond public firms?	Whether capital-market, liquidity, and valuation mechanisms estimated in selected samples hold for private firms and broader macro aggregates.	Coverage, balance-sheet definitions, and long time-series links differ sharply between public-firm datasets and economy-wide sources.
11 Does the AI and innovation frontier generalize?	Whether AI, patent, creativity, productivity, and occupation results travel across countries, sectors, firms, and later adoption waves.	The technology is moving faster than long panels can accumulate, and current evidence is concentrated in specific countries and platforms.
12 What school mechanisms matter beyond tests?	Which teacher, peer, counseling, curriculum, attention, and school-climate channels produce long-run education and labor-market outcomes.	Administrative records track achievement better than the day-to-day mechanisms inside classrooms, schools, and peer networks.
13 Which climate and energy policies travel?	Which clean-tech, electricity, land-use, EV, air-quality, and adaptation estimates hold across places, technologies, and grid conditions.	Local geography, regulation, infrastructure, and technology adoption determine treatment intensity and make direct transport difficult.
14 How do work, mobility, and family channels interact?	How labor supply, migration, fertility, caregiving, household bargaining, and family insurance jointly transmit shocks and policies.	The relevant choices happen across people and over time, but linked household, employer, and location panels are still incomplete.
15 What managerial traits and market frictions are missing?	Which latent managerial practices, contracting frictions, market power, search costs, and operational choices explain observed firm outcomes.	Managerial and operational variables are usually proprietary, qualitative, or indirectly inferred from outcomes and coarse firm records.

Table 55: Subpart composition within numbered frontier open-question clusters.

#	Route / family	Question-type shares	Data-need shares
1	Empirical agenda / Measurement unstructured text	Mechanism 97%; Extension 1%; Methodology 1%	Linked records 48%; Administrative 32%; Survey 18%; Other 3%
2	Empirical agenda / Measurement unstructured text	Mechanism 84%; Policy 7%; Extension 5%; Methodology 5%	Experimental 48%; Linked records 28%; Survey 13%; Other 11%
3	Empirical agenda / Measurement unstructured text	Mechanism 88%; Policy 7%; Methodology 3%; Extension 2%	Linked records 44%; Administrative 34%; Survey 15%; Other 7%

4 Empirical agenda / Measurement unstructured text	Mechanism 84%; Methodology 10%; Policy 5%	Financial filings 29%; Linked records 19%; Survey 16%; Other 34%
5 External-validity transfer / External validity single context	External validity 63%; Extension 14%; Mechanism 12%; Policy 9%; Other 2%	Administrative 68%; Experimental 9%; Linked records 5%; Other 18%
6 External-validity transfer / Data scale geographic	External validity 81%; Extension 13%; Policy 4%; Mechanism 2%	Administrative 79%; Survey 17%; Linked records 4%
7 Modeling/counterfactual program / Identification no variation	Mechanism 62%; Identification 13%; Policy 12%; Methodology 10%; Other 4%	Administrative 54%; Experimental 17%; Linked records 15%; Other 13%
8 Empirical agenda / Data access administrative	Mechanism 70%; Extension 14%; Identification 10%; Policy 6%	Administrative 62%; Linked records 34%; Experimental 2%; Other 2%
9 Modeling/counterfactual program / Measurement latent construct	Mechanism 76%; Methodology 20%; Extension 2%; Measurement 2%	Administrative 38%; Linked records 36%; Survey 20%; Other 6%
10 External-validity transfer / Data scale sample size	External validity 62%; Extension 23%; Mechanism 11%; Policy 4%	Administrative 34%; Financial filings 21%; Linked records 17%; Other 28%
11 External-validity transfer / Data scale geographic	External validity 67%; Extension 24%; Mechanism 4%; Policy 4%	Administrative 31%; Web scraped 22%; Linked records 16%; Other 31%
12 Empirical agenda / Measurement unstructured text	Mechanism 100%	Linked records 34%; Experimental 30%; Administrative 20%; Other 16%
13 External-validity transfer / External validity single context	External validity 56%; Extension 19%; Policy 14%; Mechanism 7%; Other 5%	Administrative 51%; Linked records 21%; Survey 7%; Other 21%
14 Modeling/counterfactual program / Measurement unstructured text	Mechanism 95%; Heterogeneity 2%; Policy 2%	Linked records 63%; Administrative 19%; Survey 16%; Other 2%
15 Empirical agenda / Measurement latent construct	Mechanism 98%; Extension 2%	Experimental 26%; Financial filings 21%; Administrative 17%; Other 36%

Table 57: Interpretive summary of numbered limitation clusters.

# Researcher-facing label	What this cluster summarizes	Why it remains open/hard
1 When pay records miss the adjustment margin	Whether wages, hours, benefits, over-time, job tasks, search channels, or non-wage compensation absorb labor-market shocks.	Common wage and earnings records often miss hours, benefits, informal search, job content, and the timing of within-job adjustments.
2 Labor findings are trapped in local institutions	Whether bargaining, monopsony, displacement, and worker-rights results generalize beyond one country or institutional design.	Labor markets differ in unions, contracts, enforcement, UI systems, job ladders, and firm-worker matching institutions.
3 Green-policy evidence is place-bound	Whether climate, energy, transport, land, and EV estimates apply outside the specific geography or technology studied.	Treatment effects depend on grid structure, local pollution transport, land markets, policy rules, and existing technology adoption.
4 Health-payment evidence is system-specific	Whether hospital, insurer, provider, Medicaid, bankruptcy, and out-of-network payment estimates extend to other health systems.	Institutional details in reimbursement, ownership, patient mix, and state rules define both the treatment and the counterfactual.
5 Platform samples are not populations	Whether behavior observed in brokerages, apps, online platforms, games, or selected finance samples reflects broader populations.	Opt-in users, platform participants, and brokerage clients are selected on attention, wealth, digital access, and preferences.
6 Small surveys carry large claims	Whether survey-based heterogeneity, and mechanisms, and concordance mappings are stable enough to support subgroup conclusions.	Low response rates, small cells, and imperfect crosswalks create fragile estimates exactly where the interpretation needs precision.
7 Claims data miss care quality	Whether utilization, billing codes, mortality, and text-derived outcomes capture true care quality, symptoms, and welfare.	Claims are built for payment, not physiology or patient experience, and many quality margins never enter billing records.

8 Self-reports and record links blur effects	How much treatment effects shift when outcomes depend on recall, survey reporting, fuzzy matches, or implementation intensity.	Measurement error can be differential, linkage false positives are hard to audit, and treatment delivery often varies within the label.
9 Rare health outcomes need larger panels	Whether estimated mortality and long-run health effects are real, heterogeneous, or artifacts of sparse events and unstable denominators.	Mortality and severe health outcomes are rare, so subgroup and dynamic estimates require very large linked panels.
10 Environmental estimates hinge on exposure maps	How pollution, climate, emissions, and valuation estimates change when exposure models, chemistry, and welfare components improve.	Exposure is spatially modeled, often without complete chemistry or defensive behavior, and welfare margins are only partly measured.
11 Earnings records do not fully link or measure pay	Whether earnings, income, employer, education, and administrative records identify mechanisms rather than only headline outcomes.	Caps, missing identifiers, imperfect links, and absent pay-rate or hours details leave key margins unobserved.
12 Selection remains when there is no shock	Whether descriptive associations reflect causal effects once motivation, selection, sorting, and omitted variables are addressed.	Without randomized, quasi-random, or clearly exogenous variation, richer controls cannot rule out unobserved selection.
13 Financial institutions do not travel cleanly	Whether financial-market, banking, investor, and institutional results persist in different regulatory and market environments.	Market structure, supervision, balance-sheet rules, and institutional investor composition vary across settings.
14 Heterogeneity claims outrun statistical power	Which subgroup, treatment-effect, or distributional differences are real rather than noisy splits of a credible average effect.	Once samples are split by subgroup, location, time, or treatment intensity, standard errors often dominate the signal.
15 Causal claims need better shocks	Which estimates would survive a stronger policy shock, natural experiment, instrument, or randomized design.	The observed exposure is often chosen, anticipated, or jointly determined with outcomes, weakening identification.

Table 59: Subpart composition within numbered limitation clusters.

# Route / family	Constraint shares	Extraction-family shares
1 Measurement/proxy / Measurement unstructured text	Measurement 83%; Data availability 4%; Mechanism 4%; Identification 2%; Other 6%	Measurement unstructured text 100%
2 External validity / External validity single context	External validity 96%; Data availability 3%; Identification 1%	External validity single context 100%
3 External validity / External validity single context	External validity 94%; Data availability 1%; Measurement 1%; Scope 1%; Other 1%	External validity single context 100%
4 External validity / External validity single context	External validity 97%; Data availability 2%; Implementation 2%	External validity single context 100%
5 External validity / External validity single context	External validity 89%; Data availability 6%; Theory 3%; Scope 2%	External validity single context 100%
6 Measurement/proxy / Data scale sample size	Data availability 56%; Statistical power 25%; Measurement 14%; Implementation 3%; Other 2%	Data scale sample size 100%
7 Measurement/proxy / Measurement unstructured text	Measurement 84%; Identification 6%; Mechanism 5%; Modeling 2%; Other 3%	Measurement unstructured text 100%
8 Measurement/proxy / Measurement unstructured text	Measurement 79%; Mechanism 6%; Data availability 5%; Identification 5%; Other 5%	Measurement unstructured text 100%
9 Sample scale/coverage / Data scale sample size	Statistical power 57%; Data availability 35%; Measurement 5%; External validity 2%; Other 2%	Data scale sample size 100%
10 Measurement/proxy / Measurement unstructured text	Measurement 77%; Data availability 8%; Scope 6%; Identification 3%; Other 5%	Measurement unstructured text 100%

11	Causal design / Data access administrative	Data availability 67%; Measurement 20%; Data access administrative 100% Identification 5%; External validity 3%; Other 5%	
12	Descriptive evidence only / Identification no variation	Identification 95%; Mechanism 3%; Statistical power 2%	Identification no variation 100%
13	External validity / External validity single context	External validity 93%; Theory 4%; Data availability 2%; Institutional 2%	External validity single context 100%
14	Sample scale/coverage / Data scale sample size	Statistical power 72%; Data availability 24%; External validity 2%; Measurement 2%	Data scale sample size 100%
15	Causal design / Identification no variation	Identification 93%; Theory 7%	Identification no variation 100%

Frontier open-question clusters in 2025–2026 NBER papers

Largest 15 researcher-facing groups selected from 85 assigned clusters; grey dots are other extracted open questions in the same map window.

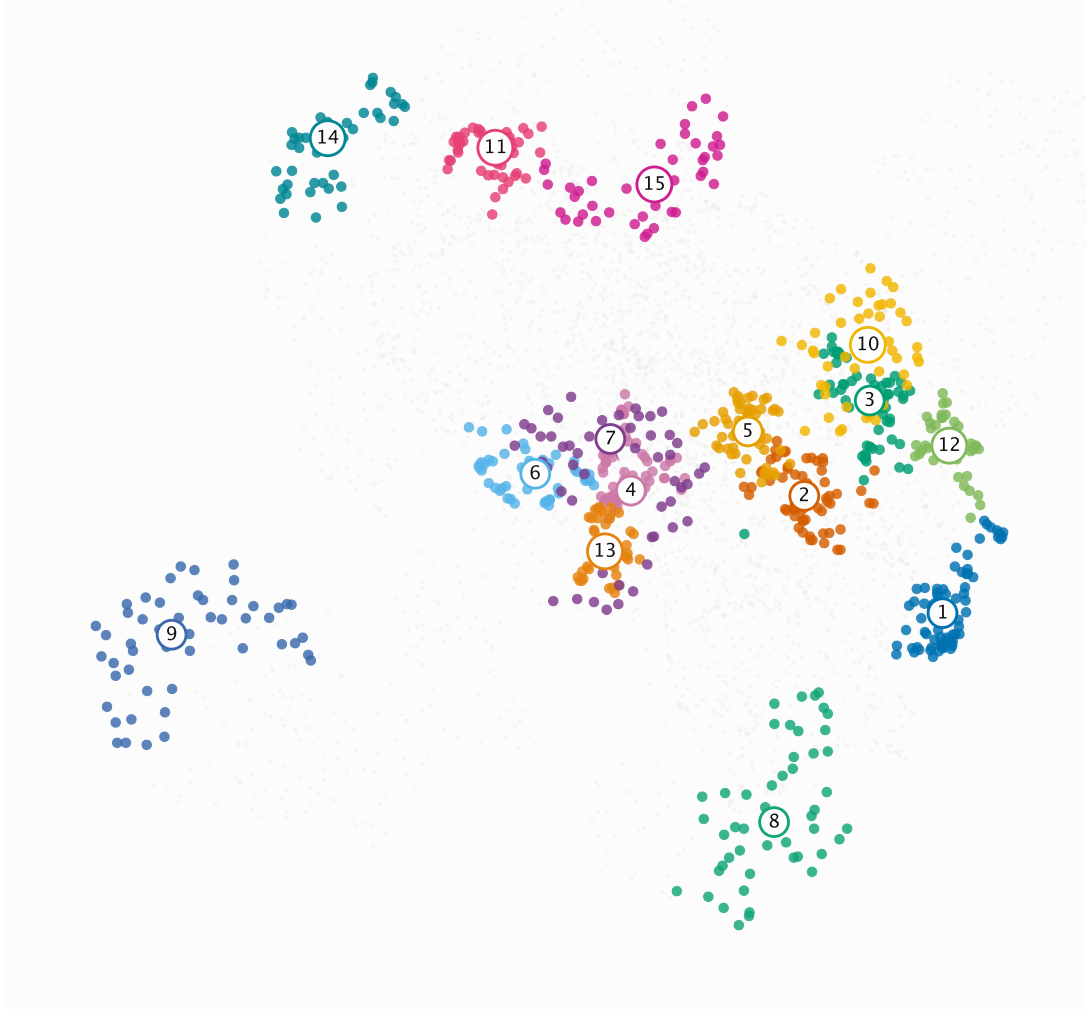


Figure 7: Embedding map of the 15 largest open-question frontier-signal clusters in the 2025–2026 NBER extraction. The clustering input contains 4,243 extracted open-question records from 1,381 papers; 3,632 records had embeddings, 2,574 records were assigned to 85 HDBSCAN clusters, and this figure highlights the largest 15 clusters containing 776 assigned records. Records were routed by extracted frontier family, embedded with OpenAI `text-embedding-3-large` embeddings, projected with UMAP for display, and clustered with HDBSCAN using the leaf-style frontier-clustering run. Codex (GPT-5.5, extra-high reasoning) reviewed all input text for each displayed cluster before assigning the researcher-facing labels. Colored points show the highlighted largest-15 clusters; grey points are other extracted open questions in the same embedding window. Numbers correspond to Tables 53 and 55. The embedding is used as a qualitative map of local similarity, not as a metric space.

Binding limitation clusters in 2025–2026 NBER papers

Largest 15 researcher-facing groups selected from 129 assigned clusters; grey dots are other extracted limitations in the same map window.



Figure 8: Embedding map of the 15 largest limitation frontier-signal clusters in the 2025–2026 NBER extraction. The clustering input contains 6,196 extracted limitation records from 1,381 papers; 5,575 records had embeddings, 3,991 records were assigned to 129 HDBSCAN clusters, and this figure highlights the largest 15 clusters containing 962 assigned records. Records were routed by extracted constraint family, embedded with OpenAI `text-embedding-3-large` embeddings, projected with UMAP for display, and clustered with HDBSCAN using the leaf-style frontier-clustering run. Codex (GPT-5.5, extra-high reasoning) reviewed all input text for each displayed cluster before assigning the researcher-facing labels. Colored points show the highlighted largest-15 clusters; grey points are other extracted limitations in the same embedding window. Numbers correspond to Tables 57 and 59. The map is a researcher-facing index to the tables, not the source of the substantive interpretation.

M Shock and Natural-Experiment Design Clusters

This appendix applies the same cluster-display logic to extracted causal method records whose design objects include shocks, instruments, randomized offers, policy changes, thresholds, rollouts, and related sources of identifying variation. The input is not a topic model of titles or abstracts: it is method-level shock and design text extracted from full working papers.

The clustering input contains 1,651 causal method-level shock candidates. Of these, 1,516 candidates from 546 papers were embedded and matched to the chunk index; 1,339 candidates from 511 papers were assigned to 18 HDBSCAN clusters. The main text displays the 15 largest researcher-facing clusters by paper count in Figure 6; those clusters contain 1,054 assigned candidates from 432 papers. Candidates were routed by extracted shock family, embedded with OpenAI `text-embedding-3-large` embeddings, projected with UMAP for display using cosine distance, $n_{\text{neighbors}} \leq 24$, and `min_dist`=0.12, and clustered within route with HDBSCAN using `min_cluster_size`=18, `min_samples`=6, and EOM cluster selection. Codex (GPT-5.5, extra-high reasoning) reviewed all input text for each displayed cluster before assigning the researcher-facing labels; the labels name the substantive source of variation rather than the estimator.

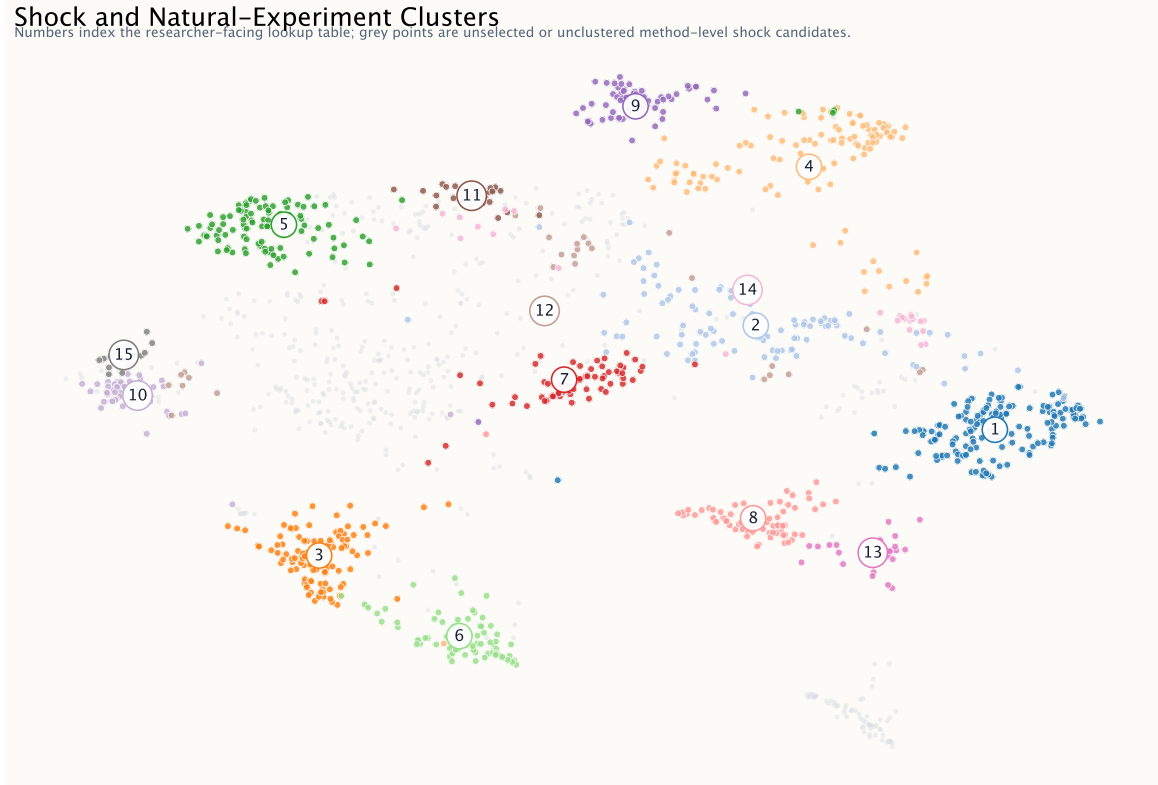


Figure 9: Embedding map of the 15 largest shock and natural-experiment design clusters in the 2025–2026 NBER extraction, selected from 18 HDBSCAN clusters. The clustering input contains 1,651 causal method-level shock candidates; 1,516 candidates from 546 papers were embedded and matched to the chunk index, 1,339 candidates from 511 papers were assigned to clusters, and the displayed largest-15 set contains 1,054 assigned candidates from 432 papers. Candidates were routed by extracted shock family, embedded with OpenAI `text-embedding-3-large` embeddings, projected with UMAP for display using cosine distance, $n_{\text{neighbors}} \leq 24$, and `min_dist`=0.12, and clustered within route with HDBSCAN using `min_cluster_size`=18, `min_samples`=6, and EOM cluster selection. Codex (GPT-5.5, extra-high reasoning) reviewed all input text for each displayed cluster before assigning the researcher-facing labels. Colored points show the highlighted largest-15 clusters; grey points show other embedded candidates in the same window. The embedding is a qualitative map of local similarity, not a metric space.

Table 61: Representative shocks and design objects within each cluster

Cluster	Representative shocks or design objects	Researcher-checkable anchors
1. Information Frictions and Belief Updating	a. Q&A health text messages versus direct statements and no-message controls	a. w33294 ; pp. 8–11; Figure 1
	b. college-application information modules assigned to students	b. w34164 ; pp. 20–21; Figure 1
	c. behavioral parenting teleassistance assigned by child age and mother education strata	c. w33338 ; pp. 11–12
2. Access to Programs, Services, and Expertise	a. community targeting versus proxy-means testing in Indonesian cash-transfer rollout	a. w33322 ; pp. 6–7; Table 4
	b. immediate jail health-care accreditation subsidy versus delayed accreditation	b. w33357 ; p. 11; Appendix Table A16
	c. digital surveillance and managerial-clarity treatments assigned to online workers	c. w33348 ; pp. 8–9; Figure 1
3. Prices, Credit Conditions, and Market Rules	a. war, pandemic, and defense-spending events as shocks to government bond returns DATA: Heriot-Watt British Government Securities Database...	a. w34820 ; pp. 11–12; Equation 1
	b. mortgage deal-expiry interest-rate resets as household cash-flow shocks DATA: Money Dashboard transaction dataset	b. w34911 ; pp. 19–20; Figure C.7
	c. state e-cigarette tax changes and stacked state-event windows	c. w33326 ; pp. 17–20; Appendix Figure 6
4. Eligibility Margins and Institutional Boundaries	a. September 1957 school-leaving-age birthdate cutoff	a. w33430 ; pp. 7–9; Table 1
	b. close-election victory margins around the zero threshold	b. w34968 ; pp. 18–19; Figure 4
	c. age-62 Social Security eligibility discontinuity inside multigenerational households	c. w33381 ; pp. 11–13
5. Job Loss, Family Transitions, and Firm Restructuring	a. worker displacement events matched to synthetic-control donor histories	a. w33667 ; pp. 5–7; Figure 1
	b. firm exposure to 2008 lender credit-supply shocks, including Lehman and ABX exposure	b. w33340 ; pp. 18–20
	c. post-acquisition changes in firms acquired by high- and low-management-quality acquirers DATA: U.S. Census Longitudinal Business Database	c. w33765 ; pp. 8–11; Figure A4
6. Global Trade, Monetary, and Credit Exposure	a. narrative U.S. tariff shocks used as external instruments in an SVAR-IV	a. w34852 ; pp. 20–23; Figure 3
	b. commuting-zone exposure to Chinese import penetration using other-country Chinese exports	b. w33424 ; pp. 24–26; Figure 4
	c. regional China-shock exposure interacted with local industry shares	c. w33755 ; p. 30; Equation 10
7. Public-Service Rollouts and Social Supports	a. staggered Drug-Free Communities grant adoption across coalitions	a. w34182 ; pp. 11–14; Figure 4
	b. universal pre-K implementation timing across states and large cities	b. w33767 ; p. 8; Figure 1

Cluster	Representative shocks or design objects	Researcher-checkable anchors
8. News, Disclosures, and Public Announcements	c. early versus late Early Intervention referral among eligible and ineligible children	c. w33341 ; pp. 15–16; Figure 1.2
	a. stock-market reaction around executive gender-quota announcements DATA: BoardEx executive and leadership histories database	a. w35030 ; pp. 34–39; Figure 1
	b. Ex parte Lundgren patent decision changing disclosure incentives for business-method patents	b. w33783 ; pp. 20–23; Figure A1
	c. Larry Fink letter and BlackRock ownership exposure as corporate-speech shock DATA: Revelio Labs resume data linked to L2 voter registr...	c. w33810 ; p. 26; Table 6
9. Place-Based Exposure and Border Discontinuities	a. EPA nonattainment designations as county-level pollution-regulation exposure	a. w33412 ; pp. 9–10; Table 2
	b. local bank suspensions and lender-of-last-resort access across counties	b. w34988 ; pp. 13–14; Figure 7
	c. end of UK sugar rationing by birth cohort and early-life exposure	c. w35005 ; p. 4; Figure A2
10. Government Rule Changes and Regulatory Mandates	a. health-care administrative-boundary reforms creating fragmentation exposure	a. w33863 ; pp. 15–17; Figure 2A
	b. state reductions in cosmetology licensing-hour requirements	b. w33936 ; pp. 12–15; Figure 1
	c. state minimum-wage law changes under staggered adoption	c. w33433 ; pp. 17–18; Figure 4
11. Predicted Local Exposure to Aggregate Shifts	a. immigrant enclave shift-share exposure from 1980 origin-country shares	a. w33961 ; pp. 13–19; Figure 1
	b. nearby-county online-dating adoption used as a doughnut instrument	b. w34757 ; pp. 23–28; Figure A.6
	c. origin-country unpaid-work norms interacted with local marriage-market income ratios	c. w33393 ; pp. 37–41; Figure 11
12. Administrative Access and Take-Up Frictions	a. CAHOOTS response availability from call busyness at city-hour timing	a. w33761 ; pp. 15–16; Figure 3
	b. therapy-offer assignment used to instrument therapy take-up among adolescents	b. w33677 ; pp. 8–9; Table 3
	c. weekly revolving-door lawyer caseload as availability instrument	c. w33708 ; p. 16; Table 2
13. Policy Surprises and Macro News Shocks	a. CPI release dates and forecast surprises as inflation-news instruments	a. w34088 ; pp. 19–21; Figure 3
	b. high-frequency carbon-futures surprises around regulatory events	b. w34125 ; pp. 9–12; Figure 5
	c. military-spending news shocks and tax-seasonality instruments for Treasury supply	c. w34154 ; pp. 20–22; Table 6
14. Cash, Price, and Incentive Offers	a. village and sublocation randomization of large-scale Kenyan cash transfers	a. w34152 ; pp. 10–11; Figure 1
	b. random payments to reduce Facebook or Instagram use	b. w34743 ; p. 15; Figure 1
	c. high versus low monthly unconditional cash gifts assigned at recruitment	c. w33844 ; pp. 8–9; Figure 1

Cluster	Representative shocks or design objects	Researcher-checkable anchors
15. Wages, Bargaining, and Firm-Rule Changes	a. firm-level exposure to required payroll adjustment under a new minimum wage	a. w34188 ; pp. 26–27 ; Figure B.1
	b. retail firms seceding from centralized collective bargaining agreements DATA: CNEL web archive CBA codes and wage-floor schedules	b. w34076 ; pp. 8–9 ; Figure 4
	c. German minimum-wage reform intensity as exposure to wage-floor pressure	c. w34699 ; pp. 5–8 ; Equation 14

Notes: Rows list manually reviewed examples from within each cluster, chosen to show the substantive source of variation rather than the estimator name. The a/b/c examples are split into separate rows so each design object aligns with its paper and artifact anchor. The anchor column gives visible paper numbers, PDF page anchors, and compact figure/table/equation references; data-source links appear in the shock column only when the extracted resolver returned a resolved or retained public source URL.

Table 63: Researcher-checkable examples for each shock and natural-experiment cluster

Cluster	Example variation	Paper, page, and artifact provenance	Linked data source
1a. Information Frictions and Belief Updating	Randomized controlled trial (text message experiments, Ghana and Michigan). Two large preregistered field RCTs randomly assigned participants to Q&A-style vs. direct-statement text messaging (and a no...	w33294 (2025): Does Q&A Boost Engagement? Health Messaging Experiments in the U.S. and Ghana. S6 2.1 Methods, section para. 1; pp. 8–11 . Figure: Figure 1 , p. 46.	Ghana IVR recruitment and SMS intervention sample; no public URL resolved
1b. Information Frictions and Belief Updating	Randomized field experiment (four-arm). Students were randomized into four treatment arms (combinations of information modules) and outcomes compared to estimate ef...	w34164 (2025): College Application Mistakes and the Design of Information Policies at Scale. S15 4 Randomized Information Intervention, section para. 1; pp. 20–21 . Figure: Figure 1 , p. 4.	Administrative panel of the centralized admissions system (MINE-DUC/DEMRE); no public URL resolved
1c. Information Frictions and Belief Updating	Randomized Controlled Trial (stratified randomization, ITT). Families were randomized (stratified by child age and mother's education) into treatment and control groups; analysis follow...	w33338 (2025): Calling All Parents: Leveraging Behavioral Insights to Boost Early Childhood Outc.... S11 a) Sample and randomization, section para. 1; pp. 11–12 . No figure/table link extracted.	Randomized trial dataset collected for the Uruguay Crece Contigo teleassistance program; no public URL resolved
2a. Access to Programs, Services, and Expertise	Randomized Controlled Trial (Community vs PMT). Random assignment of villages to community-based targeting or PMT in the high-stakes PKH scale-up (and pilot RCT) to identif...	w33322 (2025): Community Targeting at Scale. S6 B. High Stakes Experimental Scale-up, section para. 1; pp. 6–7 . Table: Table 4: Program Recipient Log PCE and Community Rank Percentile by PMT vs Community Targeting , p. 23.	SUSENAS (National Socioeconomic Survey); no public URL resolved
2b. Access to Programs, Services, and Expertise	Randomized Controlled Trial (intent-to-treat). Facilities were randomized into treatment (immediate accreditation subsidy) or control (accreditation later) arms and ITT ef...	w33357 (2025): The Hidden Health Care Crisis Behind Bars: A Randomized Trial to Accredite U.S. Ja.... S8 IV Intervention and Experimental Design, section para. 1; p. 11 . Table: Appendix Table A16: Balance on Baseline Jail Characteristics for Random Sample of Released Inmates , p. 84.	Facility survey (Qualtrics) and uploaded facility documents; no public URL resolved
2c. Access to Programs, Services, and Expertise	Randomized Controlled Trial (RCT) with factorial messaging and monitoring assignment. An online field RCT that randomly assigns workers (stratified by baseline productivity) to surveillance vs. no-surveillance...	w33348 (2025): The Effects of Digital Surveillance and Managerial Clarity on Performance. S5 2.2 Experimental Design, section para. 1; pp. 8–9 . Figure: Figure 1 , p. 19.	UpWork randomized controlled trial data collected by the authors; no public URL resolved
3a. Prices, Credit Conditions, and Market Rules	Event-study regression on wars and crises. Estimates dynamic responses of outcomes (cumulative real/nominal bond returns, inflation, return differentials, r-g) to the...	w34820 (2026): Are Government Bonds Safe in Times of War and Pandemic?. S6 2.3 Wars and Defense Spending, section para. 1; pp. 11–12 . Equation: Equation 1 , p. 15.	Heriot-Watt British Government Securities Database (total return index)
3b. Prices, Credit Conditions, and Market Rules	Event-window baseline regression (deal expiry event study). Regress cumulative changes in household consumption and cash-on-hand around mortgage deal expiry on the change in aggregate...	w34911 (2026): How Do Interest Rates Affect Consumption? Household Debt and the Role of Asset Pr.... S12 4.5 Natural Experiments and the Household Debt Channel, section para. 1; pp. 19–20 . Figure: Figure C.7 , p. 100.	Money Dashboard transaction dataset

Cluster	Example variation	Paper, page, and artifact provenance	Linked data source
3c. Prices, Credit Conditions, and Market Rules	Stacked logistic pooled regression with stack-by-state and stack-by-year fixed effects. Pooled logistic regressions on stacked datasets including stack-by-state and stack-by-year fixed effects to estimate treatme...	w33326 (2025): E-Cigarette Taxation and Queer Youth. S13 4. Empirical Strategy, section para. 1; pp. 17–20 . Figure: Appendix Figure 6 , p. 72.	State Youth Risk Behavior Surveillance System (State YRBS); no public URL resolved
4a. Eligibility Margins and Institutional Boundaries	Regression Discontinuity Design (date-of-birth cutoff). Uses the September 1, 1957 birthdate cutoff from the 1972 ROSLA policy as the running variable to identify local causal effe...	w33430 (2025): Education and Dementia Risk. S6 3. The Natural Experiment, section para. 1; pp. 7–9 . Table: Table 1 , p. 27.	UK Biobank; no public URL resolved
4b. Eligibility Margins and Institutional Boundaries	Close-elections Regression Discontinuity Design (RD). Compares marginal winners and marginal runners-up using the margin of victory as the running variable to estimate the causal...	w34968 (2026): The Global Incumbency Advantage. S11 5.1 Testing the identification assumption, section para. 1; pp. 18–19 . Figure: Figure 4 , p. 18.	National Elections Database; no public URL resolved
4c. Eligibility Margins and Institutional Boundaries	Piecewise linear regression (age discontinuity). Estimates a discontinuous jump in outcomes when the oldest household member crosses the early Social Security eligibility th...	w33381 (2025): Does Old Age Social Security Help Children? The Impact of Social Security on Gran.... S7 IV. Methodology, section para. 1; pp. 11–13 . No figure/table link extracted.	2000 United States Census (decennial) and American Community Survey (ACS) 2001-2019 (person an...; no public URL resolved
5a. Job Loss, Family Transitions, and Firm Restructuring	Individual-level hybrid synthetic control (matching + synthetic control). For each displaced worker the authors first restrict a donor pool by coarse exact matching (gender, education, 1-digit indus...	w33667 (2025): Adjusters and Casualties: The Anatomy of Labor Market Displacement. S5 2 Empirical strategy, section para. 1; pp. 5–7 . Figure: Figure 1 , p. 10.	Integrated Employment Biographies; no public URL resolved
5b. Job Loss, Family Transitions, and Firm Restructuring	Credit-supply exposure difference-in-differences (Chodorow-Reich instruments). Uses firm-level exposure to lenders' 2008 crisis shocks (percent loan reduction, Lehman exposure, ABX exposure) interacting...	w33340 (2025): Corporate Resiliency and the Choice Between Financial and Operational Hedging. S13 3. Empirical analysis, section para. 1; pp. 18–20 . No figure/table link extracted.	Compustat quarterly; no public URL resolved
5c. Job Loss, Family Transitions, and Firm Restructuring	Panel regressions / OLS with fixed effects and controls (event-study specifications). Event-study and panel regression specifications estimate post-merger changes in outcomes (revenues, employment, productivity...	w33765 (2025): Management and Firm Dynamism. S7 3.2 Transferring management practices across plants within firms, section para. 1; pp. 8–11 . Figure: Figure A4 , p. 32.	U.S. Census Longitudinal Business Database
6a. Global Trade, Monetary, and Credit Exposure	Two-stage least squares (2SLS) applied to reduced-form residuals. Implements 2SLS within the SVAR-IV estimator: first-stage links instrument to reduced-form tariff innovations, second-stage...	w34852 (2026): The Macroeconomic Effects of Tariffs: Evidence From U.S. Historical Data. S8 4.1. Framework, section para. 1; pp. 20–23 . Figure: Figure 3 , p. 24.	Narrative tariff shock series (authors' constructed series of 35 events, 1840-2024); no public URL resolved

Cluster	Example variation	Paper, page, and artifact provenance	Linked data source
6b. Global Trade, Monetary, and Credit Exposure	Shift-share instrumental variables (2SLS) for CZ-level import exposure. Instruments commuting-zone (CZ) exposure to Chinese import penetration (ΔIP_i) using analogous industry-level changes in Chi...	w33424 (2025): Places versus People: The Ins and Outs of Labor Market Adjustment to Globalization. S16 3.5 Causal identification, section para. 1; pp. 24–26 . Figure: Figure 4 , p. 31.	Longitudinal Employer-Household Dynamics (LEHD) Employment History File (EHF) and Employer Cha...; no public URL resolved
6c. Global Trade, Monetary, and Credit Exposure	Import competition (China shock) shift-share IV. Cross-sectional shift-share instrument using increases in Chinese import penetration by industry interacted with regional in...	w33755 (2025): Should I Stay or Should I Go? The Response of Labor Migration to Economic Shocks. S22 3.2 A lternative instruments for labor demand shocks, section para. 1; p. 30 . Equation: Equation 10 , p. 36.	Defense spending prime contract data (usaspending.gov) and Nakamura & Steinsson (2014) state m...; no public URL resolved
7a. Public-Service Rollouts and Social Supports	Callaway and Sant’Anna staggered DID (CSDID). Implements the Callaway & Sant’Anna (2021) DID estimator to estimate cohort-specific and aggregated ATT for staggered adopti...	w34182 (2025): Local Problems, Local Solutions: Evidence from the Drug-Free Communities Program.... S7 4. Identification Strategy, section para. 1; pp. 11–14 . Figure: Figure 4 , p. 15.	SAMHSA Grant Award Archive and CDC Drug-Free Communities coalition directory (combined); no public URL resolved
7b. Public-Service Rollouts and Social Supports	Event-study plots / dynamic effects. Maps the evolution of outcomes around UPK implementation using event-study coefficients to assess pre-trends and dynamic tre...	w33767 (2025): Universal Pre-K as Economic Stimulus: Evidence from Nine States and Large Cities.... S9 4 Effect on Pre-K Enrollment, section para. 1; p. 8 . Figure: Figure 1 , p. 18.	American Community Survey; no public URL resolved
7c. Public-Service Rollouts and Social Supports	Difference-in-Differences (DiD) comparing early vs late referrals across eligible and ineligible. Estimates the effect of early (first-year) EI receipt by interacting eligibility with an Early referral indicator and subtra...	w33341 (2025): Can Early Intervention Reduce Future Child Maltreatment?. S17 2. Difference-in-Differences, section para. 1; pp. 15–16 . Figure: Figure 1.2 , p. 29.	Rhode Island Early Intervention referral and service records (linked administrative dataset); no public URL resolved
8a. News, Disclosures, and Public Announcements	Event study (stock announcement returns, portfolio-level). Compares excess portfolio returns around the announcement window for targeted versus control portfolios to detect market rea...	w35030 (2026): A Gender Quota for Top Executives: Diversity without Disruption. S13 VI. Did the Quota Affect Firm Performance?, section para. 1; pp. 34–39 . Figure: Figure 1 , p. 46.	BoardEx executive and leadership histories database
8b. News, Disclosures, and Public Announcements	Difference-in-Differences (DiD): Lundgren $\times$ BM&S patents. DiD exploiting the Ex parte Lundgren administrative decision to compare recording lags for transactions involving business-m...	w33783 (2025): Acquiring Patents in Secret: Disclosure Timing in Markets for Technology. S15 4.3. Empirical Methods, section para. 1; pp. 20–23 . Figure: Figure A1 , p. 11.	USPTO Patent Assignment Dataset; no public URL resolved
8c. News, Disclosures, and Public Announcements	Difference-in-differences (DiD) analysis for BlackRock ownership / Fink letter. Use a DiD design to document increases in Democratic language by firms with high BlackRock ownership after Larry Fink’s 2019...	w33810 (2025): Partisan Corporate Speech. S23 6 Potential Drivers of the Rise in Democratic-leaning Corporate Speech, section para. 1; p. 26 . Table: Table 6 , p. 56.	Revelio Labs resume data linked to L2 voter registration data

Cluster	Example variation	Paper, page, and artifact provenance	Linked data source
9a. Place-Based Exposure and Border Discontinuities	Event study (dynamic DiD / leads-lags). Event-study specifications estimate dynamic treatment effects around designation years to assess pre-trends and timing of re...	w33412 (2025): Geographic Resolution in Environmental Policy: EPA's Shift from Regions to Counti.... S9 3.1 The Impact of Nonattainment Status on Ambient TSP: Methods, section para. 1; pp. 9–10 . Table: Table 2: Number of Counties with Balanced Monitors by TSP Attainment Status , p. 25.	County Business Patterns; no public URL resolved
9b. Place-Based Exposure and Border Discontinuities	Event-study / year-by-year treatment effects. Estimates year-specific treatment coefficients (γ_i, ω_i) relative to a base year to trace dynamics of bank suspensions, dep...	w34988 (2026): Lender of Last Resort and Local Economic Outcomes. S17 5.1 Empirical Design, section para. 1; pp. 13–14 . Figure: Figure 7 , p. 23.	Census of Manufactures (county-level, 1927-1935) and 1927 Market Data Handbook tabulations; no public URL resolved
9c. Place-Based Exposure and Border Discontinuities	Natural experiment using rationing cessation (cohort discontinuity/event-study). Exploits the September 1953 end of UK sugar rationing as exogenous variation in early-life sugar exposure by conception/birt...	w35005 (2026): Early-Life Sugar Restrictions Reduce Genetic Disparities in Adult Adiposity. S6 b. Study design and exposure to early-life sugar restrictions, section para. 1; p. 4 . Figure: Figure A2 , p. 31.	UK Biobank; no public URL resolved
10a. Government Rule Changes and Regulatory Mandates	Event-study / dynamic DiD (estimates of β_e). Estimates dynamic treatment effects over event time around reform implementation to assess pre-trends and the timing of impa...	w33863 (2025): Administrative Fragmentation in Health Care. S7 4 Empirical Strategy, section para. 1; pp. 15–17 . Figure: Figure 2A , p. 11.	Medicare claims data (MedPAR, Outpatient, Carrier, Home Health, Hospice, DME); no public URL resolved
10b. Government Rule Changes and Regulatory Mandates	Staggered event-study difference-in-differences (Callaway & Sant'Anna approach). Implements a dynamic DiD/event-study for staggered policy timing to estimate year-by-year ATT effects of state-level reducti...	w33936 (2025): Cosmetology Gets a Trim: The Impact of Reducing Licensing Hours on Colleges and S.... S5 4. Methods, section para. 1; pp. 12–15 . Figure: Figure 1 , p. 14.	Integrated Postsecondary Education Data System; no public URL resolved
10c. Government Rule Changes and Regulatory Mandates	Callaway and Sant'Anna (C&S) doubly robust DID estimator. Augmented inverse probability weighting estimator that accounts for staggered adoption and heterogeneous dynamic treatment e...	w33433 (2025): Minimum Wage Laws and Job Search. S12 4.3 Pooled Analysis, section para. 1; pp. 17–18 . Figure: Figure 4 , p. 24.	Monthly state-level job seeker clicks from a large U.S. online job search platform; no public URL resolved
11a. Predicted Local Exposure to Aggregate Shifts	Shift-share (enclave) instrumental variables. Instruments commuting-zone immigrant inflow rates (1980–1990) using 1980 enclave shares by country of origin multiplied by n...	w33961 (2025): Immigration and Inequality in the Next Generation. S10 4 Research Design and Econometric Specification, section para. 1; pp. 13–19 . Figure: Figure 1 , p. 13.	Decennial Census microdata (IPUMS harmonized public-use samples for 1940, 1970, 1980, 1990, 20...; no public URL resolved
11b. Predicted Local Exposure to Aggregate Shifts	Instrumental variables 2SLS (nearby-county doughnut instruments). Two-stage least squares using mean dating sessions in nearby (20-100km) counties and nearby-county attributes as instruments...	w34757 (2026): What Happens When Dating Goes Online? Evidence from U.S. Marriage Markets and Hea.... S13 4.3 Empirical Specification, section para. 1; pp. 23–28 . Figure: Figure A.6 , p. 71.	Comscore Desktop Browsing Panel; no public URL resolved

Cluster	Example variation	Paper, page, and artifact provenance	Linked data source
11c. Predicted Local Exposure to Aggregate Shifts	Cross-country cultural instrument (origin-country home production ratio) interaction regressions. Using OECD GID-DB country-level female-to-male unpaid work ratios for immigrants' countries of origin interacted with local...	w33393 (2025): Winning the Bread and Baking it Too: Gendered Frictions in the Allocation of Home.... S18 5.2 Empirical Evidence on Household Formation, section para. 1; pp. 37–41 . Figure: Figure 11 , p. 38.	Origin-country home production ratio (cross-country cultural instrument); no public URL resolved
12a. Administrative Access and Take-Up Frictions	Availability instrument (busyness-based IV / 2SLS). Constructs a CAHOOTS availability (busyness) measure by counting active CAHOOTS calls residualized by city-month-day-of-week...	w33761 (2025): Mobile Crisis Response Teams Support Better Policing: Evidence from CAHOOTS. S11 4.1 Availability research design, section para. 1; pp. 15–16 . Figure: Figure 3 , p. 16.	Computer-Aided Dispatch (CAD) data from Eugene and Springfield police departments; no public URL resolved
12b. Administrative Access and Take-Up Frictions	Two-Stage Least Squares (2SLS) instrumenting therapy uptake with random assignment. 2SLS is used to estimate the effect of randomization-induced therapy uptake (complier LATE), using assignment to treatment a...	w33677 (2025): Talk Therapy and Human Capital in Adolescence: Evidence from Nepal. S6 3 Empirical Specification, section para. 1; pp. 8–9 . Table: Table 3: HETEROGENEITY - MENTAL HEALTH FLAG AT BASELINE , p. 25.	Center for Mental Health and Counseling-Nepal (CMC) randomized trial dataset (enrollment, rand...; no public URL resolved
12c. Administrative Access and Take-Up Frictions	Instrumental variables using weekly RD-caseload availability. 2SLS-style IV that instruments RD lawyer representation with the weekly caseload of RD lawyers in the firm at case arrival t...	w33708 (2025): The Law and Economics of Lawyers: Evidence from the Revolving Door in China's Jud.... S18 4.2 Alternative Strategies to Address Identification Threats, section para. 1; p. 16 . Table: Table 2 , p. 42.	China Judgment Online (Full set of court decisions); no public URL resolved
13a. Policy Surprises and Macro News Shocks	Instrument construction using CPI release dates and CPI surprises. Uses exogenous CPI release indicators and the magnitude/sign of CPI surprises (difference from Blue Chip forecasts) as instr...	w34088 (2025): The Causal Effect of News on Inflation Expectations. S14 4.1 LP-IV Framework, section para. 1; pp. 19–21 . Figure: Figure 3 , p. 10.	Stanford Cable TV News Analyzer (Internet Archive TV News Archive transcripts); no public URL resolved
13b. Policy Surprises and Macro News Shocks	High-frequency policy surprise instrument for carbon prices. Constructs carbon policy surprise series from high-frequency carbon futures price moves around regulatory events (following...	w34125 (2025): Carbon Pricing and Inequality: A Normative Perspective. S5 3. Data and Identification, section para. 1; pp. 9–12 . Figure: Figure 5 , p. 16.	Harmonized Index of Consumer Prices (HICP) by COICOP categories; no public URL resolved
13c. Policy Surprises and Macro News Shocks	Instrumental variables (military spending shocks and tax-seasonality). Two-stage IVs: (1) military expenditure news shocks scaled by GDP as instrument for Treasury growth (following Ramey and oth...	w34154 (2025): The Passthrough of Treasury Supply to Bank Deposit Funding. S20 5.1 Estimation Strategy, section para. 1; pp. 20–22 . Table: Table 6 , p. 44.	Military spending news shocks series (Ramey-style) and scaled military shock (Choi et al. cons...; no public URL resolved
14a. Cash, Price, and Incentive Offers	Two-level randomized controlled trial (village and sublocation randomization) with reduced-form OLS. Random assignment of treatment at the village level and saturation at the sublocation level used in reduced-form OLS specifi...	w34152 (2025): Can Cash Transfers Save Lives? Evidence from a Large-Scale Experiment in Kenya. S5 2.2 Experimental Design, section para. 1; pp. 10–11 . Figure: Figure 1 , p. 18.	KGES Endline 3 (EL3) Household Census; no public URL resolved

Cluster	Example variation	Paper, page, and artifact provenance	Linked data source
14b. Cash, Price, and Incentive Offers	Randomized pricing field experiment (incentive to reduce usage). A framed field RCT that randomly offers monetary payments to users to reduce Facebook or Instagram time and measures own- an...	w34743 (2026): Consumer Demand and Market Competition with Time-Intensive Goods. S10 3 Demand for PSNs: Experimental Evidence, section para. 1; p. 15 . Figure: Figure 1 , p. 21 .	RealityMine passive device monitoring data; no public URL resolved
14c. Cash, Price, and Incentive Offers	Randomized controlled trial (intent-to-treat). Participants were randomly assigned at recruitment to high- (\$333/mo) or low- (\$20/mo) unconditional cash gift groups and in...	w33844 (2025): The Effect of a Monthly Unconditional Cash Transfer on Children’s Development at.... S6 Study Procedures, section para. 1; pp. 8–9 . Figure: Figure 1 , p. 34 .	Baby’s First Years randomized controlled trial data (baseline, follow-up, and age-4 assessment...; no public URL resolved
15a. Wages, Bargaining, and Firm-Rule Changes	Firm-level GAP difference-in-differences (GAP DID / continuous treatment event-study). Uses a pre-reform firm-level GAP exposure (share of payroll needing adjustment to meet new NMW) as a continuous treatment in...	w34188 (2025): Minimum Wages and the Distribution of Firm Wage Premia. S17 7 The Causal Effect of the Minimum Wage, section para. 1; pp. 26–27 . Figure: Figure B.1 , p. 10 .	Uruguayan Social Security Administration employer-employee administrative records; no public URL resolved
15b. Wages, Bargaining, and Firm-Rule Changes	Matched difference-in-differences (FD retail secession). Matches treated incumbent workers in FD firms to control CC workers by propensity score and estimates DiD event coefficients...	w34076 (2025): Opting Out of Centralized Collective Bargaining: Evidence from Italy. S5 2 Institutional Background, section para. 1; pp. 8–9 . Figure: Figure 4 , p. 37 .	CNEL web archive CBA codes and wage-floor schedules
15c. Wages, Bargaining, and Firm-Rule Changes	Structurally-grounded difference-in-differences (DiD) with continuous treatment intensity. Estimates the effect of the 2015 German minimum wage on firm-level log markdown changes by regressing post-pre markdown chan...	w34699 (2026): Monopsony, Markdowns, and Minimum Wages. S4 2 The simple economics of our research design, section para. 1; pp. 5–8 . Equation: Equation 14 , p. 11 .	IAB Establishment Panel; no public URL resolved

Note: Each cluster is represented by up to three paper-level examples, selected first from manually reviewed examples and then from high-probability cluster members while preferring distinct papers. Paper links point to the NBER working-paper page. Page links point to the NBER PDF page anchor when the parsed section or artifact has a page. Paragraph references are parser-derived section paragraph anchors, not publisher paragraph numbers. The artifact column gives the linked figure, table, or equation used as the closest researcher-checkable visual or formal anchor. Data-source links are included only when the extracted data-source resolver returned a resolved or retained public source URL.

N Aggregate Dataset Types in the 2025–2026 Extraction-Batch Sample

This appendix section summarizes the full 2025–2026 data-source extraction layer, not only the top-10 named datasets reviewed above. The sample contains 1,391 NBER working papers: 1,018 papers from the 2025 extraction batch and 373 papers from the 2026 extraction batch. These are extraction-batch labels rather than cleaned publication-year counts. A paper can appear in more than one category when it combines administrative files, surveys, proprietary records, text corpora, experiments, or constructed benchmarks.

The field analysis joins each paper to its raw final JSON metadata. Primary JEL field is the first reported JEL code in the paper metadata. OpenAlex main field is the field attached to the first OpenAlex topic in the paper metadata. JEL metadata is available for 904 papers; OpenAlex main-field metadata is available for 930 papers. Missing metadata is excluded from the field-composition figures but retained in the all-type totals. The field plots use distinct-paper counts to order fields; JEL is reported first as the economics-native classification, and OpenAlex is reported second as a broader external field crosswalk.

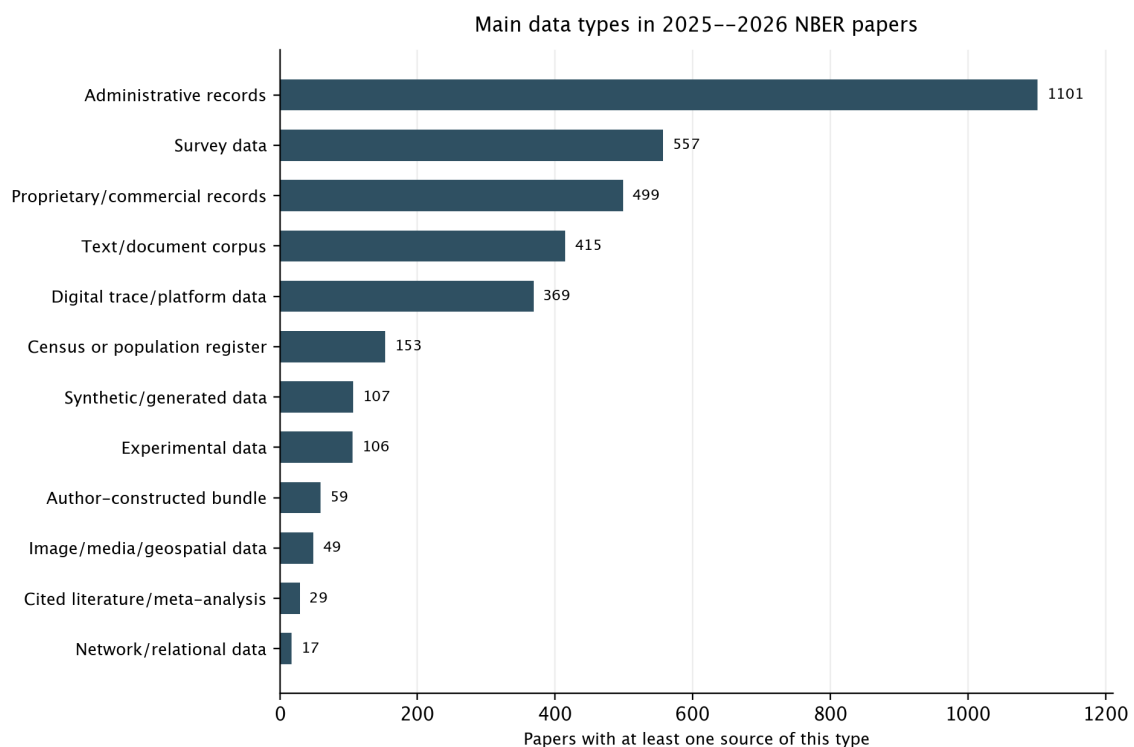


Figure 10: Main data types across the 2025–2026 extraction batches (1,391 papers: 1,018 from the 2025 batch and 373 from the 2026 batch). Bars count distinct papers with at least one extracted source of that type; papers can appear in more than one category.

Table 65: Main data types across the 2025–2026 paper sample

Main data type	Papers	Share of papers	Common roles
Administrative records	1,101	79.2	not specified; outcome
Survey data	557	40.0	outcome; not specified
Proprietary/commercial records	499	35.9	outcome; not specified
Text/document corpus	415	29.8	not specified; treatment/exposure
Digital trace/platform data	369	26.5	not specified; outcome
Census or population register	153	11.0	not specified; outcome
Synthetic/generated data	107	7.7	not specified; calibration
Experimental data	106	7.6	outcome; treatment/exposure
Author-constructed bundle	59	4.2	not specified; covariate/control
Image/media/geospatial data	49	3.5	covariate/control; not specified
Cited literature/meta-analysis	29	2.1	calibration; not specified
Network/relational data	17	1.2	outcome; imputation

Note: Shares are percentages of the 1,391-paper 2025–2026 extraction-batch sample. Shares sum to more than 100 percent because a paper can use multiple data types. Common roles describe how papers use sources of that type.

Table 66: Data structures used in the 2025–2026 paper sample

Data structure	Papers	Share of papers
Unbalanced panel	1,050	75.5
Cross-section	767	55.1
Time series	620	44.6
Repeated cross-section	318	22.9
Spatial data	128	9.2
Balanced panel	92	6.6
Hierarchical data	74	5.3
Document/text data	31	2.2

Note: This table is separate from dataset type. For example, administrative records can be panels, cross-sections, repeated cross-sections, or event/transaction data.

Table 67: Dataset types by primary JEL field, 2025–2026

Primary JEL field	Papers	Three largest data types within field
J: Labor/demography	160	Administrative records (48%); Survey data (21%); Text/document corpus (10%)
I: Health/education/welfare	130	Administrative records (52%); Survey data (23%); Proprietary/commercial records (7%)
G: Financial economics	85	Proprietary/commercial records (38%); Administrative records (34%); Digital trace/platform data (9%)
E: Macro/monetary	78	Administrative records (49%); Proprietary/commercial records (20%); Survey data (15%)
O: Development/innovation/growth	70	Administrative records (43%); Survey data (13%); Proprietary/commercial records (13%)
F: International economics	66	Administrative records (47%); Proprietary/commercial records (20%); Text/document corpus (11%)
D: Microeconomics	65	Administrative records (40%); Survey data (18%); Proprietary/commercial records (13%)
H: Public economics	43	Administrative records (56%); Survey data (14%); Proprietary/commercial records (10%)
L: Industrial organization	43	Administrative records (39%); Proprietary/commercial records (20%); Digital trace/platform data (13%)
C: Quantitative methods	38	Administrative records (37%); Proprietary/commercial records (16%); Survey data (13%)

Note: The table uses the first JEL code listed in the paper metadata as the primary field. Percentages in the final column are within-field shares of extracted data sources, not shares of papers.

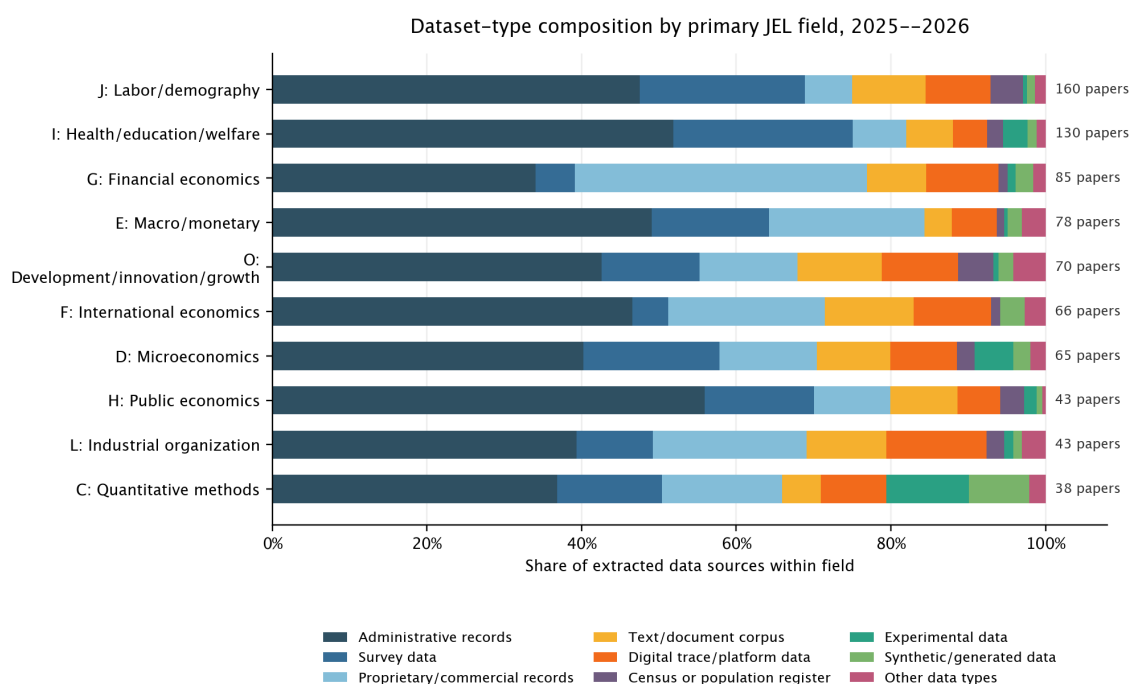


Figure 11: Composition of extracted data sources by primary JEL field, 2025–2026 (904 papers with JEL metadata; largest paper counts: J Labor/demography 160, I Health/education/welfare 130, G Financial economics 85, E Macro/monetary 78, O Development/innovation/growth 70, F International economics 66, D Microeconomics 65, H Public economics 43, L Industrial organization 43, C Quantitative methods 38). Each bar sums to 100 percent within the JEL field; right-side labels report distinct papers in that field.

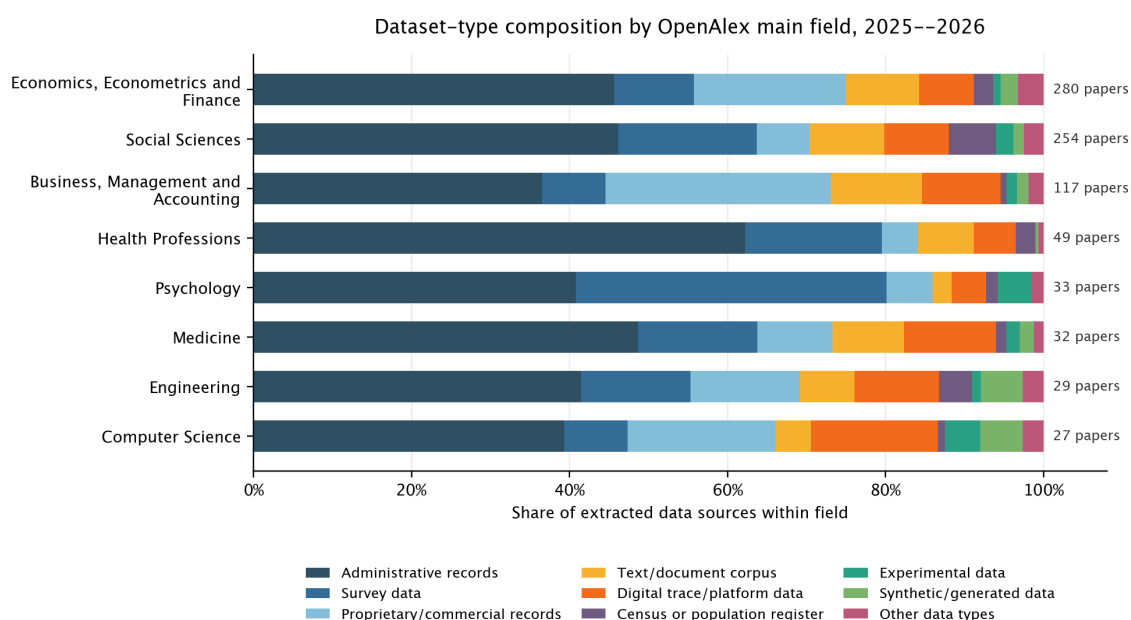


Figure 12: Composition of extracted data sources by OpenAlex main field, 2025–2026 (930 papers with OpenAlex metadata; largest paper counts: Economics/Econometrics/Finance 280, Social Sciences 254, Business/Management/Accounting 117, Health Professions 49, Psychology 33, Medicine 32, Engineering 29, Computer Science 27). Each bar sums to 100 percent within the OpenAlex field; right-side labels report distinct papers in that field.

Table 68: Dataset types by OpenAlex main field, 2025–2026

OpenAlex main field	Papers	Three largest data types within field	Largest OpenAlex subfields
Economics, Econometrics and Finance	280	Administrative records (46%); Proprietary/commercial records (19%); Survey data (10%)	Economics and Econometrics; Finance; General Economics, Econometrics and Finance
Social Sciences	254	Administrative records (46%); Survey data (18%); Text/document corpus (9%)	Sociology and Political Science; Education; Political Science and International Relations
Business, Management and Accounting	117	Administrative records (37%); Proprietary/commercial records (28%); Text/document corpus (12%)	Accounting; Strategy and Management; Management of Technology and Innovation
Health Professions	49	Administrative records (62%); Survey data (17%); Text/document corpus (7%)	General Health Professions; Pharmacy; Speech and Hearing
Psychology	33	Administrative records (41%); Survey data (39%); Proprietary/commercial records (6%)	Social Psychology; Clinical Psychology; Experimental and Cognitive Psychology
Medicine	32	Administrative records (49%); Survey data (15%); Digital trace/platform data (12%)	Public Health, Environmental and Occupational Health; Pediatrics, Perinatology and Child Health; Epidemiology
Engineering	29	Administrative records (41%); Survey data (14%); Proprietary/commercial records (14%)	Mechanical Engineering; Electrical and Electronic Engineering; Industrial and Manufacturing Engineering
Computer Science	27	Administrative records (39%); Proprietary/commercial records (19%); Digital trace/platform data (16%)	Artificial Intelligence; Information Systems; Computational Theory and Mathematics

Note: OpenAlex fields are paper-level metadata, not dataset labels. They should be read as where in the literature a type of data source is being used.

O Citation-Role Examples

Table 69: Human-reviewed examples of citation-role labels

Role	Citing paper and cited work	Example use in the local citation context	Signal for cited work
Literature survey	w34812 citing Currie (2001)	The cited work appears inside a list of prior preschool and Head Start literature summaries.	Field inclusion; low evaluative signal alone.
Prior evidence	w34733 citing Lin (2024)	The citing sentence attributes a concrete empirical finding about artists posting less after AI image generators are introduced.	Substantive prior result.
Theoretical foundation	w33896 citing Esteban and Ray (1994)	The cited framework supplies the concept of antagonism used by the current paper.	Conceptual building block.
Method justification	w34389 citing Cengiz et al. (2019)	The current paper estimates a stacked event study following the cited approach.	Methodological uptake.
Data or measurement	w33323 citing Ruggles et al. (2024)	The citation anchors Census and ACS samples sourced through IPUMS.	Data provenance or measurement support.
Contextual motivation	w34415 citing Woodward (1955)	The citation supplies historical context for Jim Crow institutions.	Problem framing; low evaluative signal alone.
Contribution framing	w34757 citing Lee (2016)	The current paper distinguishes its contribution from the cited study’s focus.	Positioning relative to prior work.
Estimate comparison	w34377 citing Raynor et al. (2021)	The current estimates of wolf-presence effects are compared with a prior percentage reduction.	Direct magnitude benchmark.
Identification strategy	w33746 citing de Chaisemartin and d’Haultfoeuille (2024)	The cited dynamic difference-in-differences estimator is the primary estimation strategy.	Causal-design building block.
Contradicts	w34208 citing Nivard et al. (2024)	The citing paper states that its findings contrast with the cited results.	Explicit contestation; not automatically a negative quality judgment.

Note: Examples come from the role-by-role human review sample in `human_reviewed.sample.annotations.csv`. They are selected for interpretive clarity, not frequency.

P Citation Quality-Signal Appendix: Cited Works, Frontiers, and Context

This appendix deepens the citation-role result in the main text using the same 2025–2026 citation-role packet. The unit is a citation link, not a paper and not a unique cited work. The purpose is to separate three layers that raw citation counts collapse: (i) which cited works receive attention, (ii) how citing authors use those works, and (iii) what topic or unresolved question the local citation context concerns. The analysis is deliberately conservative: all tables below use the second-stage reviewed outputs, exclude noisy matched-title cited-work keys from main claims, and count open-question language only when it is anchored to cited work, prior evidence, or a named literature.

P.1 Interpretive argument, critique, and analysis plan

The argument is not that every citation is a clean vote on quality. That would repeat the main weakness of raw citation counts. Citation indicators are useful partly because they aggregate scholarly attention, but their meaning is heterogeneous and theory-laden: some references are central antecedents, some are rhetorical positioning, and some are routine background documentation (Aksnes et al., 2019; Shu & Jia, 2026). The role taxonomy is therefore a way to recover structure inside the count, not a claim that the count itself is pure.

The strongest critique of raw citation counts is old and empirically persistent. Moravcsik and Murugesan (1975) distinguished organic from perfunctory citations and found a large perfunctory share; Krampen et al. (2007) reached a similar conclusion in psychology. In the NBER packet, the largest role is `literature_survey`, roughly the same order of magnitude as this perfunctory baseline. That is a validation point as much as a limitation: the classifier is recovering the fact that many citations are real but low-signal acknowledgments rather than substantive uses of a cited work.

A second literature makes the same distinction in computational terms. Valenzuela et al. (2015) frame the task as identifying meaningful rather than incidental citations, and Pride and Knoth (2017) study features for importance screening from full text. The broader citation-function literature supplies taxonomies and benchmark tasks, from early function classification (Teufel et al., 2006) through citation frames (Jurgens et al., 2018), SciCite’s background/method/result scaffold (Cohan et al., 2019), and later surveys and meta-analyses (Bertin & Atanassova, 2024; Kunnath et al., 2021; Zhang et al., 2025). The gap for this paper is disciplinary and inferential: economics uses citation counts as quality proxies, but the proxy is rarely opened up into what the citing author is doing with the cited paper.

This matters for economics because citation-based evaluation is already part of how the field measures scholarly influence. Hamermesh (2018) surveys citation measurement, uses, and impacts in economics, while Bornmann and Wohlrabe (2019) show why field normalization matters. Those studies work mostly at the level of counts. The role profile adds a missing semantic layer: two cited works with the same number of incoming links can have very different profiles if one is cited mainly as background and the other is cited as method, identification infrastructure, estimate benchmark, or contradiction.

The extraction method itself also needs a critique. Recent LLM studies show that citation-intent classification remains unstable: proprietary and open models can disagree on multi-class intent, fine-tuning helps but does not remove the need for validation, and ensembles can outperform single classifiers (Fogelson et al., 2025; Jiang, 2025; Koloveas et al., 2025). For that reason, the tables below are not presented as fully automated truth. They use conservative second-stage reviewed labels, exclude noisy cited-work matches from central claims, and separate role, cited-work identity, context domain, and open-question language rather than forcing a single brittle topic field.

The descriptive plan follows from that critique. First, show the role distribution and its 2025–2026 stability to establish the empirical baseline. Second, aggregate incoming links to a resolved cited work and report its role profile, including an evaluative-intensity share. Third, inspect high-signal roles separately, especially method uptake, identification-strategy uptake, estimate comparison, and contradiction. Fourth, validate role meaning against document structure, because citation location is known to carry information about importance and rhetorical function (Bertin et al., 2016; Safer & Tang, 2009). Fifth, keep context topics as conditioning axes rather than as role labels, following the logic of citation context analysis for literature review and theory development (Anderson & Lemken, 2023). The tables below implement the cited-work, frontier-language, and context-axis parts of that plan.

P.2 Cited-work resolver audit and role profiles

The cited-work profile is the core quality-signal object. A raw citation count treats all links to a cited work as equivalent. The profile instead asks whether those links are mostly background mentions, method uptake,

identification-strategy use, estimate comparisons, or contestation. The main obstacle is bibliographic resolution. The local citation-role layer is more mature than the cited-work profile layer. The profile tables below are therefore exploratory and should be read as diagnostics over DOI or clean-key and conservative author-year matches, not as final corpus-wide citation-network results. The current matched-title field is not reliable enough for final profiles, so the profile table below uses DOI keys when available and conservative author-year keys otherwise.

Table 70: Cited-work resolver audit for the 2025–2026 citation-role packet

Resolver status	Keys	Citations	Use in analysis
DOI or clean key	423	597 (0.5%)	Final-safe
Conservative author-year	43,049	70,358 (58.6%)	Main descriptive with caveat
Ambiguous et al. key	16,991	30,840 (25.7%)	Exploration only
Rejected/noisy title key	10,798	18,269 (15.2%)	Excluded from main claims

Note: The current `matched_title` field is too noisy for final cited-work profiles. The main tables therefore use DOI keys where available and conservative author-year keys with caveats; ambiguous `et al.` keys and rejected title keys are shown only for audit.

Table 71: Resolved cited-work profiles cited by at least 10 processed papers

Cited work	Citations	Papers	Lit. survey	Method	ID strategy	Evaluative
Callaway & Sant’Anna (2021)	180	87	23.9%	55.6%	9.4%	67.2%
Goodman-Bacon (2021)	76	66	32.9%	22.4%	40.8%	63.2%
Sun & Abraham (2021)	91	60	30.8%	42.9%	18.7%	62.6%
Acemoglu & Restrepo (2020)	74	31	20.3%	6.8%	6.8%	18.9%
Hsieh & Klenow (2009)	62	30	30.6%	19.4%	0.0%	22.6%
Ward (2023)	54	14	59.3%	14.8%	1.9%	18.5%

Note: Evaluative citations are method-justification, identification-strategy, estimate-comparison, and contradiction roles. Rows use only resolved keys marked `use` or `use_with_caveat` and are ordered by citing-paper count, not raw local-link count. The table reports both links and citing papers to avoid treating repeated local citations from one or two papers as corpus-wide influence. Cited-work names with hyperlinks resolve to the published DOI.

Table 71 shows the interpretive gain from role composition. Callaway and Sant’Anna (2021), Sun and Abraham (2021), and Goodman-Bacon (2021) are not merely frequently cited; they are cited mainly as method or identification infrastructure. By contrast, other highly cited works have larger background or prior-evidence components. The difference is exactly what a quality-signal analysis should preserve: the same citation total can represent very different forms of intellectual use.

Table 72: High evaluative-intensity cited works with at least ten citation links

Cited work (with DOI link)	Citations	Lit. survey	Method	ID strategy	Evaluative
Walters (2024)	14	0.0%	100.0%	0.0%	100.0%
Olea & Pflueger (2013)	14	7.1%	85.7%	7.1%	92.9%
Auray & Eyquem (2025)	12	0.0%	83.3%	0.0%	91.7%
Arcidiacono & Miller (2011)	11	0.0%	90.9%	0.0%	90.9%
Westfall & Young (1993)	11	0.0%	90.9%	0.0%	90.9%
Wooldridge (2021)	14	14.3%	85.7%	0.0%	85.7%
Romano & Wolf (2005)	12	0.0%	83.3%	0.0%	83.3%
Staiger & Stock (1997)	12	16.7%	0.0%	83.3%	83.3%

Note: These rows illustrate why raw counts and role composition should be separated: some moderately cited works are cited almost entirely as method or identification infrastructure. Hyperlinked names resolve to the published DOI.

P.3 Open questions, limitations, and frontier-setting citations

Some citations define what remains unresolved. These should not be read as negative citations by default. Often the cited literature is being treated as important enough to set the frontier: it establishes a result, exposes a measurement problem, or makes an external-validity question visible. The table below counts only anchored language, meaning that generic future-work phrases are excluded unless the sentence ties the open question to prior work, cited evidence, or a named literature.

Table 73: Anchored open-question, limitation, and frontier-setting citation language

Category	Citations	Share	Main roles	Interpretation
Measurement gap	132	51.8%	literature survey (67); data or measurement (22); method justification (17)	cited data or measures are incomplete or imperfect
External-validity question	39	15.3%	literature survey (18); method justification (11); contextual motivation (4)	prior result may not generalize
Limitation of prior work	39	15.3%	literature survey (22); prior evidence (8); contribution framing (3)	important prior work has scope, data, or design limits
Gap after prior work	28	11.0%	contextual motivation (12); literature survey (11); prior evidence (2)	prior work exists but a question remains
Future work	16	6.3%	literature survey (10); method justification (3); contribution framing (2)	cited work motivates a future extension
Unresolved mechanism	1	0.4%	literature survey (1)	fact or result exists but mechanism remains unclear

Note: Only rows with cited-work- or literature-anchored language are counted. Unanchored future-work wording is excluded from this table.

P.4 Context domain and multi-axis topic coding

The surrounding topic of a citation is not the same object as the citation role. A paper can cite a method paper in a labor context, a health context, or a macro-finance context; collapsing these into one topic field would lose the distinction between *how* the cited work is used and *where* it is used. The context layer should therefore remain multi-axis: context domain, outcome or channel, context mode, method subtopic, data subtopic, comparison subtopic, and open-question category.

Table 74: Strict context-domain assignments around citation links

Context domain	Citations	Share of strict domain links
Labor and inequality	7,276	20.4%
Firms, IO, and innovation	5,176	14.5%
Macro, finance, and banking	5,136	14.4%
Health, demography, and insurance	4,662	13.1%
Education and human capital	3,048	8.6%
Behavior, information, and households	2,583	7.2%
Trade, spatial, urban, and housing	1,950	5.5%
Environment, energy, and agriculture	1,859	5.2%
Development, politics, and institutions	1,809	5.1%
Public finance, tax, and transfers	1,453	4.1%
Law, crime, and regulation	690	1.9%

Note: Domain labels are based on strict second-stage evidence and are separate from citation role. Manual-review-required domain rows are excluded, so these shares describe strict domain assignments rather than all citation links.

The practical rule for the paper is therefore: use citation roles for the use-function of the cited work, use cited-work profiles for quality-signal composition, and use the context/topic layer only as a separate conditioning axis. This prevents a citation to a methods paper in a labor paper from being mistaken for a labor contribution by the cited work itself, and it prevents an open-question citation from being misclassified as criticism when it is actually frontier setting.

Table 75: Strict method-subtopic cues around citation links

Subtopic	Citations	Share of strict subtopic links
ML, text, and image prediction	1,876	24.0%
Randomized, lottery, and experiment designs	1,077	13.8%
Structural, calibration, and GE models	1,024	13.1%
IV and shift-share designs	836	10.7%
Fixed effects, AKM, and panel designs	808	10.4%
DiD and event-study designs	665	8.5%
Inference, bootstrap, and power	470	6.0%
Partial compliance, LATE, and bounds	467	6.0%

Note: Counts require sentence-level method evidence in the local citation context. These are not full-method counts for the paper. Rows show the largest displayed categories; shares use the full strict-assignment denominator, so displayed rows need not sum to 100%.

Table 76: Strict data-subtopic cues around citation links

Subtopic	Citations	Share of strict subtopic links
Survey and questionnaire data	2,773	42.3%
Firm, patent, and procurement data	1,179	18.0%
Census, ACS, and historical linked records	993	15.2%
Financial market and portfolio data	494	7.5%
Administrative, registry, tax, and claims records	364	5.6%
Education, school, and test records	359	5.5%
Geospatial, remote-sensing, and weather data	162	2.5%
Web platform and job-posting data	136	2.1%

Note: Counts require a named data source, record type, measurement object, or source-use cue in the local citation context. Rows show the largest displayed categories; shares use the full strict-assignment denominator, so displayed rows need not sum to 100%.

Table 77: Strict comparison-subtopic cues around citation links

Subtopic	Citations	Share of strict subtopic links
Consistent sign or same direction	5,003	47.0%
Larger or smaller magnitude	2,456	23.1%
Policy cost-benefit or marginal comparison	944	8.9%
External benchmark or calibration	893	8.4%
Null, mixed, or weak evidence	541	5.1%
Contrast or contradiction	475	4.5%
Measurement disagreement	322	3.0%

Note: Counts require explicit comparison, benchmark, null, contrast, or disagreement language around the citation. Rows show the largest displayed categories; shares use the full strict-assignment denominator, so displayed rows need not sum to 100%.

Q AI Peer-Review Hybrid Query Trace

This appendix documents the query trace behind the AI peer-review case study in Section 6.4. The purpose is not to reproduce an agent chat transcript. It is to show the auditable unit of work: a free-form referee concern is decomposed into sub-questions; each sub-question is routed through structured SQL filters, semantic retrieval, and graph traversal; and the returned evidence is converted into a specific review action.

The worked example starts from Cui et al.’s field experiments on GitHub Copilot (10.2139/ssrn.4945566). The enrichment pass checked the focal-paper extraction, an AI-paper corpus of 128 unique papers, a narrow semantic index of 3,816 AI-paper snippets, and the broader NBER chunk index. The query layers played different roles. SQL-style filters restricted the search to structured paper fields such as method family, outcome type, citation role, artifact links, and DOI. Semantic retrieval found nearby language when papers used different terms for the same issue. Graph traversal then followed directed edges among claims, methods, findings, tables, figures, limitations, open questions, and citation roles.

Table 78: Free-tier review points and corpus-enriched edit suggestions

Free-tier review point	Corpus-enriched edit suggestion with source links
Treatment parameter and estimand slippage. The review correctly flags that Cui et al. often describe the 26.08% estimate as a usage effect, while Section 4.1 instruments an adoption dummy and early Microsoft data sometimes use registration. Because treatment content differs across Microsoft, Accenture, and Anonymous Company, the relevant complier population is site-specific.	Move the LATE interpretation into the main framing. The headline should read as a precision-weighted summary of site-specific encouragement-induced adoption effects, not as a generic contemporaneous use effect. Add a site-by-site table with instrument, encouragement content, adoption definition, assignment level, first-stage timing, and likely complier group. The language is supported by treatment-effect weighting in Building Bridges, randomized encouragement in AI in Charge, and employer encouragement/training margins in Mind the Gap and Still Waters, Rapid Currents.
Time variation and weighted IV. The review notes that W-IV puts identifying weight on periods with larger adoption gaps. Footnote 15 already says time-varying effects change the estimand, Appendix D.4 notes Copilot changed during the study period, and Table 9 shows economically large differences between unweighted IV and W-IV.	Ask the paper to define W-IV in economic terms as an adoption-gap-weighted average of site-period complier effects. The revision should explain why this is the preferred substantive estimand when calendar time, model quality, and learning may vary. It should also point readers to the weight distribution and the IV/W-IV contrast. The methods analogy is not identical, but Harvesting Differences-in-Differences is useful for explaining why heterogeneous and dynamic treatment effects require explicit weighted-estimand language; Building Bridges supplies the broader IV/LATE weighting vocabulary.
Spillovers within collaborative teams. The review raises interference: software work is collaborative, assignment occurs at different levels across sites, and clustering at the assignment level does not decide whether estimates are direct individual effects or mixtures of direct and team effects.	Replace the abstract SUTVA concern with a concrete diagnostic. If available, estimate or report treated share of team, repository, reporting line, or code-review network. If not available, state whether site-specific estimates should be interpreted as individual direct effects, team-rollout effects, or mixtures. The benchmark is Shifting Work Patterns, which uses teammate treatment and coworker exposure, plus The Cybernetic Teammate, which makes individual-versus-team AI production central.
Pull requests and completed tasks. The review says pull requests are useful workflow telemetry, but the manuscript’s “completed tasks” language may overstate what the outcome measures. Builds rise strongly, build success is mixed, and quality evidence is Microsoft-only and conditional on positive pull-request activity.	Reframe the main result as pull-request or workflow throughput unless the paper validates PR scope, complexity, and quality across treatment status and sites. Useful checks would include files changed, lines changed, review rounds, reopened PRs, reverts, incident or bug-fix flags, test-suite outcomes, issue/ticket linkage, and story points if available. This edit is sharpened by richer measurement comparators: Generative AI at Work for workplace productivity and customer outcomes, Jagged Frontier for quantity/quality/correctness, Noy-Zhang for time and quality, GitHub Copilot developer productivity for task completion with passing tests, and The Cybernetic Teammate for expert-rated quality.

Free-tier review point	Corpus-enriched edit suggestion with source links
Pooling the site-specific IV estimates. The review correctly notes that the pooled 26.08% estimate is a precision-weighted average across distinct interventions, outcome contexts, and complier populations. The number is useful, but it is not one common causal parameter for high-skilled labor.	Call the pooled coefficient a descriptive precision-weighted summary across implementations. Add a forest plot or table with site-specific first stages, site-specific LATEs, assignment level, outcome definition, and leave-one-site-out or random-effects summaries if feasible. The motivation is the same LATE and treatment-weighting logic in Building Bridges; the implementation heterogeneity is made more concrete by adoption-environment evidence in Mind the Gap and Still Waters, Rapid Currents.
Dynamic estimates and learning. The detailed free-tier comment says Appendix C/Table 6 is hard to interpret because it does not clearly state which experiment or sample underlies the dynamic W-IV estimates, nor how the split into first-three-month and later exposure indicators is instrumented and weighted.	Ask for a compact dynamic-specification note: sample, excluded instruments, first-stage strength for each exposure indicator, weighting scheme, and whether the two endogenous exposure variables are separately identified by the same assignment variation. The text should avoid interpreting longer-run effects only as learning; sustained-use selection, task selection, and Copilot model updates are also plausible. The interpretation can draw on dynamic-treatment cautions from Harvesting Differences-in-Differences and workplace-learning evidence in Generative AI at Work.

Table 79: Sub-questions and hybrid queries behind the AI peer-review example

Review sub-question	SQL or structured-filter layer	Semantic-retrieval layer	Graph and edge traversal
Is the headline estimate a use effect, an adoption effect, or an encouragement-induced adoption effect?	Filter focal-paper method, finding, treatment, and artifact fields for usage , adoption , registration , instrument , LATE , complier , encouragement , and D.it . Search AI-adoption records for employer tool provision, training, encouragement, and rollout.	<i>Generative AI workplace field experiment treatment uptake compliance adoption first use registration employer encouragement training complier population.</i>	Traverse focal-paper treatment and adoption nodes to IV method nodes, then to Table 1, Figure 2, Section 4.1, footnotes, and the main pull-request finding. Follow topic and citation-role edges to adoption comparators such as Mind the Gap and Still Waters, Rapid Currents .
What parameter does the weighted-IV estimator target when effects vary over time or across sites?	Filter method and artifact fields for weighted , W-IV , weights , unweighted , first-stage , adoption gap , time-varying , dynamic , and estimand . In the full NBER index, restrict to IV, LATE, staggered adoption, and dynamic-treatment language before vector ranking.	<i>Instrumental variables LATE complier average treatment effect weighted IV estimator first-stage adoption gaps time-varying treatment effects dynamic treatment staggered rollout weak instrument weights reweighting estimand.</i>	Traverse from the W-IV method node to Table 3, Table 9, Figure 11, and Appendix D.4 limitations. Follow method-family edges and semantic neighbors to Building Bridges and Harvesting Differences-in-Differences .
Could treated developers affect controls through teams, repositories, code review, or coworker learning?	Filter AI workplace experiments and full-NBER chunks for spillover , interference , SUTVA , peer effect , team , teammate , network , contamination , cluster , and collaborat .	<i>Spillovers interference SUTVA treated control units peer effects teams team-level assignment clustered randomized experiment workplace collaboration shared knowledge code review teammates network treatment contamination.</i>	Traverse team-assignment, coworker-exposure, outcome, and method edges in the focal paper and comparator papers. The strongest comparator edges came from Shifting Work Patterns and The Cybernetic Teammate .
Do pull requests measure completed tasks, workflow throughput, task scope, or quality-adjusted output?	Filter outcome, finding, table, and figure fields for productivity , quality , output , task , throughput , performance , completion , grader , telemetry , pull request , build success , commits , and workflow .	<i>AI productivity outcome measurement output quantity versus quality task completion task granularity pull requests commits builds success rate code quality throughput trial and error subjective quality graders logs telemetry workplace performance.</i>	Traverse outcome nodes to Table 3, Table 4, Appendix D.3/D.4, and quality-measure limitations. Follow artifact and finding edges to Generative AI at Work , Jagged Frontier , Noy-Zhang , and the GitHub Copilot developer-productivity benchmark.

Review sub-question	SQL or structured-filter layer	Semantic-retrieval layer	Graph and edge traversal
What does the pooled estimate represent when sites differ in assignment, encouragement, outcomes, and compliance?	Filter focal and comparator records for pooled, precision-weighted, site-specific, external validity, heterogeneity, Microsoft, Accenture, and Anonymous.	<i>Multi-site field experiment pooled IV estimate site-specific treatment effect precision-weighted average implementation heterogeneity external validity complier groups.</i>	Traverse site-specific finding nodes to pooled-finding nodes and then to design-heterogeneity limitations. Follow citation-role and method edges to papers that distinguish local effects, implementation effects, and cross-study summaries.
Does the dynamic analysis identify learning, selection into sustained use, task selection, or model-quality changes over calendar time?	Filter focal records for first three months, learning, dynamic, Table 6, Appendix C, more than three months, calendar time, and model updates.	<i>Learning to use generative AI dynamic treatment effects exposure duration post-adoption selection calendar time model improvement worker task selection.</i>	Traverse dynamic-effect finding nodes to W-IV method nodes, timing caveats, and Appendix D.4 limitations. The graph distinguishes learning claims from alternative explanations: selection, task mix, and Copilot model vintage.
Is a methods comment already supported by the focal paper’s own citations, or does it need a stronger comparison?	Filter full-NBER records for method family and citation role: event study, difference-in-differences, IV, LATE, stacked event study, method justification, and identification strategy.	<i>Staggered adoption event-study estimator dynamic difference-in-differences stacked event study identification strategy citation method justification.</i>	Follow citation-role edges to local contexts where cited work is used as method justification or identification strategy rather than background. Examples include w34389 using Cengiz et al. (2019) and w33746 using de Chaisemartin and d’Haultfoeuille (2024).

Table 80: Evidence returned by the query trace and the resulting review action

Issue	Focal-paper anchors	Comparator evidence	Review action
Treatment parameter	Table 1, Figure 2, Section 4.1, footnotes 12–13, and the main IV finding.	Building Bridges ; AI in Charge ; Mind the Gap ; Still Waters , Rapid Currents .	Ask the authors to label the headline object as site-specific encouragement-induced adoption LATEs, not a generic use effect.
Weighted-IV estimand	Section 4.1, footnote 15, Table 3, Table 9, Figure 11, and Appendix D.4.	Harvesting Differences-in-Differences ; Building Bridges .	Ask for an economic definition of the W-IV estimand as an adoption-gap-weighted average of site-period complier effects.
Team spillovers	Table 1 assignment levels, Table 3 main estimates, and collaborative workflow outcomes.	Shifting Work Patterns , especially teammate-treatment and coworker-share evidence; The Cybernetic Teammate .	Convert a generic interference concern into a concrete diagnostic: treated share of team, repository, reporting line, or code-review network.
Outcome measurement	Table 3 pull requests, commits, builds, and build success; Table 4 Microsoft-only quality proxies; Appendix D.3/D.4 site comparability caveats.	Generative AI at Work ; Jagged Frontier ; Noy–Zhang ; GitHub Copilot developer-productivity benchmark ; The Cybernetic Teammate .	Reframe the result as workflow or pull-request throughput unless PR scope, complexity, and quality are validated across treatment status and sites.
Pooling and external validity	Table 3 precision-weighted pooling note and Appendix D.4 design, timing, training, and company-context caveats.	Workplace and methods comparators showing treatment-effect and implementation heterogeneity.	Present the pooled estimate as a descriptive precision-weighted summary across implementations, not as one common productivity parameter for high-skilled work.
Dynamic exposure	Table 6 dynamic effects, Appendix C, and Appendix D.4 timing/model-update caveats.	Dynamic-treatment and learning-adoption analogies from the broader methods and AI-workplace corpus.	Clarify the sample, excluded instruments, and W-IV weighting for short-run versus longer-run exposure; avoid interpreting dynamics only as learning.

Reusable Query Templates

The SQL layer first narrows the candidate set. The semantic layer then ranks candidates within that set by similarity to an embedded query. This example is the methods query used to find IV, LATE, weighting, and dynamic-treatment analogies:

```
WITH cand AS (
  SELECT
    c.paper_id,
    pm.title,
    c.entity_type,
    c.entity_id,
    c.text,
    c.embedding
  FROM chunks c
  LEFT JOIN paper_metadata pm USING (paper_id)
  WHERE c.embedding IS NOT NULL
    AND regexp_matches(
      lower(c.text),
      '(instrument|iv|2sls|late|complier|first.stage|weighted|weighting|staggered|dynamic|treatment effect|weak instrument|estimand|reweight)'
    )
  LIMIT 250000
)
SELECT
  paper_id,
  title,
  entity_type,
  entity_id,
  array_cosine_similarity(embedding, ?::FLOAT[3072]) AS similarity,
  text
FROM cand
ORDER BY similarity DESC
LIMIT 28;
```

The semantic query supplied to the embedding model for that SQL-filtered search was:

```
instrumental variables LATE complier average treatment effect weighted IV estimator first-stage
adoption gaps time-varying treatment effects dynamic treatment staggered rollout weak
instrument weights reweighting estimand
```

The graph layer then converts retrieval into a review decision. In pseudocode, each review issue follows the same pattern:

```
start with review_issue
anchor review_issue to focal-paper claim, method, outcome, or caveat nodes
follow edges:
  claim -> method
  method -> finding
  finding -> table_or_figure
  method/finding -> limitation
  limitation -> open_question
  focal topic -> related-paper topic
  focal citation -> citation-role context
return:
  focal source anchor
  comparator paper and role
  whether the comment should be kept, revised, split, or dropped
```

This is why the review becomes more specific after corpus enrichment. The agent is not only retrieving papers that sound related; it is asking whether each comment is supported inside the focal-paper graph, whether related papers supply method precedent or measurement context, and what concrete edit would make the manuscript clearer.

R Taxonomy Definitions

The pipeline uses controlled vocabularies for classifying methods, data sources, identification rigor, and theory components. These taxonomies are stored as JSON configuration files and injected into prompts at runtime.

Table 81: Identification Rigor Taxonomy

Level	Description & Assumption	Canonical Examples & Tests
Randomized Experiment	RCT, field experiment, lab experiment with random assignment <i>Assumption:</i> Random assignment ensures independence of treatment and potential outcomes	STAR class size experiment (Krueger 1999); Oregon Health Insurance Experiment (Finkelstein et al. 2012); Moving to Opportunity (Chetty, Hendren, Katz 2016) <i>Tests:</i> balance tests, attrition analysis, compliance rates, ITT vs. LATE
Strong Quasi-Experimental	DID, RDD, IV with validated instruments, synthetic control <i>Assumption:</i> Design-specific: parallel trends (DID), continuity (RDD), exclusion restriction (IV), no anticipation (SC)	Card and Krueger (1994) minimum wage DID; Lee (2008) close-election RDD; Angrist and Krueger (1991) quarter-of-birth IV <i>Tests:</i> parallel trends, McCrary density test, first-stage F-statistic, pre-treatment RMSPE, placebo tests
Weak Quasi-Experimental	Fixed effects, matching, propensity score, selection on observables <i>Assumption:</i> Selection on observables / conditional independence assumption	Abowd, Kramarz, Margolis (1999) worker-firm FE; Dehejia and Wahba (1999) propensity score matching; Hainmueller (2012) entropy balancing <i>Tests:</i> balance tables, common support, sensitivity analysis (Oster bounds), covariate balance after matching
Structural Estimation	Structural model with calibration/estimation (BLP, dynamic programming) <i>Assumption:</i> Model-specific functional form and exclusion restrictions	Berry, Levinsohn, Pakes (1995) demand estimation; Rust (1987) dynamic discrete choice; Heckman and Singer (1984) duration models <i>Tests:</i> overidentification test (J-statistic), out-of-sample fit, comparison to reduced-form estimates, counterfactual validation
Stated Preference / Choice Experiment	DCE, contingent valuation, survey-based preference elicitation <i>Assumption:</i> Respondents truthfully reveal preferences; no hypothetical bias	Exxon Valdez contingent valuation (Carson et al. 2003); Conjoint analysis of health insurance attributes; Discrete choice experiments for environmental valuation <i>Tests:</i> scope tests, consequentiality tests, hypothetical bias correction, internal consistency

Table 81 – continued

Level		Description & Assumption	Canonical Examples & Tests
Descriptive Correlational	/	OLS without causal claim, descriptive statistics, correlations <i>Assumption:</i> None claimed; documents associations or patterns	Piketty and Saez (2003) top income shares; Autor, Katz, Kearney (2008) wage inequality trends; Chetty et al. (2014) intergenerational mobility atlas <i>Tests:</i> robustness to alternative definitions, sample composition checks
Theoretical Analytical	/	Mathematical model, game theory, mechanism design without empirical estimation <i>Assumption:</i> Model assumptions (rationality, equilibrium concept, functional forms)	Myerson (1981) optimal auction design; Milgrom and Stokey (1982) no-trade theorem; Acemoglu and Autor (2011) task-based framework <i>Tests:</i> comparative statics, boundary case analysis, numerical simulations

Table 82: Method Family Taxonomy (91 types in 9 groups)

Group (count)	Method Families
Causal / Quasi-Experimental (16)	difference_in_differences, triple_difference, regression_discontinuity, regression_kink, instrumental_variables, synthetic_control, synthetic_difference_in_differences, matching, event_study, bunching, rct_experiment, lagged_dv_panel, ipw_reweighting, pre_trend_detrending, reduced_form_ols, other_causal
Structural (10)	structural_estimation, structural_discrete_choice, structural_dynamic, simulated_method_of_moments, generalized_method_of_moments, maximum_likelihood_structural, calibration, comparative_statics, bayesian_state_space, other_structural
Regression / Panel (12)	fixed_effects, random_effects, dynamic_panel, ols_regression, quantile_regression, nonparametric, logit_probit, censored_regression, count_model, factor_model_regression, factor_extraction, other_regression
Discrete Choice (8)	discrete_choice, multinomial_choice, ordered_choice, conditional_logit, latent_class_model, mixed_logit, nested_logit, other_discrete_choice
Selection / Bounds (4)	selection_correction, control_function, bounds_estimation, other_selection
Decomposition (5)	variance_decomposition, oaxaca_blinder, shift_share_decomposition, growth_accounting, other_decomposition
Measurement / Variable Construction (12)	classification_model, classification_tagging, prediction_model, record_linkage, traditional_nlp, clustering, content_classification, imputation_method, index_construction, willingness_to_pay, survey_experiment, other_measurement
Machine Learning (8)	ml_prediction, ml_causal, causal_forest, double_machine_learning, ml_heterogeneity, ml_selection, language_model, other_ml
Other (16)	review_synthesis, qualitative_support, survey_support, specification_test, counterfactual_simulation, benchmark_comparison_model, numerical_pricing, bootstrap_inference, continuous_time_markov, fixed_point_computation, descriptive, simulation, time_series, vector_autoregression, cointegration, other

Table 83: Method Purpose Classification (all method families)

Purpose	Count	All Method Families
Causal	20	bunching, causal_forest, difference_in_differences, double_machine_learning, event_study, instrumental_variables, ipw_reweighting, lagged_dv_panel, matching, ml_causal, ml_heterogeneity, other_causal, pre_trend_detrending, rct_experiment, reduced_form_ols, regression_discontinuity, regression_kink, synthetic_control, synthetic_difference_in_differences, triple_difference
Decomposition	5	growth_accounting, oaxaca_blinder, other_decomposition, shift_share_decomposition, variance_decomposition
Descriptive	4	descriptive, qualitative_support, review_synthesis, survey_support
Estimation	29	bootstrap_inference, bounds_estimation, censored_regression, cointegration, conditional_logit, control_function, count_model, discrete_choice, dynamic_panel, factor_model_regression, fixed_effects, latent_class_model, logit_probit, mixed_logit, multinomial_choice, nested_logit, nonparametric, ols_regression, ordered_choice, other, other_discrete_choice, other_regression, other_selection, quantile_regression, random_effects, selection_correction, specification_test, time_series, vector_autoregression
Measurement	17	classification_model, classification_tagging, clustering, content_classification, factor_extraction, imputation_method, index_construction, language_model, ml_prediction, ml_selection, other_measurement, other_ml, prediction_model, record_linkage, survey_experiment, traditional_nlp, willingness_to_pay
Structural	16	bayesian_state_space, benchmark_comparison_model, calibration, comparative_statics, continuous_time_markov, counterfactual_simulation, fixed_point_computation, generalized_method_of_moments, maximum_likelihood_structural, numerical_pricing, other_structural, simulated_method_of_moments, simulation, structural_discrete_choice, structural_dynamic, structural_estimation

Prompt-routing note. The main text reports first-run display categories for method-purpose records: measurement, causal, descriptive, structural parameter, structural, decomposition, and theoretical. The published method-family taxonomy below separates **estimation** from **measurement**, while some production prompts in the first corpus run used a narrower routing enum and sent non-causal regression and association models through the broader **measurement** branch before display normalization. The next release will reconcile the prompt routing, method-family taxonomy, and display purpose taxonomy so regression, measurement, data-

construction, theoretical, and descriptive components use one explicit mapping.

Table 84: Data Collection Method Taxonomy

Collection Method	Sub.	Subtypes (keyword indicators)
Administrative	7	government records (tax, ssa, cms); healthcare records (claims, ehr, electronic health); education records (transcripts, testing, nces); employment records (payroll, ui, unemployment insurance); financial records (credit, banking, financial); legal records (court, patents, contracts); other admin
Survey	5	household survey (psid, cps, acs); establishment survey (asm, mops, establishment); specialized survey (nhis, nlsy, national health); custom survey (author-designed, custom); other survey
Experimental	5	field experiment (field experiment, rct, randomized); lab experiment (lab experiment, laboratory); online experiment (a/b test, online experiment); natural experiment (natural experiment, quasi-experimental); other experimental
Observational Digital	5	platform data (social media, marketplace, platform); transaction data (purchases, trades, transactions); clickstream (web behavior, clickstream); geolocation (gps, cell tower, geolocation); other digital
Text Document	8	news media (news, newspaper, media); social media text (tweets, posts, social media text); corporate filings (10-k, earnings calls, sec filings); legal documents (court opinions, legislation, legal); academic text (papers, patents, academic); government text (congressional, federal register, government); user reviews (reviews, user reviews); other text
Image Media	5	satellite imagery (satellite, satellite imagery); street imagery (street view, street imagery); document images (scanned, historical documents); medical images (medical imaging, x-ray, mri); other media
Network Relational	4	social network (social network); economic network (supply chains, trade network); organizational network (organizational); other network
Synthetic Generated	3	simulation based (simulation, monte carlo); ml generated (ml generated, synthetic data); other synthetic

Table 86: Theory Component Types

Type	Description
<code>assumption</code>	Maintained hypothesis or axiom (e.g., rationality, CRRA preferences, iid shocks)
<code>definition</code>	Formal definition of a concept, variable, or equilibrium notion
<code>proposition</code>	Formal statement derived from assumptions, with proof
<code>theorem</code>	Major analytical result with proof (stronger than proposition)
<code>lemma</code>	Auxiliary result used to prove a proposition or theorem
<code>corollary</code>	Direct consequence of a theorem or proposition
<code>mechanism</code>	Causal channel through which an effect operates (e.g., moral hazard, selection)
<code>model_structure</code>	Core model specification: agents, timing, technology, information structure
<code>equilibrium_condition</code>	Conditions characterizing equilibrium (FOCs, market clearing, fixed points)
<code>comparative_static</code>	How equilibrium outcomes change with parameters
<code>identification_argument</code>	Argument for why model parameters are recoverable from data
<code>boundary_condition</code>	Terminal, initial, or boundary conditions constraining the solution
<code>prediction</code>	Testable prediction derived from the model
<code>counterfactual</code>	What-if scenario evaluated through the model (e.g., policy removal)
<code>welfare_implication</code>	Welfare consequence (e.g., deadweight loss, consumer surplus change)
<code>policy_implication</code>	Normative or design recommendation from model results
<code>limitation</code>	Acknowledged scope restriction or simplifying assumption
<code>open_question</code>	Unresolved theoretical question identified by the paper
<code>other</code>	

Table 85: Citation Function Label Taxonomy (10 roles)

Role	Definition	ACL-ARC origin	Graph edge
<code>prior_evidence</code>	Cited paper produced a specific empirical result the current paper benchmarks against, confirms, or extends	background	Yes
<code>method_justification</code>	Provides the estimator, regression specification, index construction, or statistical procedure used	uses	Yes
<code>identification_strategy</code>	Source or validation of the causal identification strategy — natural experiment, IV relevance/exclusion argument, or RD validity claim	uses	Yes
<code>theoretical_foundation</code>	Provides the theoretical model, mechanism, or conceptual framework the paper builds on, tests, or inherits assumptions from	extends	Yes
<code>estimate_comparison</code>	Current paper explicitly compares its estimates numerically to the cited paper’s estimates — to validate, reconcile, or contest them	compares_or_contrasts	Yes
<code>data_or_measurement</code>	Source of the dataset, variable construction method, index, score, or measurement instrument used	economics_extension	Yes
<code>contribution_framing</code>	Paper positions its contribution relative to this literature strand — ‘we contribute to X by doing Y’, or ‘unlike X we find Y’	extends	Yes
<code>contextual_motivation</code>	Establishes policy relevance, a stylized fact, or motivating empirical regularity; often government reports or survey papers	motivation	No
<code>literature_survey</code>	Appears in an enumerative list of related work with no specific knowledge transfer — ‘see also’, ‘among others’, ‘examples include’	background	No
<code>contradicts</code>	Results or conclusions conflict with the cited work	compares_or_contrasts_subset	Yes

Note: Labels are assigned to a local citation context, not to a cited work globally. The extraction records one primary role per citation link so role counts are additive. A cited work can therefore have multiple roles across a paper or across the corpus when different citing contexts use it in different ways.

The following prompts are used verbatim at each pipeline stage. Stage 2 (methodology blueprint) and Stage 3 (entity extraction) prompts define the full output schema and controlled-vocabulary constraints. Enrichment-stage prompts (Stages 4.76–4.78) are targeted single-field passes. All prompts are injected with the taxonomy vocabularies from Appendix R at runtime.

S Extraction Prompts

Each pipeline stage uses structured prompts to instruct the LLM. System prompts define the extraction task, required fields, and output schema. User prompts inject the paper text and context from prior stages. Template variables (e.g., `{{paper_id}}`) are filled at runtime.

S.1 Stage 2: Methodology Blueprint

Model: GPT-5-mini

S.1.1 Architect — Methodology Blueprint (System Prompt)

```

You are an Expert Econometrician creating a Methodology Blueprint.

CRITICAL: Output valid JSON only. No LaTeX. Keep responses concise.

YOUR ROLE: Identify WHAT methods exist and WHERE they are discussed.
Detailed extraction happens in later stages - you just create the roadmap.

## TASK 1: PAPER CLASSIFICATION

Return BOTH:

- paper_type: empirical | descriptive | theoretical | mixed
- paper_genre: empirical | descriptive | theoretical | mixed | review_survey | conceptual_methodological
               | benchmarking_comparative | unknown

- empirical: Uses data to estimate causal effects, test hypotheses, or measure relationships using
              econometric methods
- descriptive: Documents patterns/facts without econometric analysis (summary stats, tabulations, graphs
              only)
- theoretical: Develops models without empirical estimation
- mixed: Combines NOVEL theoretical model development with empirical estimation. Use ONLY when a paper
          develops a new structural/theoretical model AND estimates it empirically. If the paper merely uses
          standard empirical methods, classify as "empirical".
- review_survey: Primarily synthesizes prior literature, evidence, or approaches rather than reporting a
                 new empirical design as the main contribution.
- conceptual_methodological: Primarily advances a framework, typology, measurement concept, or
                             methodological argument, even if it discusses empirical examples.
- benchmarking_comparative: Primarily compares datasets, methods, measurement systems, or model families
                             as the main paper contribution.

If a paper is a standard empirical/theoretical/descriptive/mixed paper, paper_genre may equal paper_type
.

If theoretical/descriptive/review_survey/conceptual_methodological with no concrete implemented methods:
    Return empty methodology_graph [].
Do NOT fabricate empirical methods just to populate the graph for non-empirical genres.

## TASK 2: METHOD IDENTIFICATION

Identify the paper's concrete empirical/theoretical methods. For each method extract:

1. **method_id**: Unique ID with paper prefix (e.g., "w32050_M1")
2. **method_name**: Name as used in the paper
3. **method_family**: Must be one of: [91 method families from taxonomy]
4. **method_purpose**: How the method is used (see below)
4.5 **usage_status**: used_in_paper | suggested_for_future | mentioned_not_exploited

```

5. **brief_description**: 1-2 sentences on what the method does in this paper

6. **source_section_ids**: Section IDs where method is described (e.g., ["S3", "S4"])

7. **method_references_hint**: Citation text for methods papers mentioned (e.g., ["Callaway & Sant'Anna (2021)"])

8. **graph_layer**: method_core | method_aux (best effort)

9. **node_priority**: main | secondary | supporting (best effort)

METHOD PURPOSE CLASSIFICATION

causal: Makes causal claims (DiD, IV, RDD, RCT, Event Study, Synthetic Control, Matching)

structural: Estimates model parameters (SMM, GMM, MLE for structural models)

measurement: Constructs variables, estimates associations, or runs non-causal empirical models (OLS, logit, FE, time-series, Tobit, selection models) without explicit design-based identification.

decomposition: Decomposes variance (AKM, Oaxaca-Blinder)

descriptive: Summary statistics, tabulations, cross-tabs, and graphs ONLY. Do NOT use "descriptive" for any regression-based analysis. If a method runs regressions (OLS, FE, logit), use "measurement" unless the paper explicitly uses causal identification (then use "causal").

CAUSAL TRIGGER RULE: If the paper explicitly uses IV/2SLS, RDD, DiD/event-study with identifying assumptions, RCT/randomized assignment, synthetic control, matching for treatment effects, or clear exogenous policy shock identification, classify that method as **causal**.

NON-CAUSAL REGRESSION RULE: Do NOT label a regression as causal merely because it studies a policy, break date, reform episode, or historical shock. OLS/logit/FE/time-series/structural-break/policy-rule regressions stay **measurement** unless the paper explicitly states an identification strategy and identifying assumption.

USAGE STATUS

used_in_paper: The paper actually implements this method and reports estimates/results from it.

suggested_for_future: The method/design/instrument is suggested for future work, later versions, or extensions the paper does NOT actually run.

mentioned_not_exploited: The paper discusses or benchmarks against the method concretely, but does not actually implement it in the paper's own analysis.

FUTURE-WORK RULE: If the text says "future work", "in subsequent versions", "we may use", "we will attempt", or otherwise describes a method not actually executed, do NOT mark it as used.

IMPORTANT: Do NOT treat the words "propose" or "proposed" by themselves as future-work cues. Papers often propose a model/estimator and then implement it in the same paper.

WHAT TO INCLUDE

Include:

- Causal identification methods (DiD, IV, RDD, etc.)
- Estimation methods (OLS, logit, probit, etc.)
- ML methods used for variable construction
- Structural estimation methods
- Review synthesis, qualitative/survey support, numerical pricing, and classification/tagging procedures when they are central to the paper's design or argument
- Central concrete future or unexploited strategies ONLY if they materially matter for understanding the paper, and then mark 'usage_status' as 'suggested_for_future' or 'mentioned_not_exploited'

WHAT NOT TO INCLUDE

Do NOT list as separate methods:

- Fixed effects alone (these are specification details within a method, not separate methods -- e.g., do NOT list "Firm FE" as a method. But DO list the regression that uses firm FE.)
- Control variables
- Standard error clustering choices
- Robustness checks that use the same method
- Methods mentioned only in the literature review with no role in the paper's own analysis

METHOD INDIVIDUATION TEST

Before finalizing your method list, check every pair of candidate methods:

Same method (list ONCE, note variants in brief_description):

- OLS with industry FE and OLS with firm FE -> ONE method
- Table 3 Col 1 (no controls) and Table 3 Col 5 (full controls) -> ONE method
- Same regression in levels vs. logs when both test the same hypothesis -> ONE method
- OLS vs. WLS of the same regression -> ONE method

Separate methods (list as distinct entries):

```

- OLS vs. 2SLS (different identification strategy)
- Conditional logit vs. latent class model (different likelihood)
- Effect on wages vs. effect on employment (different dependent variable)
- IV using rainfall vs. IV using distance (different instrument)

RULE: If two estimators have different likelihoods or different identifying assumptions,
they are separate methods. If they only differ in which controls/FE/weights are included,
they are the same method.

### ML / ALGORITHMIC METHODS

If ML/AI methods are used (Random Forest, Neural Networks, LLMs, word embeddings), include them in
ml_methods_detected:
- method_id, method_name, analytical_role, creates_variable
- analytical_role values: variable_construction | prediction | causal_estimation | text_analysis |
  image_analysis
- creates_variable: Name of variable created (e.g., "sentiment_score", "predicted_income")
- For LLMs: note the model used if mentioned (GPT-4, BERT, etc.)

If algorithmic methods are used even outside classic ML, include them in algorithmic_methods_detected:
- record linkage / name matching / inventor disambiguation
- named entity recognition / text vectorization / TF-IDF / cosine similarity
- clustering / hierarchical clustering / k-means / Ward linkage
- content-based classification / text-based indicators
- automatic valuation models or other algorithmic scoring systems
- Use the same fields: method_id, method_name, analytical_role, creates_variable
- analytical_role values may also include: linkage | clustering | content_classification | measurement
... [58 additional lines omitted]

```

S.1.2 Architect — Methodology Blueprint (User Prompt Template)

```

PAPER TEXT:
{{ context_str }}

OUTPUT SCHEMA (keep response concise - detailed extraction happens in later stages):
{
  "paper_type": "empirical | descriptive | theoretical | mixed",
  "paper_genre": "empirical | descriptive | theoretical | mixed | review_survey |
    conceptual_methodological | benchmarking_comparative | unknown",
  "paper_type_reasoning": "Brief explanation of classification",
  "title": "Paper title",
  "year": "YYYY",

  "methodology_graph": [
    {
      "method_id": "{{paper_id}}_M1",
      "method_name": "Paper-specific method name",
      "method_family": "Must be from provided list",
      "method_purpose": "causal | structural | measurement | decomposition | descriptive",
      "usage_status": "used_in_paper | suggested_for_future | mentioned_not_exploited",
      "brief_description": "1-2 sentences on what the method does",
      "source_section_ids": ["S3", "S4"],
      "method_references_hint": ["Author & Author (Year)"]
    }
  ],

  "ml_methods_detected": [
    {
      "method_id": "{{paper_id}}_ML1",
      "method_name": "Random Forest",
      "analytical_role": "variable_construction | prediction | causal_estimation | text_analysis |
        image_analysis",
      "creates_variable": "variable_name_created"
    }
  ],

  "algorithmic_methods_detected": [
    {
      "method_id": "{{paper_id}}_A1",

```

```

    "method_name": "Inventor name matching algorithm",
    "analytical_role": "linkage | clustering | content_classification | measurement",
    "creates_variable": "variable_name_created"
  }
],

"main_topics": ["Topic 1", "Topic 2"],
"research_questions": ["Main research question(s)"],
"contribution_claims": [
  {
    "claim_id": "{{paper_id}}_C1",
    "claim_text": "Concise statement of a claimed paper contribution",
    "contribution_type": "methodological | empirical | theoretical | data",
    "supported_by_finding_ids": [],
    "supported_by_method_ids": [],
    "source_section_ids": ["S1", "S2"],
    "anchor_start": "First few words",
    "anchor_end": "Last few words"
  }
],
"abstract_summary": "1-2 sentence summary of main contribution",
"geographic_focus": "Primary geographic setting",
"time_focus": "Primary time period studied",

"theory_components": [
  {
    "component_id": "{{paper_id}}_T1",
    "component_type": "assumption | definition | proposition | theorem | lemma | corollary | mechanism
    | model_structure | equilibrium_condition | comparative_static | identification_argument |
    boundary_condition | prediction | counterfactual | welfare_implication | policy_implication |
    limitation | open_question | other",
    "component_text": "Concise statement of the theoretical component",
    "source_section_ids": ["S2", "S3"],
    "anchor_start": "First few words",
    "anchor_end": "Last few words"
  }
]
}

```

INSTRUCTIONS:

1. Classify paper type first
- 1.5. Also return paper_genre. Use review_survey for literature reviews/surveys, conceptual_methodological for framework or methodology essays, and benchmarking_comparative for papers whose main contribution is comparison/benchmarking across datasets, methods, or systems.
2. Identify executed methods first; only include concrete future or unexploited strategies if central, and then set 'usage_status' accordingly
- 2.5. Do NOT treat the words "propose" or "proposed" alone as evidence of future work. If the paper proposes and executes a model/estimator in the same paper, use 'used_in_paper'.
3. For each method, note which sections discuss it (source_section_ids)
4. Include citation hints for methods papers (just the citation text)
5. Return ONLY valid JSON (no markdown, no LaTeX)
6. If paper_type is theoretical or mixed, include theory_components. Otherwise return theory_components: [].
7. Future-work IVs, proposed natural experiments, and unimplemented extensions are NOT 'usage_status="used_in_paper"'.

NOTE: Do NOT extract detailed identification_design, validity tests, or findings. Those will be extracted by specialized modules in later stages.

S.2 Stage 3a: Detailed Method Extraction

Model: GPT-5-mini

S.2.1 Method Extraction — Causal Purpose

```
## METHOD PURPOSE: CAUSAL
```

This method makes CAUSAL CLAIMS. You MUST extract the identification_design block.

IDENTIFICATION DESIGN (REQUIRED):

Extract the complete causal identification strategy:

1. **CAUSAL CHAIN**:

- variation_source: What provides exogenous variation (instrument, shock, running variable, policy)
- variation_source_type: instrument | running_variable | policy_shock | exogenous_shock
- treatment_variable: SHORT name (1-5 words) for what is directly treated -- the unit or variable that receives the treatment (e.g. "store", "individual", "firm", "municipality", "bond price"). This is NOT the instrument or policy shock (that is variation_source) and NOT the outcome. In IV: instrument -> treatment_variable -> outcome. In DiD: policy shock -> treatment_variable -> outcome. Use null if there is no distinct treatment variable (e.g. pure descriptive, time-series model).
- target_mechanism: What the variation source affects

2. **VALIDITY**:

- identifying_assumption: One-sentence testable assumption (e.g., "parallel trends")
- validity_defense: How authors argue the assumption holds
- threats_acknowledged: List of acknowledged threats to identification
- assumption_tested: true | false (did the paper empirically test the identifying assumption?)
- assumption_test_type: statistical_test | verbal_defense | indirect_evidence | not_addressed
- what_would_test_it: data/measurement needed for a direct test (ONLY when assumption_tested=false)

3. **STATISTICAL TESTS** (Extract what IS REPORTED):

Look for and extract EXACT VALUES:

- IV: First-stage F-statistic (flag if < 10)
- RDD: Bandwidth value, McCrary test result (p-value)
- DiD: Parallel trends test, pre-trend F-statistic, event study coefficients
- Event Study: Pre-event coefficient joint test
- Synthetic Control: Pre-treatment RMSPE

For each test:

```
{{
  "test_name": "parallel_trends | f_statistic | mccrary | overid | etc.",
  "test_statistic": "EXACT reported value (e.g., 'F=25.3')",
  "test_result": "What the test shows",
  "passes_threshold": true | false | null,
  "interpretation": "Brief interpretation"
}}
```

4. **DIAGNOSTIC STATS** -- Extract into a top-level 'diagnostic_stats' object:

****ROUTING RULE**:** Always prefer a typed block over 'open_ended_stats'. Use 'open_ended_stats' ONLY for stats that do not fit any typed block below.

- CORRECT: $R^2=0.43$ -> '{"ols_panel_stats": {"r_squared": 0.43}}'
- CORRECT: Conley SE radius 500km -> 'open_ended_stats' (no typed block exists)
- WRONG: putting R^2 , F-stat, or parallel-trends p-value in 'open_ended_stats'

Always extract if present:

- 'ols_panel_stats.r_squared', 'ols_panel_stats.within_r_squared', 'ols_panel_stats.overall_f_stat'
- 'ols_panel_stats.n_clusters' (integer -- how many clusters SE is clustered on)

****IV/2SLS**** ('method_family = instrumental_variables'):

- 'iv_stats.first_stage_f_stat' -- F-stat from first-stage regression (Staiger-Stock threshold: $F > 10$).

Look in: table notes, footnotes, and appendix tables ("Table A3: First-Stage Results").

- 'iv_stats.kleibergen_paap rk_stat' -- preferred over F under heteroskedasticity
- 'iv_stats.partial_r_squared_first_stage' -- explanatory power of instruments in first stage
- 'iv_stats.n_instruments', 'iv_stats.n_endogenous' (integers)
- If $n_{\text{instruments}} > n_{\text{endogenous}}$ (overidentified):
'iv_stats.sargan_p_value', 'iv_stats.hansen_j_p_value'

****DiD / Event Study**** ('method_family = difference_in_differences | event_study'):

- 'did_stats.parallel_trends_f_stat', 'did_stats.parallel_trends_p_value'
(joint F/chi2 test that pre-period coefficients are jointly zero)
- 'did_stats.pre_period_coefficients' -- numeric array:
[{"period": -3, "coef": -0.02, "se": 0.01, "p": 0.4}, ...]
- 'did_stats.callaway_santanna_used', 'did_stats.sun_abraham_used',
'did_stats.borusyak_jaravel_spiess_used', 'did_stats.de_chaisemartin_dhaultfoeuille_used' (booleans)
- 'did_stats.event_window_leads', 'did_stats.event_window_lags' (integers)

```

**RDD** ('method_family = regression_discontinuity'):
- 'rdd_stats.bandwidth_numeric' (number), 'rdd_stats.bandwidth_unit' (string)
- 'rdd_stats.bandwidth_method' ("ik_optimal" | "cct" | "manual" | "cross_validation")
- 'rdd_stats.kernel_type' ("triangular" | "rectangular" | "epanechnikov")
- 'rdd_stats.polynomial_order_int' (integer)
- 'rdd_stats.mccrary_p_value' -- p-value for McCrary (2008) density continuity test
- 'rdd_stats.effective_n_left', 'rdd_stats.effective_n_right' (integers within bandwidth)
- 'rdd_stats.bandwidth_50pct_result' ("significant" | "not_significant" | null)

**Synthetic Control** ('method_family = synthetic_control'):
- 'synth_control_stats.pre_treatment_rmspe', 'synth_control_stats.post_treatment_rmspe'
- 'synth_control_stats.rmspe_ratio' (post/pre)
- 'synth_control_stats.n_donor_units', 'synth_control_stats.permutation_p_value'

**Matching** ('method_family = matching | psm'):
- 'matching_stats.max_standardized_mean_diff_after' (max SMD across all covariates after matching)
- 'matching_stats.balancing_test_p_value'
- 'matching_stats.common_support_fraction'

**ANYTHING ELSE** (only if no typed block fits) -- add to 'open_ended_stats':
{"stat_name": "Conley spatial SE radius", "stat_value": "500km", "stat_numeric": 500,
 "threshold_context": null, "passes_threshold": null,
 "stat_category": "standard_error_spec", "source_location": "Table 3 note"}}
Categories: 'instrument_strength | pre_trend | model_fit | balance | spatial | other'

**Return null (not 0) for any field not explicitly reported.**

5. **METHOD-SPECIFIC FIELDS**:
Depending on method_family, extract additional details:

- **IV**: instrument_type, exclusion_argument, relevance_argument, first_stage_f_stat
- **RDD**: running_variable, cutoff, bandwidth_method, bandwidth_value, polynomial_order,
  mccrary_test_result
- **DiD**: staggered (true/false), estimator (twfe|callaway_santanna|sun_abraham|
  de_chaisemartin_dhaultfoeuille|borusyak_jaravel_spiess|wooldridge_extended_twfe|other),
  never_treated_control, parallel_trends_test, event_study_pretrends
- **Event Study**: reference_period, event_window, binned_endpoints, pre_event_test
- **Synthetic Control**: control_units, pre_treatment_periods, pre_treatment_rmspe,
  weights_optimization
- **Bunching**: bunching_variable, notch_vs_kink, counterfactual_distribution_method (polynomial|
  parametric|other), excluded_range, excess_mass_estimate
- **Matching/PSM**: matching_method (nearest_neighbor|kernel|caliper|cem|entropy_balancing),
  caliper_value, common_support_enforced, balance_test_result
- **Triple Difference**: triple_interaction_term, within_comparison, between_comparison,
  placebo_group

### EFFECT TYPE: treatment_effect

```

S.2.2 Method Extraction — Structural Purpose

```

## METHOD PURPOSE: STRUCTURAL

This method estimates STRUCTURAL MODEL PARAMETERS.

### CRITICAL: Set identification_design to null

### STRUCTURAL SPECIFICATION (REQUIRED):

Extract the economic model being estimated:

1. **Model Description**:
- model_description: What economic model is being estimated
- parameters_estimated: List of parameters being recovered (e.g., "discount factor", "risk aversion")
- parameters_externally_set: [
  {
    "parameter_name": "...",
    "parameter_value": "...",
    "source_citation": "Author (Year)",

```

```

        "source_context": "Context of source estimate",
        "sensitivity_tested": true | false,
        "context_specific": true | false
    }
]
- targeted_moments: List of moments targeted in estimation
- untargeted_moments: Optional list of moments used for out-of-sample validation
- estimation_method: SMM | GMM | MLE | calibration | simulation
- moments_or_likelihood: What is used for estimation (moments matched, likelihood function, etc.)

2. **Identification**:
- identifying_restrictions: Economic assumptions that identify parameters
- exclusion_restrictions: Any exclusion restrictions used

3. **STRUCTURAL FIT STATS** -- Extract into 'structural_fit_stats':
- 'structural_fit_stats.estimate_method': "SMM" | "GMM" | "MLE" | "calibration"
- 'structural_fit_stats.n_targeted_moments' (integer)
- 'structural_fit_stats.n_parameters_estimated' (integer)
- 'structural_fit_stats.moments_matched_fraction' (0-1, from reported fit table)
- 'structural_fit_stats.j_statistic', 'structural_fit_stats.j_p_value'
- 'structural_fit_stats.log_likelihood'
- 'structural_fit_stats.calibrated_moments':
    [{"moment_name": "...", "data_value": X, "model_value": Y, "matches": true}]
- 'structural_fit_stats.untargeted_moments_validated' (boolean)
- 'structural_fit_stats.untargeted_moments':
    [{"moment_name": "...", "data_value": X, "model_value": Y}]
- 'structural_fit_stats.counterfactual_validation_pass' (boolean)

Also look for and record in 'model_fit[]':
- J-statistic for overidentification tests
- Likelihood ratio tests
- Comparison to reduced-form estimates

For each test in model_fit:
{{
    "test_name": "j_statistic | overid_test | likelihood_ratio | other",
    "test_statistic": "Reported value",
    "interpretation": "What it shows about model fit"
}}

4. **THEORY / WELFARE STATS** (for calibration/theoretical methods) -- Extract into '
    theory_validation_stats':
- 'theory_validation_stats.calibration_discipline':
    "fully_calibrated" | "partially_calibrated" | "purely_theoretical"
- 'theory_validation_stats.externally_calibrated_parameters':
    [{"parameter_name": "discount_factor", "value": "0.96", "source_citation": "Author (Year)"}]
- 'theory_validation_stats.sensitivity_tests_reported' (boolean)
- 'theory_validation_stats.key_sensitivity_parameters':
    [{"parameter": "sigma", "baseline": 2.0, "sensitivity_range": "1.5-3.0", "result_change_pct": 8}]
- 'theory_validation_stats.welfare_decomposition':
    [{"channel": "price effect", "value": 1.2, "share_of_total_pct": 45, "unit": "percent"}]
- 'theory_validation_stats.comparative_statics':
    [{"variable": "tax rate", "direction": "increase", "magnitude_claim": "10% reduction in output"}]

5. **Counterfactuals**:
- counterfactual_validation: How model is validated
- policy_simulations: If model is used for policy analysis

IMPORTANT: For structural models, always look for counterfactual exercises.
These are often the main contribution (e.g., "removing the policy would reduce welfare by X%").
Extract each counterfactual as a separate entry in policy_simulations.

**Return null (not 0) for any field not explicitly reported.**

### EFFECT TYPE: structural_parameter

```

S.2.3 Method Extraction — Measurement Purpose

```
## METHOD PURPOSE: MEASUREMENT
```

```

This method is NON-CAUSAL. It may be used for:
- classification or prediction to construct variables, OR
- non-causal empirical estimation that documents associations/patterns without a design-based
  identification strategy.

### CRITICAL: Set identification_design to null
### This is NOT a causal method!

### MEASUREMENT SPECIFICATION (REQUIRED):

Extract the measurement or non-causal estimation task:

1. Measurement Task:
  - classification_task: What variable, relationship, or empirical quantity is being measured/
    classified/predicted
  - target_variable: The outcome being predicted or estimated
  - predictor_variables: Features/regressors used
  - model_type: logistic | probit | random_forest | neural_network | llm | other
  - training_procedure: How the model is trained or estimated. For standard regressions with no train/
    test split, say so briefly or set null.
  - measurement_novelty: novel_method | existing_method_new_context | standard_approach
  - measurement_builds_on: Citation text for prior measurement approach (if applicable)

2. Validation (Extract what IS REPORTED):
  Look for validation or fit metrics when reported:
  - Accuracy rate, AUC, F1 score, Precision, Recall
  - Train/test split, cross-validation, holdout validation
  - Human validation (inter-coder reliability, human-ML agreement)
  - For standard non-causal regressions with no predictive validation, use 'validation_type = "none"'
    and leave metrics empty.

  {
    "validation_type": "train_test_split | cross_validation | holdout | none",
    "validation_metrics": [
      {
        "metric_name": "accuracy | auc | f1 | precision | recall",
        "metric_value": "Reported value (e.g., 0.85)",
        "interpretation": "What this tells us"
      }
    ],
    "human_validation": "If compared to human coding",
    "human_agreement": "Agreement rate with human coders"
  }

3. Downstream Use:
  - use_in_analysis: How the measured or predicted quantity is used in subsequent analysis
  - linked_method_ids: Which methods use this constructed variable

### EFFECT TYPE: classification_accuracy for prediction/classification tasks; association for non-causal
  regression or estimation tasks

```

S.2.4 Method Extraction — Decomposition Purpose

```

## METHOD PURPOSE: DECOMPOSITION

This method DECOMPOSES variance or gaps into components.

### CRITICAL: Set identification_design to null

### DECOMPOSITION SPECIFICATION (REQUIRED):

Extract what is being decomposed:

1. Decomposition Details:
  - total_decomposed: What is being decomposed (wage gap, variance, inequality, etc.)
  - decomposition_method: AKM | Oaxaca_Blinder | Kitagawa | Variance_decomposition | other
  - components_identified: [
    {

```

```

        "component_name": "Name (e.g., worker FE, firm FE, sorting)",
        "share_of_total": "Percentage or fraction",
        "interpretation": "What this component represents"
    }
]

2. **Technical Details**:
- bias_corrections: ["Limited mobility bias correction", etc.]
- standard_errors: "How uncertainty is quantified"
- residual_share: "Share not explained by components"

3. **Interpretation**:
- economic_interpretation: "What the decomposition reveals"

### EFFECT TYPE: decomposition_share

```

S.2.5 Method Extraction — Descriptive Purpose

```

## METHOD PURPOSE: DESCRIPTIVE

This method documents ASSOCIATIONS or PATTERNS without causal claims.

### CRITICAL: Set identification_design to null

### ESTIMATION SPECIFICATION:

Extract regression details:

1. **Estimation Method**:
- estimation_method: OLS | logit | probit | quantile | panel_FE | other
- purpose_in_paper: "Documenting patterns | robustness check | auxiliary regression"
- interpretation: "How authors describe results (correlation, association, pattern)"

2. **Specification**:
- controls_included: List of control variables
- fixed_effects: {"unit_fe": true/false, "time_fe": true/false, "other_fe": "description"}
- standard_errors: {"clustering_level": "...", "method": "robust/clustered/bootstrapped"}

3. **Caveats**:
- caveats: "Any caveats authors mention about interpretation"
- why_not_causal: "Why this is not interpreted causally"

### EFFECT TYPE: association

```

S.2.6 Method Extraction — Common Fields (All Purposes)

```

## METHOD INDIVIDUATION -- READ THIS FIRST

This is the single most consequential extraction decision. Two specifications are the
SAME METHOD if they test the same hypothesis using the same identification strategy and
differ only in fixed effects, control variables, or sample restrictions. They become
SEPARATE METHODS only if they use a different instrument, a different treatment variable,
a different outcome variable, or a fundamentally different identification approach.

**Same method (record as specification_variants, NOT separate method entries):**
- A table with 6+ columns varying only FE structures (industry FE, firm FE, bond FE, etc.)
- Adding/removing control variables without changing the core estimation strategy
- Excluding a subset of observations for robustness (e.g., dropping crisis years)

**Separate methods (extract as distinct method entries):**
- The instrument changes (stock return IV vs. industry return IV -- two methods)
- The identification approach changes (OLS vs. 2SLS -- always separate)
- An interaction term tests a distinct moderating hypothesis (price ? coverage vs. price ? volatility)
- The sample restriction tests a distinct prediction (short-maturity only vs. long-maturity only, when
  the point is to test a maturity-specific prediction)

```

****When in doubt:**** Would the paper's argument change if this specification were removed? If yes, it is a separate method. If no, it is a specification variant.

COMMON FIELDS (Extract for all methods):

1. **Research Context:**

- research_question_addressed: What question does this method answer?
- identification_concern: What threat to validity does it address?
- key_assumption: Main identifying assumption or modeling assumption

2. **Argumentative Role & Hierarchy:**

- argumentative_role: What role does this method play in the paper's argument?
Choose ONE: "baseline" | "prediction_test" | "causal_identification" | "mechanism_test" | "heterogeneity_analysis" | "robustness" | "auxiliary"
 - baseline: Establishes the primary empirical fact (first table, main specification)
 - prediction_test: Tests a specific prediction derived from theory or a formal model
 - causal_identification: Establishes causality via exogenous variation (IV, DiD, RDD, RCT)
 - mechanism_test: Tests a proposed causal channel or mediating pathway
 - heterogeneity_analysis: Tests whether treatment effects vary across subgroups, dimensions, or conditions -- including triple differences, interaction-based heterogeneity, and subsample regressions asking "for whom?" or "under what conditions?".
CRITICAL: Use this for ANY method that splits the sample or adds an interaction term to ask "is the effect different for group X?". Do NOT use auxiliary or robustness for these.
 - robustness: Confirms main results survive alternative specifications, samples, or estimators
 - auxiliary: Measurement, data description, construction of variables used by other methods
- builds_on: Method ID that this method extends or validates (e.g., "M1"). Set to null for baseline or auxiliary methods.
Example: An IV method builds_on the baseline OLS it addresses endogeneity for. A mechanism test builds_on the main finding it tests a channel for.

2.5 **Scope, Spillovers, and Mechanism Links:**

- equilibrium_scope: What equilibrium scope does this method assume or test?
Choose ONE: "partial_equilibrium" | "general_equilibrium" | "pe_test" | "structural_ge" | "not_applicable" | "other: [description]"
 - partial_equilibrium: Standard PE assumption; no GE feedback. DEFAULT for reduced-form causal.
 - general_equilibrium: Explicitly accounts for market-clearing feedback across markets.
 - pe_test: Tests whether spillovers contaminate a PE estimate (geographic placebo, donut-hole).
 - structural_ge: Full structural model that solves for a GE (trade model, spatial equilibrium).
 - not_applicable: Descriptive, measurement, or decomposition -- PE/GE distinction irrelevant.Leave null if unresolvable from text.
- spillover_design: Set to null if spillovers are not mentioned for this method.
If spillovers ARE discussed for this method, extract:
 - {
 - "spillover_present": true,
 - "spillover_direction": "within_unit | cross_unit | market_level | geographic",
 - within_unit: Spillovers within the treated unit (e.g. within-firm contagion)
 - cross_unit: Treated units affect untreated units in same market
 - market_level: Price/quantity equilibrium effects that touch all units
 - geographic: Spatial spillovers across geographically adjacent units
 - "spillover_handling": "excluded_by_design | estimated_directly | bounded | tested_placebo | acknowledged_unaddressed"
 - excluded_by_design: Design prevents spillovers (isolated markets, no-overlap geographies)
 - estimated_directly: Spillover magnitude IS a main result in the paper
 - bounded: Paper provides bounds on spillover contamination
 - tested_placebo: Spillover placebo test run on untreated nearby units
 - acknowledged_unaddressed: Mentioned as limitation only, no test
- tests_mechanism_ids: Leave as []. This will be cross-referenced from proposed_mechanisms in post-processing.

3. **Specification Variants (if this method has multiple specifications in the same table):**

- specification_variants: [
 - {
 - "column": "Table 3, Col 1",
 - "fixed_effects": ["industry", "year"],
 - "coefficient": "19.529",
 - "se": "2.499",
 - "significance": "p < 0.01"]

Record each column/specification that varies FE, controls, or sample within the same method.
Omit this field if there is only one specification.

```

- preferred_specification: "Table 3, Col 6" | "not indicated"
  Authors signal preferred specs by: discussing them most, placing them in the final column,
  or using them as the baseline for subsequent analysis. If unclear, set "not indicated".
- preferred_rationale: "Most demanding FE structure" (brief reason, or null)

4. Implementation Details:
- dependent_variable: {"name": "...", "description": "..."}
- key_regressors: [{"name": "...", "description": "...", "expected_sign": "positive|negative|ambiguous"}]
- controls: ["List of control variables"]
- fixed_effects: {"unit": true/false, "time": true/false, "other": "description"}
  NOTE: Fixed effects are implementation details of methods, not separate methods.
  Extract them here but do not create separate method entries for FE.
  Exception: If the paper's PRIMARY identification relies on within-unit variation
  from FE (e.g., within-firm, within-student), note this in identification_design.
- standard_errors: {"clustering_level": "firm/state/etc", "method": "robust/clustered/bootstrapped"}
**REQUIRED**: clustering_level MUST be filled whenever clustering is used. Check TABLE NOTES and
FOOTNOTES first -- papers almost always state clustering there (e.g., "Standard errors clustered at
state level in parentheses"). Common values: "firm", "state", "county", "individual", "school", "
household", "market". The unit of treatment is almost always the clustering level. Use null only
when the paper uses robust but unclustered SEs.
- sample_restrictions: ["Any sample restrictions applied"]

5. Method References:
- method_references: [
  {
    "citation_text": "Author (Year)",
    "method_variant": "What variant they cite (e.g., 'staggered DiD')",
    "context": "How the citation is used",
    "citation_role": "uses_method"
  }
]
- citation_role: Choose EXACTLY ONE label from:
prior_evidence | method_justification | identification_strategy | theoretical_foundation |
estimate_comparison | data_or_measurement | contribution_framing | contextual_motivation |
literature_survey | contradicts
Classify the PRIMARY role this citation plays:
method_justification=borrows their estimator/specification,
identification_strategy=source or validation of the causal identification design (IV/RD/DiD),
data_or_measurement=uses their dataset, index, or variable construction,
theoretical_foundation=inherits their model or theoretical assumption,
... [48 additional lines omitted]

```

S.2.7 Method Extraction (User Prompt Template)

```

Target Method: "{{ method_name }}"
Canonical Name: {{ canonical_name }}
Method ID: {{ method_id }}
Method Type: {{ method_type }}
Method Family: {{ method_family }}
Method Purpose: {{ method_purpose }}
Description: {{ description }}
Source Section IDs: {{ source_section_ids | join(", ") }}

{% if method_references_hint %}
Method References Mentioned: {{ method_references_hint | join("; ") }}
{% endif %}

---

## EXTRACTION TASK:

Extract detailed information for this method from the paper text below.

{% if method_purpose == "causal" %}
**IMPORTANT**: This is a CAUSAL method. You MUST extract the identification_design block with:
- variation_source and variation_source_type
- identifying_assumption and validity_defense
- assumption_tested, assumption_test_type, and what_would_test_it (if untested)

```

```

- validity_tests (extract ALL reported statistical tests with exact values)
- method_specific fields based on method_family
{% elif method_purpose == "structural" %}
**IMPORTANT**: This is a STRUCTURAL method. Set identification_design to null.
Extract the structural_specification block instead, including parameters_externally_set and targeted/
    untargeted moments when reported.
{% elif method_purpose == "measurement" %}
**IMPORTANT**: This is a NON-CAUSAL measurement/estimation method. It may construct variables, classify/
    predict outcomes, or estimate associations without a design-based identification strategy. Set
    identification_design to null.
Extract the measurement_specification block, including measurement_novelty and measurement_builds_on. If
    the method is an association/regression model rather than ML prediction, describe the empirical
    quantity being measured and set validation/training fields to "none" or null when not reported.
{% elif method_purpose == "decomposition" %}
**IMPORTANT**: This is a DECOMPOSITION method. Set identification_design to null.
Extract the decomposition_specification block.
{% else %}
**IMPORTANT**: This is a DESCRIPTIVE/ESTIMATION method. Set identification_design to null.
Extract the estimation_specification block.
{% endif %}

## REQUIRED COMMON FIELDS:

For all methods, extract:
- research_question_addressed: What question does this method answer?
- identification_concern: What threat to validity or research challenge does it address?
- key_assumption: Main assumption (identifying assumption for causal, modeling assumption for others)
- argumentative_role: "baseline" | "prediction_test" | "causal_identification" | "mechanism_test" | "
    robustness" | "auxiliary"
    Use this as the method's primary role in the paper's argument. Causal status is encoded
    separately, so a causal baseline should be labeled by its argumentative use when there is
    ambiguity; future schemas may allow multiple role labels.
- builds_on: Method ID this extends/validates (e.g., "M1"), or null for baseline/auxiliary
- specification_variants: Array of specification variant objects (omit if only one specification)
- preferred_specification: "Table X, Col Y" or "not indicated"
- preferred_rationale: Brief reason for preference (or null)
- dependent_variable: {"name": "...", "description": "..."}
- key_regressors: [{"name": "...", "description": "...", "expected_sign": "positive|negative|ambiguous
    "}]
- controls: List of control variables
- fixed_effects: {"unit": true/false, "time": true/false, "other": "..."}
- standard_errors: {"clustering_level": "...", "method": "..."}
- sample_restrictions: Any sample restrictions
- method_references: Citations to methodological papers
- follows_variant: "standard | modified | novel"
- source_section_ids: Section IDs where method is described
- anchor_start: First 5-10 words of relevant passage (VERBATIM)
- anchor_end: Last 5-10 words of relevant passage (VERBATIM)
- method_components: Optional reusable sub-techniques inside the method. Include when the method uses
    algorithmic, numerical, or data-processing pieces such as TF-IDF, cosine similarity, record linkage
    , clustering, PCA preprocessing, or Bayesian updating. For each component extract: component_kind,
    component_type, component_name, role_in_parent_method, creates_variable, feeds_method_ids,
    source_section_ids, anchor_start, anchor_end. Omit this field if there are no meaningful sub-
    techniques to record.

---

## PAPER TEXT:

{{ paper_text }}

---

Return ONLY valid JSON matching the EmpiricalMethod schema. No markdown code blocks.

```

S.3 Stage 3b: Findings, Limitations, and Open Questions

Model: GPT-5-mini

S.3.1 Findings Extraction (System Prompt)

```
Extract EMPIRICAL FINDINGS, LIMITATIONS, and OPEN QUESTIONS from this economics paper.

PAPER ID: [paper_id]

CRITICAL: Return valid JSON only. No markdown code blocks.

## METHODS USED (from prior extraction) - Link findings to these:

[methods from Stage 3a]
{paper_mode_note}

## EFFECT TYPE GUIDE:

Match the effect type to the identification strategy:

- **IV/2SLS** -> LATE (Local Average Treatment Effect for compliers)
- **DiD/Event Study/TWFE** -> ATT (Average Treatment on Treated)
- **Callaway-Sant'Anna** -> group_time_ATT
- **Sun-Abraham** -> cohort_ATT
- **Sharp RD** -> local_RD
- **Fuzzy RD** -> local_LATE
- **Matching/PSM** -> ATT
- **RCT** -> ATE (Average Treatment Effect)
- **Causal Forest/Double ML** -> CATE (Conditional ATE)
- **Double ML (average)** -> ATE
- **Structural Model** -> structural_parameter
- **Decomposition** -> decomposition_share
- **OLS/Descriptive** -> association (DEFAULT. Use "association" unless the paper employs an explicit
  identification strategy like IV, DiD, RDD, or RCT. Fixed effects alone do NOT make OLS causal. Only
  use ATE if authors explicitly claim causal identification through a quasi-experimental design.)
- **Bunching** -> elasticity
- **Quantile Regression** -> QTE (Quantile Treatment Effect)

## VARIABLE NAMING CONVENTIONS:

For treatment_canonical and outcome_canonical, use standardized names that enable cross-paper linkage.
The canonical name should be the BROADEST RECOGNIZABLE CONCEPT -- something another economist would
recognize as a standard variable.

COMMON TREATMENTS: minimum_wage, class_size, health_insurance, immigration, trade_exposure,
school_voucher, cash_transfer, tax_rate, pollution_exposure, teacher_quality, incarceration,
unemployment_insurance, college_attendance, parental_leave, food_stamps, income_tax, sales_tax,
property_tax, tariff, exchange_rate, monetary_policy, fiscal_stimulus, subsidy, deregulation,
carbon_pricing, vaccination, drug_policy, social_security, sanctions, foreign_aid, microfinance,
housing_policy

COMMON OUTCOMES: employment, wages, earnings, test_scores, graduation_rate, mortality, hospitalization,
crime, firm_entry, productivity, consumption, savings, gpa, college_enrollment,
labor_force_participation, poverty_rate, patent_count, hours_worked, investment, revenue,
bankruptcy, fertility, marriage, migration, emissions, gdp, inequality, social_mobility, welfare,
willingness_to_pay, insurance_coverage, housing_prices, birth_weight

Use these exact names when applicable. For novel treatments/outcomes, use the BROADEST standard concept
(2-3 words max), NOT a paper-specific compound. Put study-specific detail in the treatment_variable
or outcome_variable field instead.

GOOD canonical names: soda_tax, financial_shock, payout_rate, teacher_salary, endowment_return,
permit_price, willingness_to_pay, ecigarette_policy, product_attributes
BAD canonical names (too specific): willingness_to_pay_attributes,
choice_share_ecigarette_vs_combustible, endowment_shock_near_benchmark,
non_price_attributes_public_use_healthier_cessation_aid

Rule of thumb: If the canonical name has more than 3 underscores, it is too specific. Decompose it into
the broadest standard concept. The treatment_variable and outcome_variable fields carry the full
paper-specific detail.

WIKIDATA SEARCH HINTS:
For each treatment_canonical and outcome_canonical, also provide a wikidata_hint -- a plain English
phrase (2-5 words) that would return the right entity on Wikidata/Wikipedia. Use a broader, more
recognizable term. Example: treatment_canonical "soda_tax" -> treatment_wikidata_hint "sugar tax policy
".
```

If the canonical name is already a common term (e.g., "employment"), set the hint to null.

NEVER use "other" as a treatment_canonical or outcome_canonical -- always name the specific variable.

ANCHOR EXTRACTION:

For each entity (finding, limitation, open question), provide anchor text to locate it:

- ****anchor_start**:** First 5-10 words of the relevant sentence/passage (VERBATIM from text)
- ****anchor_end**:** Last 5-10 words of the relevant sentence/passage (VERBATIM from text)
- Together these define a span that can be used for context extraction

EXTRACTION INSTRUCTIONS:

1. EMPIRICAL FINDINGS

Use 'causal_findings' for design-based causal estimates.

Use 'associational_findings' for non-design empirical results that still belong in the paper's main evidence, including:

- predictive accuracy / forecast performance
- benchmarking or model-comparison results
- descriptive gradients, elasticities, or correlations without explicit design-based identification
- decomposition or accounting results that summarize empirical patterns rather than structural counterfactuals
- classifier / validation / out-of-sample performance findings

Do NOT drop a real empirical result just because it is not causal.

1A. CAUSAL FINDINGS

For each empirical result that reports a causal estimate:

- ****finding_id**:** Unique ID (F1, F2, etc.)
- ****statement**:** One-sentence finding statement in plain language. It must describe the estimate produced by ****method_id****. Do NOT call a reduced-form, intent-to-treat, or bounding result "IV" unless the linked method actually estimates IV/2SLS/fuzzy RD.
- ****method_id**:** Which method produced this finding (must match methods list)
- ****method_name**:** Name of the method
- ****effect_type**:** Use the guide above to match method to effect type
- ****causal_claim_strength**:** causal | associational (set to "associational" for OLS/FE/logit patterns without explicit design-based identification)
- ****effect_type_reasoning**:** Brief explanation (e.g., "DiD identifies ATT under parallel trends")
- ****treatment_variable**:** The treatment/policy/intervention
- ****outcome_variable**:** The outcome being measured
- ****causal_direction**:** positive | negative | null | mixed
- ****point_estimate**:** REQUIRED when reported. Extract the EXACT coefficient, percentage, or magnitude as written (e.g., "0.15", "15%", "-2.3 percentage points", "\$500"). NEVER leave null if the paper reports a number in text or tables. Check the referenced table/column for the specific value.
- ****estimate_unit**:** log_points | pp | percent | std_dev | units | dollars
- ****standard_error**:** Extract ONLY when this finding corresponds to one specific reported estimate. Look for parenthesized values below/beside coefficients in tables, "(SE)", "standard error", "s.e.". Extract as numeric (e.g., 0.023). If the finding is a narrative summary spanning multiple estimates /tables/outcomes, set null.
- ****confidence_interval**:** Extract as string "[lower, upper]" (e.g., "[-0.05, 0.12]") ONLY when the interval belongs to this specific estimate. If the finding is a summary roll-up or the interval belongs to just one example inside a broader narrative finding, set null.
- ****p_value**:** Extract significance level as numeric (e.g., 0.01, 0.05) ONLY when it applies to this specific estimate. Look for: p<, p=, asterisks in tables (***=0.01, **=0.05, *=0.10), "significant at the X% level". If the finding summarizes multiple estimates, set null.
- ****is_significant**:** true | false
- ****significance_level**:** "p<0.01" | "p<0.05" | "p<0.10" | "not_significant"
- ****estimate_interpretation**:** "Treatment increases outcome by X%"
- ****practical_significance**:** Include this object ONLY if authors discuss magnitude (see structure below). Omit entirely if no magnitude discussion.
- ****is_main_result**:** true if this is a primary result, false if robustness/auxiliary
- ****graph_layer**:** input | method_core | method_aux | intermediate_result | main_result | robustness | limitation | future_work
- ****node_priority**:** main | secondary | supporting
- ****is_intermediate_result**:** true | false

PRACTICAL_SIGNIFICANCE OBJECT (extract whenever authors interpret magnitude):

Authors often translate estimates into practical terms. LOOK FOR phrases like:
 "equivalent to...", "corresponds to...", "about X% of...", "similar in magnitude to...", "comparable to ...", "implies...", "this means..."

When found, create the **practical_significance** object:

```

'''json
{
  "effect_translation": "0.1 std dev equals $500/year in earnings",
  "baseline_comparison": "Equivalent to 2 additional years of schooling",
  "policy_relevance": "A 10pp policy change would increase outcomes by 5%",
  "magnitude_benchmark": "medium",
  "magnitude_rationale": "Similar to class size effects in Krueger (1999)"
}
'''
... [510 additional lines omitted]

```

S.3.2 Findings Extraction (User Prompt Template)

```

    {% if methods_summary %}
## METHODS USED IN THIS PAPER:

{% for method in methods_summary %}
- {{ method.method_id }}: {{ method.method_name }} ({{ method.method_purpose }})
  Purpose: {{ method.brief_description or method.description or "N/A" }}
{% endfor %}
{% endif %}

---

{% if paper_type == "descriptive" or paper_extraction_mode == "descriptive_only" or not
  has_used_causal_methods %}
## PAPER MODE: DESCRIPTIVE / NON-DESIGN EMPIRICAL

- Do not force causal findings when the linked methods are not explicit identification designs.
- For OLS/FE/logit/time-series/policy-rule/break-date regressions without an explicit design, use
  associational language.
- Leave 'identification_opportunities', 'potential_instruments', and 'proposed_mechanisms' empty unless
  the paper clearly states them.
{% endif %}

---

{% if paper_extraction_mode == "descriptive_only" %}
## PAPER EXTRACTION MODE: DESCRIPTIVE-ONLY

- Prioritize descriptive and associational results, limitations, open questions, and data-grounded
  takeaways.
- Do not force causal findings, instruments, or mechanisms unless the paper states them explicitly.
- Prefer 'associational_findings' for non-design empirical results.
- It is acceptable for 'causal_findings' to contain findings with 'effect_type = "association"' when
  needed, and the pipeline will route them later.
- 'causal_claim_strength' should reflect the paper's wording only, not your guess about the true
  identification strength.
{% elif extraction_mode == "main" %}
## EXTRACTION MODE: MAIN FINDINGS ONLY

- Prioritize headline causal estimates from the main tables/figures.
- Return 'causal_findings', 'associational_findings', and 'null_findings' when present.
- ALWAYS extract 'proposed_mechanisms' if the paper discusses causal channels, mediators, or
  transmission mechanisms -- even in main mode.
- Keep 'heterogeneity_findings', 'robustness_checks', 'limitations', 'open_questions', '
  identification_opportunities', and 'potential_instruments' empty unless absolutely central.
- Do NOT duplicate the same estimate across multiple findings.
{% elif extraction_mode == "secondary" %}
## EXTRACTION MODE: SECONDARY EVIDENCE ONLY

- Prioritize 'heterogeneity_findings', 'null_findings', 'robustness_checks', 'limitations', '
  open_questions', 'proposed_mechanisms', 'identification_opportunities', and 'potential_instruments'
  .

```

```

- Include 'causal_findings' or 'associational_findings' ONLY if needed to link secondary content and
  ONLY when not duplicating the main pass.
- Do NOT repeat headline estimates already extracted in the main pass.
{% endif %}

{% if prior_findings_summary %}
---

## MAIN PASS ESTIMATES ALREADY EXTRACTED (DO NOT REPEAT):

{% for item in prior_findings_summary %}
- {{ item.finding_id or "F?" }} | method={{ item.method_id or "?" }} | treatment={{ item.treatment or
  "?" }} | outcome={{ item.outcome or "?" }} | table={{ item.table_reference or "?" }} | estimate={{
  item.point_estimate or "?" }}
{% endfor %}

Secondary pass rule: do not emit a causal_finding that matches the same method+treatment+outcome+table+
estimate.
{% endif %}

---

## PAPER TEXT:

{{ paper_text }}

---

## EXTRACT AS JSON:

{
  "causal_findings": [
    {
      "finding_id": "F1",
      "statement": "Class size reduction increases test scores.",
      "method_id": "[ID from methods list above]",
      "method_name": "Method name",
      "effect_type": "ATT | LATE | ATE | CATE | local_RD | group_time_ATT | cohort_ATT |
structural_parameter | decomposition_share | association | elasticity | QTE",
      "causal_claim_strength": "causal | associational",
      "effect_type_reasoning": "DiD identifies ATT under parallel trends",
      "treatment_variable": "Treatment name",
      "outcome_variable": "Outcome name",
      "causal_direction": "positive | negative | null | mixed",
      "point_estimate": "0.15 or 15%",
      "estimate_unit": "log_points | pp | percent | std_dev | units | dollars",
      "standard_error": 0.03,
      "confidence_interval": "[0.09, 0.21]",
      "p_value": 0.01,
      "is_significant": true,
      "significance_level": "p<0.01 | p<0.05 | p<0.10",
      "estimate_interpretation": "Treatment increases outcome by X%",
      "economic_magnitude": "Equivalent to Y or null",
      "is_main_result": true,
      "graph_layer": "main_result",
      "node_priority": "main",
      "is_intermediate_result": false,
      "is_heterogeneity": false,
      "subgroup": null,
      "time_horizon": "long_run | short_run | dynamic",
      "equilibrium_scope": "partial_equilibrium | general_equilibrium | structural_ge | not_applicable |
null",
      "tests_prediction_id": null,
      "prediction_confirmed": "confirmed | partially_confirmed | disconfirmed | not_claimed | null",
      "anchor_start": "First few words",
      "anchor_end": "Last few words",
      "table_reference": "Table 3, Column 4",
      "source_section_id": "S5",
      "source_section_ids": ["S5"],
      "data_source_ids": [{"paper_id }_D1"],
      "sample_description": "Full sample",
      "observation_count": "50000",
      "treatment_canonical": "minimum_wage",

```

```

    "treatment_wikidata_hint": null,
    "outcome_canonical": "employment",
    "outcome_wikidata_hint": null,
    "outcome_domain": "labor | education | health | trade | finance | public_finance | development |
io | environment | macro | other: [describe domain]",
    "magnitude_benchmarks": [
      {
        "benchmark_type": "share_of_mean | interquartile_effect | ratio | dollar_value |
comparison_to_prior | standard_deviation_move",
        "calculation": "1 SD increase in treatment raises outcome by X% of baseline mean",
        "baseline_value": "Baseline mean = Y"
      }
    ],
    "parent_finding_id": null,
    ... [298 additional lines omitted]

```

S.4 Stage 3c: Data Sources

Model: GPT-5-mini

S.4.1 Data Sources Extraction (System Prompt)

```

Extract DATA SOURCES from this economics paper.

PAPER ID: [paper_id]

CRITICAL: Return valid JSON only. No markdown code blocks.
CRITICAL JSON RULES:
- Do not emit pseudo-type annotations such as "(int)", "(str)", "(float)", "(bool)", "(list)", "(dict)".
- Every value must be plain JSON (number, string, boolean, null, object, or array).
- Do not include comments or explanatory text.

IMPORTANT: All entity IDs must be prefixed with the paper_id for global uniqueness.
- Format: [paper_id]_D1, [paper_id]_D2, etc.
{methods_block}
{contribution_block}

## DATA CLASSIFICATION TAXONOMY:

### COLLECTION METHOD (primary classification):

1. **administrative**: Government records, tax data, social security
   - Subtypes: government_records, tax_filings, social_security, regulatory_filings, court_records

2. **survey**: Household/firm surveys, questionnaires
   - Subtypes: household_survey, firm_survey, establishment_survey, longitudinal_panel

3. **experimental**: RCT data, lab experiments
   - Subtypes: field_experiment, lab_experiment, natural_experiment_data

4. **observational_digital**: Web scraping, digital traces, sensors
   - Subtypes: web_scraping, social_media, mobile_app, sensor_iot, transaction_digital

5. **text_document**: Text corpora, documents, reports
   - Subtypes: news_articles, corporate_filings, policy_documents, academic_papers, legal_documents

6. **image_media**: Satellite imagery, photos, videos
   - Subtypes: satellite_imagery, street_view, photographs, video_recordings

7. **network_relational**: Network data, relationships
   - Subtypes: social_network, supply_chain, patent_citations, coauthor_network

8. **synthetic_generated**: Simulated or ML-generated data
   - Subtypes: simulation, bootstrap, imputed, ml_generated

9. **census_complete**: Complete enumeration/census data (not sampled)
   - Subtypes: population_census, economic_census, business_register

```

```

10. **proprietary_commercial**: Licensed commercial data products
    - Subtypes: financial_data, credit_bureau, marketing_analytics, healthcare_claims

### DATA STRUCTURE:

- cross_section: One observation per unit at one point in time
- time_series: One unit observed over many periods
- panel_balanced: Multiple units over time, all periods observed
- panel_unbalanced: Multiple units over time, some missing periods
- repeated_cross_section: Different samples at different times
- hierarchical: Nested structure (students in schools in districts)
- spatial: Geographic/location data
- network: Relational data with nodes and edges

### OBSERVATION UNIT:

- individual: People, workers, patients
- household: Families, households
- firm: Companies, businesses
- establishment: Plants, stores, branches
- geographic: Counties, states, zip codes
- transaction: Purchases, trades, contracts
- document: Papers, articles, filings
- other: Specify if different

### LINKED DATA:
If the paper links/merges multiple administrative or survey datasets:
- Create a SEPARATE data_source entry for each original source
- Document the linkage in the data_construction section
- Note the linkage key/method (SSN match, fuzzy name match, geographic merge, etc.)

## SURVEY-SPECIFIC FIELDS (only if collection_method = survey):

- **survey_response_format**: open_ended | closed_categorical | likert_ordinal | numeric_continuous |
  mixed_format
- **survey_mode**: capi (computer-assisted personal) | cati (phone) | cawi (web) | mail_paper |
  mixed_mode
- **survey_sampling**: simple_random | stratified | cluster | multistage | convenience | quota | census
- **survey_experiment_type**: conjoint | vignette | discrete_choice_experiment | contingent_valuation |
  list_experiment | none
  (Use when the survey ITSELF contains an embedded experiment)

## TEXT-SPECIFIC FIELDS (only if collection_method = text_document):

- **text_annotation_status**: human_expert | human_crowdsourced | llm_labeled | hybrid_human_llm |
  existing_metadata | unlabeled

## WELL-KNOWN DATASETS (use these canonical_names when matched):

US: ACS, ATUS, BRFSS, CCD, CPS, CRDC, HCUP, HRS, IPEDS, IPUMS, LEHD, MEPS, NHANES, NHIS, NLSY79, NLSY97,
    NSDUH, PISA, PSID, QWI, SAIPE, SIPP, SSA, TIMSS
International: LISS, BHPS, SOEP, ECHP, LFS, SHARE, EU-SILC

If the paper uses one of these, set canonical_name to the acronym above.

## MACHINE LEARNING LINKAGE:

If ML/LLM methods were applied to this data source:
- **ml_methods_applied**: List of ML method IDs (e.g., ["paper_id_ML1"])
- **llm_methods_applied**: List of LLM method IDs (e.g., ["paper_id_LLM1"])
- **variables_created_by_ml**: List of variable names created (e.g., ["sentiment_score", "topic_label"])

## DOWNSTREAM USAGE:

- **used_in_empirical_methods**: List of method IDs that use this data (e.g., ["paper_id_M1", "paper_id_M2"])
- **source_usage_status**: used_in_paper | suggested_for_future | mentioned_not_exploited
  - used_in_paper: the paper actually uses this source in an executed empirical method
  - suggested_for_future: the source is described as useful for future work / later versions but is not used now
  - mentioned_not_exploited: the source is concretely discussed but not actually used in the paper's own estimation

```

- Do NOT key this only on words like "propose" or "proposed". Many papers propose a measure or dataset and then use it. Use explicit future-work or non-use context.
- ****source_scope****: focal_paper | cited_literature | review_methodology_discussion | hypothetical_example
 - focal_paper: the focal paper directly uses or constructs this source
 - cited_literature: this row describes a dataset from another paper or cited study, not the focal paper's own data
 - review_methodology_discussion: this is a teaching/example dataset used to explain methods in a review/chapter
 - hypothetical_example: this is a hypothetical or illustrative source, not real focal data
- ****evidence_basis****: direct_use | appendix_data_note | literature_review | calibration_target | discussion_only
- ****source_granularity****: atomic_source | constructed_bundle | derived_measure_only
 - atomic_source: one concrete external source a replicator could locate
 - constructed_bundle: an author-built bundle/panel/index assembled from multiple named sources

... [211 additional lines omitted]

S.4.2 Data Sources Extraction (User Prompt Template)

```
## PAPER TEXT:

{{ paper_text }}

---

## EXTRACT AS JSON:

Strict output rules:
- Return JSON only (no prose, no markdown fences).
- Do not include pseudo-type tokens such as '(int)', '(str)', '(float)', '(bool)', '(list)', '(dict)'.

{
  "data_sources": [
    {
      "data_source_id": "{{ paper_id }}_D1",
      "name": "Full official name of dataset",
      "short_name": "Acronym or null",
      "canonical_name": "Standardized acronym if well-known (e.g., CPS, ACS, PSID) or null",
      "alternative_names": ["Alternate names used in paper"],
      "source_identity": "Concrete source a replicator can locate/request",
      "derived_variables": [
        {
          "var_name": "constructed_var_1",
          "var_label": "Constructed variable label",
          "construction": "How the variable was built from the raw source",
          "unit": null,
          "used_in_methods": ["{{ paper_id }}_M1"],
          "is_new_variable_created_by_authors": true,
          "variable_contribution_type": "constructed_index",
          "linked_contribution_claim_ids": ["{{ paper_id }}_C1"]
        }
      ],
      "is_new_dataset_created_by_authors": true,
      "data_contribution_type": "existing_dataset",
      "linked_contribution_claim_ids": ["{{ paper_id }}_C1"],
      "doi_or_registry_id": "DOI/registry id or unresolvable_from_text",
      "administering_organization": "Organization or unresolvable_from_text",
      "in_paper_citation": "Exact citation used in paper for this source",
      "identity_unknown_reason": "If canonical identity is unclear, explain why; otherwise null",

      "collection_method": "survey",
      "collection_subtype": "household_survey",
      "data_structure": "panel_unbalanced",
      "observation_unit": "individual",

      "survey_response_format": "mixed_format",
      "survey_mode": "cawi",
      "survey_sampling": "stratified",
      "survey_experiment_type": "none",
    }
  ]
}
```

```

"text_annotation_status": null,

"geographic_scope": "Country/region",
"time_period": "Start year - End year",
"sample_size": 125000,

"ml_methods_applied": [{"paper_id"}_ML1],
"llm_methods_applied": [{"paper_id"}_LLM1],
"variables_created_by_ml": ["sentiment_score"],

"used_in_empirical_methods": [{"paper_id"}_M1],
"source_usage_status": "used_in_paper",
"source_scope": "focal_paper",
"evidence_basis": "direct_use",
"source_granularity": "atomic_source",
"component_source_labels": ["Underlying Source 1", "Underlying Source 2"],
"explicit_focal_use": true,
"support_chunk_ids": ["C12", "C18"],
"source_anchor_type": "main_text",
"is_appendix_example": false,
"is_cited_study_data": false,
"used_in_tables_figures": ["Table 2", "Figure 1"],
"linkage_to": [{"paper_id"}_D2],
"linked_sample_size": 2502,
"access_tier": "public",
"role_in_paper": "primary_outcome_source",
"data_role_in_paper": "focal_estimation_input",
"measurement_task": "outcome_measurement",
"why_this_source": "Why this source was selected over alternatives",
"how_used_in_empirics": "How this source is operationally used in identification/measurement",
"ml_usage_justification": "If ML is used, explain why ML was needed for this source; else null",

"data_references": [
  {
    "citation_text": "Ruggles et al. (2020)",
    "data_source_name": "IPUMS-CPS",
    "context": "We use data from..."
  }
],

"source_section_ids": ["S3", "S4"],
"anchor_start": "First few words",
"anchor_end": "Last few words",

"_post_processing": {
  "resolved_citations": null,
  "registry_match": null,
  "cluster_id": null,
  "embedding_id": null
}
},

"data_construction": [
  {
    "construction_id": [{"paper_id"}_DC1],
    "construction_type": "merge",
    "description": "How variables were constructed",
    "input_sources": [{"paper_id"}_D1, [{"paper_id"}_D2],
    "output_variable": "Name of constructed variable",
    "target_variables": ["employment_rate"],
    "methodology": "Merge",
    "rule_or_formula": "Real wages = nominal wages / CPI",
    "parameters_text": "Winsorized at the 1st and 99th percentiles",
    "transformation_steps": ["logged outcome", "deflated to 2005 dollars"],
    "missing_data_handling": ["Dropped missing wage observations"],
    "outlier_handling": ["Winsorized top and bottom 1%"],
    "weighting_details": ["Applied person weights"],
    "normalization_steps": ["Standardized within year"],
    ... [30 additional lines omitted]
  }
]

```

S.5 Stage 3 (Theory): Theory Component Extraction

Model: GPT-5-mini

S.5.1 Theory Component Extraction (System Prompt)

```
Extract THEORETICAL COMPONENTS from this economics paper.

PAPER ID: [paper_id]

CRITICAL: Return valid JSON only. No markdown code blocks.

Extract the key theoretical building blocks and keep each component concise.
Do not invent components that are not supported by text.

## COMPONENT TYPES

Use one of:
- assumption
- definition
- proposition
- theorem
- lemma
- corollary
- mechanism
- model_structure
- equilibrium_condition
- comparative_static
- identification_argument
- boundary_condition
- prediction
- counterfactual
- welfare_implication
- policy_implication
- limitation
- open_question
- other

If a component type is plausible but outside this list, use lower_snake_case.

## EXTRACTION RULES

1. component_id format: [paper_id]_T1, [paper_id]_T2, ...
2. component_text should be short and faithful to paper wording.
3. source_section_ids must map to section IDs in provided paper text.
4. Include anchor_start and anchor_end when the component is stated in a localized sentence or paragraph
.
5. Prefer fewer high-quality components over many weak ones.
6. Add optional semantic hints when the paper makes them explicit:
  - proposition_class for propositions, theorems, comparative statics, or predictions
  - mechanism_class for mechanisms or channels
  - implication_type for policy, welfare, counterfactual, or prediction implications
  - open_question_type for future-work or unresolved-question components
  - scenario_tag for named scenarios, reforms, or counterfactual labels

## OUTPUT JSON SCHEMA

{{
  "theory_components": [
    {{
      "component_id": "[paper_id]_T1",
      "component_type": "assumption",
      "component_text": "Agents maximize expected utility.",
      "proposition_class": null,
      "mechanism_class": null,
      "implication_type": null,
      "open_question_type": null,
      "scenario_tag": null,
      "source_section_ids": ["S2", "S3"],
      "anchor_start": "Agents maximize",
      "anchor_end": "expected utility."
    }}
  ]
}}
```

```

    }}
  ]
}}

```

Return ONLY valid JSON.

S.5.2 Theory Component Extraction (User Prompt Template)

```

    PAPER ID: {{ paper_id }}

PAPER TEXT:
{{ paper_text }}

Extract theoretical components according to the system instructions.
Return ONLY valid JSON matching:
{
  "theory_components": [
    {
      "component_id": "{{paper_id}}_T1",
      "component_type": "assumption | definition | proposition | theorem | lemma | corollary | mechanism
      | model_structure | equilibrium_condition | comparative_static | identification_argument |
      boundary_condition | prediction | counterfactual | welfare_implication | policy_implication |
      limitation | open_question | other",
      "component_text": "Concise component statement",
      "proposition_class": "optional concise subtype",
      "mechanism_class": "optional concise subtype",
      "implication_type": "optional concise subtype",
      "open_question_type": "optional concise subtype",
      "scenario_tag": "optional named scenario or reform tag",
      "source_section_ids": ["S2", "S3"],
      "anchor_start": "First few words",
      "anchor_end": "Last few words"
    }
  ]
}

```

S.6 Stage 4: Within-Paper Synthesis and Entity Linking

Model: GPT-5-mini

S.6.1 Within-Paper Synthesis and Entity Linking (System Prompt)

```

    WITHIN-PAPER ENTITY LINKING for [paper_id]

Create a knowledge graph linking entities EXPLICITLY mentioned in this paper.

DO NOT suggest or generate new ideas - ONLY extract relationships stated in the paper.

## METHODS EXTRACTED:

[extracted entities from prior stages]

## FINDINGS EXTRACTED:

[extracted entities from prior stages]

## LIMITATIONS IDENTIFIED:

[extracted entities from prior stages]

## OPEN QUESTIONS:

[extracted entities from prior stages]

```

```

## DATA SOURCES:

[extracted entities from prior stages]
{theory_section}
## IDENTIFICATION OPPORTUNITIES:

[extracted entities from prior stages]

## TABLE / EQUATION / FIGURE EVIDENCE:

[extracted entities from prior stages]

---

## TASK: Link these entities based on what the paper EXPLICITLY states

### 1. ENTITY EDGES (REQUIRED)

Create edges ONLY for relationships explicitly described in the paper text.
Every edge must have evidence quoting or paraphrasing the paper.

### 2. RESEARCH GAPS (FROM PAPER ONLY)

Extract research gaps ONLY if the paper explicitly states them in:
- "Future work could..."
- "We cannot address..."
- "A limitation of our approach..."
- "This leaves open the question..."

DO NOT generate gaps that aren't explicitly stated.

### 3. REPLICATION REQUIREMENTS (FROM PAPER ONLY)

Extract what the paper explicitly states is needed to replicate the analysis.

### 4. IDENTIFICATION RIGOR ASSESSMENT

Classify this paper's highest identification rigor level based on the methods ACTUALLY USED (not merely
discussed):

- "experimental": RCT, field experiment, lab experiment with random assignment
- "quasi_experimental_strong": DID, RDD, IV with validated instruments, synthetic control, event study
  with clear exogenous treatment
- "quasi_experimental_weak": Fixed effects, matching, propensity score without quasi-experimental design
- "structural": Structural model estimated via MLE, GMM, simulated methods
- "stated_preference": DCE, contingent valuation, choice experiment
- "descriptive": OLS associations, descriptive statistics without causal claim
- "theoretical": Mathematical model, proofs, propositions without empirical estimation

Provide confidence (0.0-1.0) and reasoning citing specific paper evidence.

### 5. STRATEGY STATUS ASSESSMENT

For each identification opportunity listed above, classify whether the paper:
- "exploited": Actually uses this as an identification strategy in estimation
- "mentioned": Discusses or motivates this strategy but does not implement it
- "proposed": Explicitly proposes this for future work

Provide confidence and reasoning for each.

### 6. METHOD FAMILY CORRECTIONS (if any)

Review the method_family assignments above. If any are incorrect based on the paper's actual methodology
, suggest corrections. Only flag genuine misclassifications -- do not change families that are
reasonable categorizations.

Return an empty array if all families are correct.

---

## OUTPUT FORMAT (JSON only):

```

```

{{
  "entity_edges": [
    {{
      "source_id": "[paper_id]_D1",
      "target_id": "[paper_id]_M1",
      "edge_type": "direct_input_to",
      "evidence": "Quote or paraphrase from paper showing this relationship",
      "confidence": 0.9
    }}
  ],

  "research_gaps_explicit": [
    {{
      "gap_id": "RG1",
      "gap_type": "external_validity | mechanism | measurement | identification | scope",
      "description": "What the paper EXPLICITLY states is a gap or limitation",
      "source_limitation_ids": ["L1"],
      "source_question_ids": ["Q1"],
      "anchor_text": "Direct quote from paper identifying this gap",
      "source_section_id": "S15"
    }}
  ],

  "replication_requirements": {{
    "data_requirements": ["List of data elements paper says are needed"],
    "identification_assumptions": ["Assumptions paper explicitly states"],
    "validity_conditions": ["Conditions paper says must hold"]
  }},

  "identification_rigor": {{
    ... [66 additional lines omitted]
  }}

```

S.6.2 Theory Paper Synthesis (System Prompt)

```

WITHIN-PAPER THEORY SYNTHESIS for [paper_id]

This is a THEORETICAL paper. Create a knowledge graph linking theory components
based on the logical structure EXPLICITLY presented in the paper.

## THEORY COMPONENTS EXTRACTED:

[extracted entities]

## EQUATION / TABLE / FIGURE EVIDENCE:

[extracted entities]

---

## TASK: Link theory components based on the paper's logical flow

### 1. ENTITY EDGES (REQUIRED)

Create edges showing how components build on each other:
- Which assumptions ground which propositions/theorems?
- Which model structures produce which equilibrium conditions?
- Which equilibrium conditions lead to which comparative statics/predictions?
- Which results have which welfare/policy implications?
- Which assumptions create which limitations?

Every edge must have evidence quoting or paraphrasing the paper.

### 2. RESEARCH GAPS (FROM PAPER ONLY)

Extract research gaps ONLY if the paper explicitly states them in:
- "Future work could..."
- "We cannot address..."
- "A limitation of our approach..."
- "This leaves open the question..."

```

DO NOT generate gaps that aren't explicitly stated.

3. REPLICATION REQUIREMENTS (FROM PAPER ONLY)

For theoretical papers: key assumptions that must hold, boundary conditions, and what extensions the paper explicitly identifies.

4. IDENTIFICATION RIGOR ASSESSMENT

Classify this paper's identification rigor level. For a theoretical paper this is typically "theoretical", but confirm based on the actual content:

- "experimental": RCT, field experiment, lab experiment with random assignment
- "quasi_experimental_strong": DID, RDD, IV with validated instruments, synthetic control
- "quasi_experimental_weak": Fixed effects, matching, propensity score
- "structural": Structural model estimated via MLE, GMM, simulated methods
- "stated_preference": DCE, contingent valuation, choice experiment
- "descriptive": OLS associations, descriptive statistics without causal claim
- "theoretical": Mathematical model, proofs, propositions without empirical estimation

Provide confidence (0.0-1.0) and reasoning.

OUTPUT FORMAT (JSON only):

```
{
  "entity_edges": [
    {
      "source_id": "[paper_id]_T1",
      "target_id": "[paper_id]_T3",
      "edge_type": "theoretically_grounds",
      "evidence": "Quote or paraphrase from paper showing this relationship",
      "confidence": 0.9
    }
  ],
  "research_gaps_explicit": [
    {
      "gap_id": "RG1",
      "gap_type": "external_validity | mechanism | measurement | identification | scope",
      "description": "What the paper EXPLICITLY states is a gap or limitation",
      "source_limitation_ids": ["T14"],
      "source_question_ids": ["T15"],
      "anchor_text": "Direct quote from paper identifying this gap",
      "source_section_id": "S10"
    }
  ],
  "replication_requirements": {
    "data_requirements": [],
    "identification_assumptions": ["Key assumptions paper explicitly states must hold"],
    "validity_conditions": ["Conditions paper says must hold for results"]
  },
  "identification_rigor": {
    "level": "theoretical",
    "confidence": 0.95,
    "reasoning": "Why this level, citing paper evidence"
  }
}
```

VALID EDGE TYPES (use only these):

- theoretically_grounds: assumption/definition grounds a proposition/theorem/model
- produces: model_structure/equilibrium produces comparative_static/prediction/implication
- qualified_by: result is qualified by limitation/boundary_condition
- motivates: limitation/open_question motivates further work
- operationalizes: one component operationalizes/implements another
- evidenced_by_equation: component is formalized in an equation
- evidenced_by_table: component references a table
- evidenced_by_figure: component references a figure

```
## CRITICAL RULES:
```

1. ONLY create edges for relationships STATED in the paper
2. Every edge MUST have evidence from paper text
3. Follow the paper's logical structure -- don't invent connections
4. DO NOT estimate "tractability" or "difficulty"

S.7 Stage 4.75: Artifact Enrichment

Model: GPT-5-mini

S.7.1 Artifact Enrichment — Table/Figure/Equation Linking

You are a research paper analyst specializing in linking tables, figures, and equations to the methods and findings that use them.

```
## Task
```

Given a list of **isolated artifacts** (tables, figures, or equations that have no edges to any method or finding) and a list of **available entities** (methods, findings, data sources), determine which entities each artifact is associated with.

```
## Rules
```

1. Only create an edge if there is clear evidence the entity references or uses the artifact.
2. Use 'edge_type' values:
 - 'evidenced_by_table' -- a finding or method references results shown in a table
 - 'evidenced_by_figure' -- a finding or method references a figure
 - 'evidenced_by_equation' -- a method or finding references an equation
3. The 'source_id' must be the entity (method_id, finding_id, or data_source_id).
4. The 'target_id' must be the artifact_id.
5. Provide brief 'evidence' text explaining the link.
6. Set 'confidence' between 0.5 and 0.95. Use higher values when the text explicitly names the artifact.

```
## Output Format
```

Return ONLY valid JSON with this structure:

```
'''json
{
  "artifact_edges": [
    {
      "artifact_id": "TAB...",
      "entity_id": "M1 or F3 etc.",
      "edge_type": "evidenced_by_table",
      "evidence": "brief explanation",
      "confidence": 0.75
    }
  ]
}
'''
```

If no links can be determined, return: {"artifact_edges": []}

S.8 Stage 4.76: Precision Enrichment

Model: GPT-5-mini

S.8.1 Precision Enrichment — SE/CI/p-value/N Fill

You are extracting statistical precision from economics regression tables.

```

Rules:
- Standard errors appear as values in parentheses ( ) directly below coefficient estimates.
- Observation counts appear as values in brackets [ ] next to estimates, or in an "Observations" row.
- Confidence intervals are rarely reported; return null if not present.
- To find the correct row: match BOTH the "outcome" field (e.g. "soda_purchases_per_trip" -> row labeled "Soda") AND the estimate value (e.g. "-6.1" -> find the cell closest to -6.1 in that row).
- Each outcome spans 3 lines in markdown: (1) coefficient, (2) SE in parentheses on the next line, (3) observation count in brackets on the third line.
- If you can identify the row with high confidence, set confidence="high". Otherwise set confidence="low" and return null.
- Return null (not 0) for any field you cannot identify with reasonable confidence.

FINDINGS NEEDING ENRICHMENT:
[JSON list of findings with null fields]

REGRESSION TABLES:
[Regression table markdown from Stage 4.7]

Return ONLY valid JSON:
{
  "findings": [
    {
      "finding_id": "F1",
      "standard_error": <number or null>,
      "confidence_interval": <[lower, upper] or null>,
      "p_value": <number or null>,
      "observation_count": <integer or null>,
      "column_matched": "<column header you used or null>",
      "row_matched": "<row label from table or null>",
      "confidence": "<high|low>"
    }
  ]
}

```

S.9 Stage 4.77: Method Diagnostic Stat Enrichment

Model: GPT-5-mini

S.9.1 Method Diagnostic Stat Enrichment — RDD/IV/DiD/Synth Stats

```

You are extracting diagnostic statistics from economics paper tables.

Method: [method_id] -- [method_name] (family: [method_family])

Fields to fill (currently null):
[null target fields]

Field descriptions:
- iv_stats.first_stage_f_stat: F-statistic FROM THE FIRST-STAGE regression of the endogenous variable on instruments. Typically reported in a dedicated "First-Stage" table or table note. MUST be explicitly labeled as an F-statistic, not an observation count. Plausible range: 1-500.
- iv_stats.kleibergen_paap_rk_stat: KP rank stat (heteroskedasticity-robust instrument strength). Plausible range: 1-500.
- iv_stats.sargan_p_value: Sargan overidentification test p-value (0-1).
- iv_stats.partial_r_squared_first_stage: Partial R-squared of instruments in first stage (0-1).
- did_stats.parallel_trends_f_stat: Joint F-stat that pre-period coefs are jointly zero. Plausible range : 0.01-50.
- did_stats.parallel_trends_p_value: P-value for joint pre-trend test (0-1).
- did_stats.pre_period_coefficients: ONLY populate when the table explicitly shows separate coefficients for distinct PRE-TREATMENT TIME PERIODS (e.g., t=-3, t=-2, t=-1 relative to treatment event). The table MUST have period labels like "-2", "-1", "Two periods before", etc. DO NOT extract from: (a) balance/covariate comparison tables (comparing treated vs. control means), (b) summary statistics tables, (c) cross-sectional regressions without a time-period dimension. If no such time-indexed pre-treatment table exists, return null (not []).
- rdd_stats.mccrary_p_value: P-value for McCrary (2008) density continuity test (0-1).
- rdd_stats.bandwidth_numeric: Bandwidth value (numeric, small positive number).

```

```

- rdd_stats.effective_n_left / effective_n_right: Obs within bandwidth on each side.
- synth_control_stats.pre_treatment_rmspe: Pre-treatment RMSPE (small positive number).
- synth_control_stats.rmspe_ratio: Post/pre RMSPE ratio (positive number).
- synth_control_stats.permutation_p_value: Permutation test p-value (0-1).
- matching_stats.max_standardized_mean_diff_after: Max SMD across covariates after matching (typically 0-0.25).
- matching_stats.balancing_test_p_value: P-value from balance test (0-1).
- ols_panel_stats.r_squared: R-squared of regression (0-1). Look for "R2", "Adjusted R2".
- ols_panel_stats.within_r_squared: Within R-squared for panel FE regression (0-1).
- ols_panel_stats.overall_f_stat: Joint F-statistic for the regression model. Plausible range: 1-10000.
  IMPORTANT: Do NOT confuse with "Observations" row (e.g., 176,527 is the sample size, not an F-stat)
- ols_panel_stats.n_clusters: NUMBER OF CLUSTER GROUPS, NOT observations. This is the count of distinct clusters (e.g., 50 states, 500 firms, 12 industries). Typically 2-5000. Values > 10,000 are almost certainly observation counts -- return null in that case.

Rules:
- Return null (not 0 and not []) for any field not explicitly reported.
- For pre_period_coefficients: return null if the table does not show separate pre-period estimates.
  Only return an array if actual coefficient values are present.
- For numeric fields: return a number, not a string.
- NEVER confuse the "Observations" row (sample size, often 10,000-1,000,000) with F-statistics or cluster counts.
- F-statistics are small numbers (typically 1-500). If a value looks like a sample size, it is NOT an F-statistic.

TABLES:
[Table markdown]

Return ONLY valid JSON:
{
  "method_id": "[method_id]",
  "diagnostic_stats": {
    "iv_stats": {},
    "did_stats": {},
    "rdd_stats": {},
    "synth_control_stats": {},
    "matching_stats": {},
    "ols_panel_stats": {}
  },
  "notes": "brief note about where values were found or why null"
}

Only include the sub-objects relevant to this method_family. Omit irrelevant sub-objects.

## SELF-CHECK -- Before finalizing your JSON:

For each numeric value you are about to return:
1. Can you point to the exact cell in the table above? (If no, return null)
2. Is the field label in the table the same as what you're extracting?
  (F-stat ? Observations; n_clusters ? sample size)
3. Is the value in the expected range?
  - F-statistics: 0.1-500 (values > 5000 are observation counts, not F-stats)
  - n_clusters: 2-10,000 (values > 50,000 are observation counts)
  - p-values: 0-1 (values > 1 are impossible)
  - R-squared: 0-1
4. For pre_period_coefficients: does the table have explicit period labels (t=-2, t=-1)?

Return null for any field that fails checks 1-4.

```

S.10 Stage 4.78: Table Structure Enrichment

Model: GPT-5-mini

S.10.1 Table Structure Enrichment — SE type, FE levels, cluster level

You are reading an economics regression table from an academic paper.

Table: [table_label]

Full table content (markdown):
[Table markdown excerpt]

Extract the following fields. Look at ALL parts of the table including:

- Column headers (specification names, sample labels)
- Row labels (variable names, panel headings)
- Parenthetical values below estimates (standard errors)
- Notes/footnotes at the bottom of the table
- Significance stars (*, **, ***)

Return ONLY valid JSON (no explanation, no markdown code blocks):

```
{
  "table_type": "regression | summary_stats | first_stage | balance | event_study | correlation |
  missingness | outlier_diagnostic | sample_construction | other",
  "column_roles": ["Brief description of what each column represents, in order. E.g. ['OLS baseline', '
  IV', 'Reduced form', 'Female subsample']"],
  "se_type": "robust | clustered | bootstrapped | OLS | not_reported | null",
  "cluster_level": "The EXACT noun entity at which SEs are clustered (e.g. 'firm', 'state', 'county', '
  individual', 'school district'). Extract only the entity name, not the full phrase. null if not
  clustered or not stated.",
  "fe_levels": ["List of fixed effects included. Use specific level names like 'firm FE', 'year FE', '
  industry-year FE'. Empty list [] if none stated."],
  "has_r_squared": true_or_false,
  "has_f_stat": true_or_false,
  "typical_n_obs": "The N (observations) value from the most common specification, as a string (e.g.
  '45,231'). null if not reported.",
  "significance_notation": "The star notation used, e.g. '* p<0.1 ** p<0.05 *** p<0.01'. null if not
  stated.",
  "variables_covered": ["Variable names or groups covered by the table. Empty list [] if unclear."],
  "notes_text": "The full text of table notes/footnotes, verbatim. null if none visible.",
  "sample_description": "Any sample restriction or weighting description from notes (1 sentence max).
  null if none.",
  "weighting_note": "Any note about weights, weighted estimates, survey weights, or weighted samples (1
  sentence max). null if none.",
  "missingness_note": "Any note about missing observations, missingness rules, or attrition handling (1
  sentence max). null if none.",
  "outlier_note": "Any note about trimming, winsorization, top-coding, or outlier exclusions (1 sentence
  max). null if none.",
  "transformation_note": "Any note about logs, normalization, deflation, rescaling, standardization, or
  constructed indices (1 sentence max). null if none."
}
```